Continuous Hidden Markov Models for Equity Returns: Heavy-Tail Emission Families and Regime-Conditional Value-at-Risk

Abdulrahman Alswaidan[1] Cade Jin[2] Jeffrey D. Varner*[1]

[1]Robert Frederick Smith School of Chemical and Biomolecular Engineering, Cornell University, Ithaca, NY 14850, USA

[2]Cornell Ann S. Bowers College of Computing and Information Science, Cornell University, Ithaca, NY 14850, USA

Abstract

Synthetic generators of daily US-equity returns support stress testing, regulatory backtesting, and downstream model training on data that cannot be released for privacy or licensing reasons, but the regime-switching route has been shaped for two decades by a negative result: low-state-count hidden Markov models with Gaussian emissions fail to reproduce the slow decay of the absolute-return autocorrelation function. We revisited that result through a continuous hidden Markov model framework in which a textbook mixture-of-eigenvalues identity recasts the failure as an effective-rank statement on the centred transition matrix, and an empirical diagnostic on the fitted estimator showed that the implied rank bound is non-binding at the cross-ticker median once enough latent states are admitted, shifting the binding axis from the temporal kernel to the marginal mixture. We developed a unified expectation-conditional-maximisation scaffold supporting four heavy-tailed emission families (Gaussian, Student-t, Laplace, and Generalised Error Distribution) under shared forward-backward recursions and quantile-based initialisation, with the maximisation step the only architectural difference between families. The empirical study covered a decade of daily SPY data, a sector-balanced 30-ticker panel, a CRSP cross-decade transfer between non-overlapping ten-year and two-year windows, and a six-asset US-equity copula extension. We built a regime-conditional Value-at-Risk that propagates the one-step-ahead state forecast through the family-appropriate predictive mixture, and validated it under the Christoffersen joint conditional-coverage test alongside an Engle-Manganelli dynamic-quantile backstop. The construction is honestly positioned against an i.i.d. bootstrap and a maximum-likelihood hidden semi-Markov benchmark that exceed it on raw single-window distributional fidelity; the continuous HMM's value lies in the structural use cases those alternatives cannot serve, namely regime-conditional risk forecasting, multi-asset copula composition, and parametric synthetic-data privacy. The scope is daily US equities under stationary out-of-sample regimes, and periodic refit is the deployment protocol for stress folds the static fit cannot bracket.

Keywords: Continuous Hidden Markov Model, Baum-Welch, Student-t Emissions, Stylized Facts, Volatility Clustering, Value-at-Risk, Synthetic-Data Generator.

*Corresponding author: `jdv27@cornell.edu`. Coauthor contacts: `aa2725@cornell.edu` (A.A.), `cj383@cornell.edu` (C.J.).

1 Introduction

Synthetic generators of equity-return time series support stress testing, regulatory backtesting, scenario design, and the training of downstream machine-learning models on data that cannot be released for privacy or licensing reasons [1–3]. The bar for a useful generator is that it reproduce, simultaneously, the canonical stylized facts of daily returns: heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the autocorrelation function (ACF) of absolute returns [4, 5]. Generalized autoregressive conditional heteroskedasticity (GARCH) models [6, 7] capture volatility clustering but not the multi-regime structure of the marginal; i.i.d. generators match the marginal but destroy temporal persistence; deep generators [8–10] reproduce some stylized facts at the cost of interpretability and reproducibility. An influential negative result has shaped two decades of work on the regime-switching route between these tiers. Rydén et al. [11] fitted Gaussian-emission hidden Markov models (HMMs) at low state counts and concluded that the geometric sojourn-time distribution of a standard Markov chain produces exponential ACF decay too fast to match the observed slow decay; Bulla and Bulla [12] restored ACF fidelity by replacing the Markov chain with a hidden semi-Markov chain at the cost of substantially increased complexity. The empirical Rydén et al. failure compounds two constraints: a temporal one (the absolute-return ACF must admit slow-decay modes) and a distributional one (the marginal mixture must carry enough components to match observed kurtosis). The closed-form mixture-of-eigenvalues identity for the absolute-return ACF over the non-unit eigenvalues of the transition matrix [13–15] gives an algebraic upper bound on the number of decay modes through the rank of the centred transition matrix, and the central question of this paper is whether that bound binds on equity-return data once enough latent states are admitted.

We instantiate this question on a decade of daily SPY exchange-traded fund (ETF) data with a multi-year held-out window, using a continuous HMM (CHMM) trained by Baum-Welch [16, 17] under quantile-based initialization in four emission variants (Gaussian, per-state Student-t in the tradition of [18, 19], Laplace, and the Generalised Error Distribution, GED) that share identical forward-backward recursions and differ only in the M-step. The empirical scope is daily US equities under stationary out-of-sample regimes: a non-equity stress test on the GLD ETF collapses under static fitting, a documented scope limit rather than a deferred follow-up. We extend the single-asset construction along four orthogonal generalisation axes: a six-fold rolling-origin walk-forward on the SPY decade, a sector-balanced 30-ticker panel spanning ten Global Industry Classification Standard (GICS) sectors, a Center for Research in Security Prices (CRSP) cross-decade transfer between a non-overlapping ten-year in-sample window and a two-year out-of-sample slice, and a six-asset US-equity Student-t copula extension that composes per-asset CHMM marginals into a joint generator through Sklar's theorem. The construction is honestly positioned: the i.i.d. bootstrap and a maximum-likelihood hidden semi-Markov model (HSMM) both exceed the CHMM on raw single-window distributional fidelity, but neither admits a regime-conditional risk forecast, a parametric multi-asset coupling, or a privacy-preserving synthetic representation. The CHMM value-add is the structural use cases those alternatives cannot serve, and the body evaluates the construction on each of them in turn.

Our contributions are empirical and architectural. First, we provide empirical evidence that the algebraic rank bound on the centred transition matrix from the bilinear identity is non-binding at the cross-ticker median once enough latent states are admitted, with the rank constraint still doing visible work on a left tail of regime-introduction tickers; this recasts the median-case Rydén et al. [11] failure as a marginal-mixture limit rather than a temporal one. Second, we develop a unified expectation-conditional-maximisation (ECM) scaffold for four emission families (Gaussian, Student-t, Laplace, GED) under shared forward-backward recursions in which the M-step is the only architectural difference, with the GED per-state shape parameter nesting Gaussian and Laplace as boundary cases. Third, we construct a regime-conditional Value-at-Risk (VaR) that propagates the one-step-ahead state forecast through the family-appropriate predictive mixture and validate it under the Christoffersen joint conditional-coverage test, with Monte Carlo power calibration and an Engle-Manganelli dynamic-quantile backstop; this is the use case that no exchangeable bootstrap or unconditional GARCH benchmark can serve. Finally, we establish an empirical scope along four orthogonal generalisation axes (rolling-origin walk-forward, sector-balanced cross-ticker panel, CRSP cross-decade transfer, six-asset Student-t copula) that bound the construction against single-window over-fitting and document the stationarity-scope limit at which periodic refit becomes the deployment protocol.

In this study, we fitted a continuous HMM with quantile-based initialisation to a decade of daily SPY returns, a sector-balanced 30-ticker panel, a CRSP cross-decade transfer between a non-overlapping ten-year in-sample window and a two-year out-of-sample slice, and a six-asset US-equity copula, under four emission families sharing identical forward-backward recursions and differing only in the M-step. We observed that the four-family panel reproduces all three symmetric Cont stylized facts at the selected operating point, that heavy-tailed emissions narrow the kurtosis gap relative to the Gaussian baseline without disturbing the shared scaffold, that the regime-conditional Value-at-Risk passes the Christoffersen joint conditional-coverage test cleanly at the standard significance level on the headline window and on the majority of rolling-origin walk-forward folds, and that the Student-t copula composes a six-asset basket with off-diagonal correlation reproduction substantially tighter than the single-index alternative. The interpretation is that the Rydén low-K failure is a marginal-mixture limit rather than a temporal one once initialisation is fixed and the emission family is allowed to carry heavy tails, that the CHMM value-add over stronger single-window distributional baselines lies on the structural use cases (regime-conditional risk forecasting, multi-asset copula composition, parametric synthetic-data privacy) rather than on raw single-window fidelity, and that periodic refit is the documented deployment recipe under the stationary out-of-sample scope.

2 Related Work

The discrete-state regime-switching tradition opens with Hamilton [20], who introduced Markov-switching to economics, and Schaller and Van Norden [21], who estimated regime-switching specifications on US monthly stock returns and documented the predictive content of the latent regime for return forecasting. The seminal evaluation against the Cont stylized facts is due to Rydén et al. [11], who fitted Gaussian-emission hidden Markov models at $K = 2$ or 3 states and showed that the geometric sojourn-time distribution of a standard Markov chain produces exponential ACF decay too fast to match the observed slow decay of absolute returns. Bulla and Bulla [12] restored fidelity with hidden semi-Markov models at substantially higher complexity, replacing the geometric sojourn with an explicit-duration family at the cost of a duration-augmented forward-backward, and Nystrup et al. [22] extends the same line to long-memory volatility. Each of these contributions sits at the same level of architectural choice as the present paper: the latent regime is a finite-state object with per-state emission densities, and what varies across the lineage is the sojourn distribution and the per-state family. The emission families that supply the per-state densities used in this work draw on the maximum-likelihood Student-t formulations of Peel and McLachlan [18] and Liu and Rubin [19], with the Generalized Error Distribution [23, 24] providing a continuous shape parameter that interpolates between Gaussian ($p = 2$) and Laplace ($p = 1$) and underlies our CHMM-GED variant in the same role that the per-state degrees of freedom ν_k plays in CHMM-t. The discrete-state predecessor of this paper, Alswaidan and Varner [3], reproduced the same three stylized facts via Laplace quantile binning plus a Poisson-jump duration mechanism, and the present continuous-emission CHMM replaces the discretization step with the four per-state emission families above under a unified ECM scaffold, recovering the slow-ACF target through the spectral mechanism developed below rather than through an explicit duration model.

Conditional-variance and stochastic-volatility models occupy a parallel architectural niche in which the latent state is the volatility process rather than a discrete regime. The GARCH framework [6, 7] encodes volatility persistence through a single decay parameter and a unimodal innovation distribution and fails distributional Kolmogorov-Smirnov on equity returns, and successive extensions have closed some gaps without changing this binding constraint: heavy-tailed innovations [25], asymmetric leverage variants (exponential GARCH, EGARCH, 26; Glosten-Jagannathan-Runkle GARCH, GJR-GARCH, 27; threshold GARCH, 28), regime-switching extensions (Markov-switching GARCH, MS-GARCH, 29, with the `MSGARCH` R package of 30 the standard reference implementation), and realised-variance frameworks [31, 32] all reintroduce structure at the variance-process layer while preserving the unimodal-innovation constraint. The continuous-latent counterpart is the stochastic-volatility family of Taylor [33], which evolves log-volatility as a latent AR(1) and proceeds by simulation-based estimators [34, 35], hidden-Markov approximations on a discretised volatility grid [36, 37], or Markov-switching SV [38] that interpolates between the two formulations. Across both families the recurring failure mode is the same: a single innovation distribution shared across volatility regimes cannot tile the empirical kurtosis target while the variance recursion already binds the temporal axis, and the regime-switching extensions reintroduce latent states only at the variance-process layer rather than at the marginal-density layer where the present CHMM lives. The architectural distinction matters for the regime-conditional Value-at-Risk construction below, because a state-conditional predictive density is available only when the latent state carries an explicit per-state emission family, and the kurtosis-matching axis empirically separates the finite-state object with heterogeneous heavy-tailed emissions from a single continuous-volatility latent (the kurtosis points of Table 8 and the extended baseline panel make this concrete).

Deep generative market models trade per-state interpretability for high-capacity unconditional fidelity. Generative adversarial network (GAN)-based generators reproduce the marginal appearance of return series but typically fail volatility-clustering tests unless architectures encode temporal dependence explicitly [39, 40], with the TimeGAN family of Yoon et al. [8], the convolutional Wasserstein GAN QuantGAN of Wiese et al. [9], and the autoregressive-diffusion approach of Rasul et al. [10] the standard references. Path-signature generators [41–44] and score-based or diffusion approaches [10, 45–47] extend the family with finer time-series structure but inherit the same trade-off: tighter visual fidelity to the observed price tape, opaque per-state semantics, no parametric copula handle for multi-asset composition, and no closed-form predictive density for downstream conditional-coverage tests. We included QuantGAN as the deep-generative row in our extended panel, where it functions as a sanity check on whether high-capacity architectures recover the same stylized facts at our sample sizes rather than as a forecasting baseline, and the present paper's contribution is grounded against parametric and non-parametric baselines with the deep row read as a negative control. The recurring failure mode across the deep family is mode collapse on the heavy-tail axis combined with the absence of an explicit state variable, so the per-state semantics that the CHMM uses for regime-conditional Value-at-Risk and copula composition are not available off the shelf, and a deep generator that wins on raw single-window distributional fidelity still does not afford the structural use cases this paper exercises.

Cross-asset synthesis, synthetic-data evaluation, and non-stationarity handling round out the literature on which the present construction draws. The Single Index Model (SIM) [48] propagates a market factor through a rank-one linear decomposition but distorts each non-market marginal, and Sklar's theorem [49, 50] supports a cleaner decomposition in which each asset's fitted marginal is preserved and only the copula is estimated, with Gaussian and Student-t copulas [51, 52] the standard family choices and rank-reordering simulation [53] the standard injection that preserves marginals exactly. Dynamic-conditional-correlation models [54] and composite-likelihood estimators for vast covariance matrices [55] are the standard alternatives at higher dimensions, and our six-asset construction sits well below the dimensions where those estimators become essential while the rank-based copula architecture composes with either at scale. Synthetic-data evaluation in this work is anchored on the framework of Stenger et al. [56], the proper-scoring-rule literature [57–59], kernel two-sample tests [60], and signature-based metrics [41, 61], which together fix the comparison axes (KS, ACF-MAE, kurtosis, CRPS, signature distance) that the body panel exercises. Finally, the walk-forward stress folds reported below are regime introductions that the in-sample distribution does not span, and their natural complements are Bayesian online change-point detection [62] and the particle-filter recursion of Fearnhead and Liu [63], both of which detect the regime introduction rather than absorbing it into a static fit; the integration of online detection with the present CHMM scaffold is left as a companion-paper direction.

3 Methods

3.1 Data and model

The single-asset analysis uses daily SPDR S&P 500 ETF (SPY) prices from January 3, 2014 to January 3, 2024 (a ten-year training window, $T_{\mathrm{IS}} = 2{,}516$), with the period from January 4, 2024 through April 20, 2026 ($T_{\mathrm{OoS}} = 572$) held out. Prices are split-adjusted but not dividend-adjusted; the data sources are described in the acknowledgments. A vendor-stitch sanity check on the 323-day Polygon vs Alpaca / IEX overlap returned zero VWAP and return differential and Kolmogorov-Smirnov (KS) $p = 1.00$, so the OoS series is single-source-equivalent on the overlap and no boundary artefact is detected. The empirical study extends beyond SPY to a sector-balanced 30-ticker panel spanning ten GICS sectors with three large-cap representatives each, a CRSP cross-decade transfer (1994 to 2004 in-sample, 2004 to 2006 out-of-sample), and a six-asset US-equity universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) used for the cross-asset copula extension. For ticker i and trading day j the annualised excess growth rate is

$$G_{i,j} \equiv \frac{1}{\Delta t} \ln\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f, \qquad \Delta t = 1/252, \quad r_f = 0, \tag{1}$$

with $P_{i,j}$ the session volume-weighted average price (VWAP). The factor $1/\Delta t$ is a constant rescaling; all simulations and metrics are evaluated on the daily-scale series.

We model G_t as a CHMM $\mathcal{M} = (\mathcal{S}, \mathbf{T}, \mathbf{F}, \boldsymbol{\pi})$ with K latent states, per-state emission density $f_k(x; \boldsymbol{\theta}_k)$ with corresponding per-state cumulative distribution function (CDF) $F_k(x; \boldsymbol{\theta}_k) = \int_{-\infty}^{x} f_k(u; \boldsymbol{\theta}_k)\, du$ used in the regime-conditional VaR construction below, transition matrix $\mathbf{T} \in \mathbb{R}^{K \times K}$, and initial distribution $\boldsymbol{\pi}$ [17]. The stationary marginal is the K-component mixture

$$f(x) = \sum_{k=1}^{K} \bar{\pi}_k\, f_k(x; \boldsymbol{\theta}_k), \qquad \bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}} \mathbf{T}, \tag{2}$$

with $\bar{\boldsymbol{\pi}}$ the stationary distribution. Existence and uniqueness require irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1), and we learn $\mathbf{T}$, $\boldsymbol{\theta}_{1:K}$, and $\boldsymbol{\pi}$ jointly by EM. The four emission families share a single forward-backward scaffold and quantile-based initialization, and the M-step is the only architectural difference (architecture diagram in Figure 1). *CHMM-N*: $f_k = \mathcal{N}(\mu_k, \sigma_k^2)$. *CHMM-t*: $f_k = t_{\nu_k}(\mu_k, \sigma_k)$ with per-state degrees of freedom $\nu_k \in [\nu_{\min}, \nu_{\max}]$. *CHMM-L*: $f_k = \mathrm{Laplace}(\mu_k, \beta_k)$

with per-state Laplace scale denoted β_k to avoid collision with the canonical Laplace-scale symbol b. *CHMM-GED*: $f_k = \mathrm{GED}(\mu_k, \alpha_k, p_k)$ with per-state shape $p_k \in [p_{\min}, p_{\max}]$ that nests Gaussian ($p_k = 2$) and Laplace ($p_k = 1$) as boundary cases.

3.2 Estimation by EM

Estimation of $(\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$ proceeds by EM in log-space. We write O_t for the observed daily excess growth rate G_t at time t and $\gamma_t(k) = \mathbb{P}(s_t = k \mid O_{1:T})$ for the smoothed posterior probability of state k at time t under the current parameter iterate, both returned by the shared forward-backward recursion. All four emission families share the same log-space forward-backward recursions and the same quantile-based initialization [64]: observations are sorted, partitioned into K equal-sized chunks, and each state's initial location and scale are seeded from the corresponding chunk; $T_{ij}^{(0)} = \pi_k^{(0)} = 1/K$. The EM loop iterates to tolerance $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with `max_iter` $= 60$, and convergence is typically reached in 20 to 40 iterations. The implementation is in Python 3.11 using NumPy 1.26 and SciPy 1.11, with all golden-section searches run to a relative tolerance of 10^{-4} and an inner iteration cap of 80. Quantile-based initialization is what makes Baum-Welch reach a non-degenerate $K \geq 3$ fit. Random initialisations standard in the 1990s often converge to overlapping local optima, which is one of the two compounding sources of the low-K failure observed by Rydén et al. [11]; the other is the spectral one developed below. Algorithmic details and the forward-backward recursion are reported in the appendix.

The four emission families differ only in the M-step. *CHMM-N (Gaussian).* The classical Baum-Welch M-step [16] is closed form:

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k)\, O_t}{\sum_t \gamma_t(k)}, \qquad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k)\, (O_t - \mu_k)^2}{\sum_t \gamma_t(k)}.$$

CHMM-t (Student-t via ECM). We adopt the ECM formulation of Peel and McLachlan [18] and Liu and Rubin [19], the same heavy-tailed innovation choice studied by Abanto-Valle et al. [37] for stochastic-volatility-in-mean models. The latent precision

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2} \tag{3}$$

is treated as an augmented E-step quantity, and the M-step updates

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \qquad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2}{\sum_t \gamma_t(k)} \tag{4}$$

have closed form conditional on ν_k. The degrees of freedom ν_k are updated by a golden-section search maximizing the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. The two-block CM step preserves the EM monotonicity guarantee on the compact bracket, so the incomplete-data log-likelihood is non-decreasing across iterations. *CHMM-L (Laplace).* The weighted maximum-likelihood estimators are in closed form: μ_k is the posterior-weighted median of the observations (weights $\gamma_t(k)$, computed by cumulative-weight bracketing with linear interpolation between neighbouring order statistics at a cumulative-weight tolerance of 10^{-8}), and $\beta_k \leftarrow \sum_t \gamma_t(k) |O_t - \mu_k| / \sum_t \gamma_t(k)$. *CHMM-GED (Generalized Error Distribution via three-stage ECM).* For $f_k(x) = \mathrm{GED}(x; \mu_k, \alpha_k, p_k)$ the M-step decomposes into three CM updates per state: a bracketed $\mu_k \leftarrow \arg\min_\mu \sum_t \gamma_t(k)\, |O_t - \mu|^{p_k}$ via golden-section over $[\mu_k^{(n-1)} - 5\alpha_k^{(n-1)}, \mu_k^{(n-1)} + 5\alpha_k^{(n-1)}]$ centred on the previous EM iterate, then the closed-form scale

$\alpha_k \leftarrow \left[(p_k/W_k)\sum_t \gamma_t(k)\,|O_t - \mu_k|^{p_k}\right]^{1/p_k}$ with $W_k = \sum_t \gamma_t(k)$, and finally a bracketed $p_k \leftarrow \arg\max_{p\in[p_{\min},p_{\max}]} \sum_t \gamma_t(k) \log f_k(O_t;\mu_k,\alpha_k,p)$ via golden-section over $[p_{\min},p_{\max}] = [0.5, 3.0]$. Each CM step is a partial maximization, so EM monotonicity holds on the compact bracket, and CHMM-N (all $p_k = 2$) and CHMM-L (all $p_k = 1$) are the boundary cases.

3.3 Evaluation

The empirical study is organised along two pipelines. *Pipeline A* fits a CHMM to each ticker independently and evaluates single-asset metrics, carrying the descriptive, state-count, headline-generator, cross-ticker, and Value-at-Risk analyses. *Pipeline B* reuses each per-asset CHMM marginal and injects cross-asset dependence through a rank-based Student-t copula [51, 52]; profile MLE selects $\nu^\star$, and the rank-reordering simulator of Iman and Conover [53] preserves each fitted CHMM marginal under Sklar's theorem [49], with SIM, Gaussian-copula, and truncated C-vine comparators reported in the appendix. Each CHMM family is benchmarked against representatives from each generator tier, all fit on the same IS window: an i.i.d. bootstrap [65] as the non-parametric distributional ceiling, Gaussian and Laplace i.i.d. generators as marginal baselines, GARCH(1,1) Gaussian [7] and Student-t [25] as conditional-variance baselines, and MS-GARCH [29] as the closest regime-switching foil. Extended GARCH-family, semi-Markov, deep-generative QuantGAN [9], stochastic-volatility, multifractal, and jump-diffusion baselines are reported in the appendix.

For each model we simulate $P = 1{,}000$ independent paths of length T following [56], under a deterministic global seed root 20260420; a secondary seed 20260422 is used for the cross-asset Pipeline-B panels and the block-bootstrap recalibration, and the seed is fixed per panel and stated in every relevant table caption. Headline metrics are the two-sample Kolmogorov-Smirnov [66, 67] pass rate at $\alpha = 0.05$, mean simulated excess kurtosis, and the absolute-return ACF mean absolute error (MAE) over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L}\sum_{\tau=1}^{L}\left|\hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau)\right|, \tag{5}$$

where $\hat{\rho}^{\text{obs}}_{|G|}(\tau)$ is the empirical absolute-return ACF at lag τ on the observed window and $\bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau)$ its path-averaged simulated counterpart over the P simulated paths. A separate Value-at-Risk panel carries the unconditional Kupiec [68] and expected-shortfall (ES) envelope diagnostics, while block-bootstrap KS recalibration, Anderson-Darling (AD), W_1, Hellinger, and continuous ranked probability score (CRPS) auxiliaries are reported in the appendix.

4 Spectral Mechanism

Let $\mathbf{v}_k$ and $\mathbf{w}_k$ denote the right and left eigenvectors of $\mathbf{T}$ at non-unit eigenvalue λ_k, and let $\sigma^2_{|G|}$ denote the stationary variance of $|G_t|$ under the marginal mixture (2). The closed-form bilinear identity for the absolute-return ACF of a stationary CHMM, $\mathbb{E}[|G_t|\,|G_{t+\tau}|] = \mathbf{m}^\top \text{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{T}^\tau\,\mathbf{m}$ with $\mathbf{m}_k = \mathbb{E}[|G_t| \mid s_t = k]$, and its eigendecomposition over the non-unit eigenvalues of $\mathbf{T}$ are folklore in the regime-switching literature (13, §22.2; 14, Ch. 3; 15). Under irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1, with $\bar{\boldsymbol{\pi}}$ the unique stationary distribution; 69) and finite per-state second moments with non-trivial location and scale contrast (Assumption 2), the spectral decomposition $\mathbf{T}^\tau = \mathbf{1}\bar{\boldsymbol{\pi}}^\top + \sum_{k=2}^{K}\lambda_k^\tau\,\mathbf{v}_k\mathbf{w}_k^\top$ subtracts the marginal product to give the ACF

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^{K} w_k\,\lambda_k^\tau, \qquad w_k = \left(\mathbf{m}^\top \text{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{v}_k\right)\left(\mathbf{w}_k^\top \mathbf{m}\right)/\sigma^2_{|G|}. \tag{6}$$

Equation (6) is a real-coefficient sum of geometric decays at the non-unit eigenvalues of $\mathbf{T}$; complex-conjugate λ_k pairs combine into damped oscillatory modes, and non-diagonalisable $\mathbf{T}$ adds polynomial-times-geometric prefactors via Jordan blocks. A self-contained derivation, the lag-zero behaviour, and the squared-return ACF are reported in the appendix, together with the formal Assumptions 1 and 2 and the cited propositions on identifiability, MLE consistency, and ECM monotonicity. Our contribution on this axis is exclusively empirical. We apply equation (6) to recast the Rydén et al. [11] low-K failure on the SPY 2014 to 2024 instance as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$, and we demonstrate empirically that the implied $K-1$ rank bound is non-binding at $K \geq 3$.

Equation (6) reframes the Rydén low-K failure as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$. At $K = 2$, equation (6) reduces to the mono-exponential form $\rho_{|G|}(\tau) = \lambda_2^\tau$ with $\lambda_2 = T_{11} + T_{22} - 1$, and slow decay is matched by tuning $T_{11} + T_{22}$. The temporal axis is therefore satisfiable at $K = 2$; the binding axis is the two-component marginal, which cannot match the empirical kurtosis target. At $K \geq 3$ the matrix admits multiple non-unit eigenvalues and the marginal mixture has enough components to match the observed tail. Empirically, the absolute-return ACF-MAE was nearly flat across $K \in \{3, 6, \ldots, 21\}$ while distributional KS rose, and a direct effective-rank diagnostic showed a single non-unit eigenvalue carrying 93.6% of the lag-1 ACF at $K = 18$ on SPY (Table 10), so the K-rank upper bound is non-binding at $K \geq 3$ on this instance. Repeating the diagnostic on the 30-ticker cross-section gave a wider distribution with cross-ticker median dominant-mode share 0.76 and minimum 0.326 on NEM (Table 11); the rank-non-binding claim is supported at the cross-ticker median rather than as a universal property of equity-return data.

5 Empirical Study

The empirical study is organised along two pipelines used throughout. Pipeline A is the single-asset CHMM fit per ticker with no cross-asset coupling and carries the descriptive, state-count, headline-generator, cross-ticker, and Value-at-Risk analyses, while Pipeline B layers a rank-based copula on top of the Pipeline-A marginals and carries the cross-asset dependence analysis.

We computed sample stylized-fact diagnostics on the SPY IS and OoS windows and observed all three Cont stylized facts on both windows (Figure 4): heavy-tailed marginal (IS excess kurtosis 7.68, OoS 5.29; Table 1, Observed row), negligible linear autocorrelation in raw G_t, and persistent volatility clustering in $|G_t|$. A stationary block bootstrap at mean block length $L = 20$ gave broadly overlapping IS and OoS 95% CIs of $[2.17, 12.40]$ and $[0.90, 8.26]$ on the kurtosis point estimates (Table 12), so kurtosis-fidelity targets should be read against the joint envelope rather than the point estimates. Having confirmed the three stylized facts on both windows, we next swept $K \in \{3, 6, 9, 12, 15, 18, 21\}$ for CHMM-N on the IS window and observed that the $K = 3$ versus $K = 6$ comparison sat inside sampling noise under both four-fold and six-fold rolling-origin cross-validation (CV; $|z| \leq 0.07$ on mean held-out log-likelihood, with the sign flipping between fold designs; full K-sweep CV panel in the appendix, Section A.10), with the Bayesian information criterion (BIC) also picking $K = 3$. Under parsimony we took $K^\star = 3$ as the body operating point; higher state counts were decisively worse on held-out criteria, and absolute-return ACF-MAE was essentially flat across the entire sweep, consistent with the spectral identity (6).

With $K^\star = 3$ in hand, we fit the four CHMM variants on the SPY IS window and observed 79.8 to 90.6% IS and 63.1 to 83.2% OoS KS against six benchmark generators and an explicit-duration semi-Markov reference (Table 1). The CHMM-N row landed at 89.7% IS and 80.5% OoS, simulated kurtosis 3.83 and 3.53, and $|G_t|$ ACF-MAE 0.0460 (on par with GARCH at 0.0485). The shared-ν Student-t row gave the cleanest heavy-tail match without a penalty hyperparameter at simulated kurtosis 4.68 and 4.46 against observed 7.68 and 5.29, with 91.9% IS and 82.1% OoS KS. The penalised CHMM-t at $\lambda = 20$ is reported as a sensitivity reference: the IS aggregate 14.91 is a $1/\nu_k$-shrinkage outcome with λ tuned at $K = 18$ and re-used at $K^\star = 3$, where the lower-bracket-pinned state carried a larger share of the three-state mixture (discussed below); CHMM-GED's per-state shape landed at 5.48 and 5.12, and CHMM-L at 5.30 and 4.85 (losing roughly 10pp on OoS KS at $K = 3$). The mixture-of-eigenvalues sum (6) reproduced the slow absolute-return ACF that the Rydén et al. [11] $K = 2$ to 3 Gaussian setup could not recover on the SPY 2014 to 2024 instance, and the raw-return ACF-MAE closed the third stylized fact at 0.0234 to 0.0240, indistinguishable from the i.i.d. baselines (Table 1, G_t column). The four CHMM variants were statistically indistinguishable on the OoS sample-CRPS [59] (CHMM-N 1.0393, CHMM-t-pen 1.0373, CHMM-L 1.0405, CHMM-GED 1.0389, against bootstrap 1.0398 and GARCH(1,1) 1.0440), so the family choice was driven by the per-row kurtosis match (Table 8); the cross-generator ranking was robust to the distributional-metric choice across KS, Anderson-Darling, Wasserstein-1, and Hellinger. The MS-GARCH benchmark plateaued at 34 to 36% IS KS in our Nelder-Mead point-estimate plug-in simulator versus the CHMM-N plug-in at 89.7% on the same simulator design, and a reference Bayesian posterior-predictive re-run via the `MSGARCH` R package of Ardia et al. [30] dropped the MS-GARCH row to 0 to 0.1%, reflecting parameter-uncertainty integration (Table 1). A six-fold rolling-origin walk-forward (Table 2) attained median OoS KS 62.1% at $K^\star = 3$, with two stress folds (W2 COVID, W4 2022 rate-hike onset) below 10% that every generator in the panel rejected; the walk-forward median is the operationally informative summary for production deployment.

We inspected the fitted CHMM-N state assignments at $K^\star = 3$ and observed clean partitioning along the conventional volatility axis (high-, low-, and moderate-vol with the natural drift signs), with the bimodal CHMM-GED $\hat{p}_k$ partition reflecting the same axis at $K = 18$ via Laplace-like states in the high-volatility tail and Gaussian-like states in the bulk (Figure 5); the latent states therefore mapped onto identifiable economic regimes at every K examined. To bound the CHMM against an alternative scaffold on the same Markov backbone, we replaced the geometric sojourn distribution with an explicit truncated Pareto and refit through the explicit-duration forward-backward of Yu [70], obtaining ML HSMM-N at $K^\star = 3$ at 98.4% IS and 91.0% OoS KS (Table 1), the strongest single-window OoS KS row in the panel among parametric generators (only the non-parametric i.i.d. bootstrap was higher at 92.1%) and beating every CHMM operating point. The CHMM and ML HSMM are best read as complementary scaffolds on the same Markov backbone, with KS, ACF, and kurtosis trade-offs that select between them by use case: the HSMM advantage on KS was offset by an absolute-return ACF-MAE matching the i.i.d. baseline level (0.0629) rather than CHMM-N's 0.0460 (Table 1), because the fitted Pareto sojourn concentrated probability mass on a single low-volatility state and the regime-driven slow-ACF mechanism that the CHMM exploits at $K \geq 3$ was muted. At $K = 18$ a discrete-Gamma-sojourn HSMM closed the ACF gap to $|G_t|$ ACF-MAE 0.0462 (Table 26), so the CHMM-vs-HSMM trade-off at high K is not strictly that CHMM wins on ACF and HSMM wins on KS; rather, the geometric-sojourn HSMM at $K^\star = 3$ won on KS with an ACF-MAE matching the i.i.d. level, and the Gamma-sojourn HSMM at $K = 18$ matched CHMM on ACF. Readers selecting a synthetic-data generator on raw OoS KS alone should prefer ML HSMM-N at $K^\star = 3$; CHMM at $K^\star = 3$ is the right choice when the structural use cases below (regime-conditional VaR and multi-asset copula composition) are in scope. The stationary block bootstrap of Politis and Romano [65] attained 99.7% IS and 92.1% OoS KS (Table 1), above every CHMM operating point, and CHMM differentiates on use cases the bootstrap cannot serve, namely regime-conditional VaR (the bootstrap is exchangeable and admits no forward state filter), multi-asset copula composition (which needs parametric marginals), and privacy or licensing constraints (every bootstrap path is a permutation of literal IS observations). A block-bootstrap KS recalibration at $L = 20$ (the appropriate null under autocorrelated daily returns; [65]) preserved the cross-generator ordering on both windows, with the absolute level dropping roughly 5pp on IS and 20 to 30pp on OoS (Table 25),

Table 1. Headline generator comparison on SPY. 1,000 simulated paths, $\alpha = 0.05$; IS $T = 2{,}516$, OoS $T = 572$. Columns: KS = Kolmogorov-Smirnov pass rate; Kurt = mean simulated excess kurtosis (observed 7.68 IS, 5.29 OoS); ACF-MAE = mean absolute error of the absolute-return / raw-return ACF over 252 lags. Bold marks the panel-best entry on each axis *within block* (block = i.i.d. baselines, GARCH family, MS-GARCH, CHMM body, co-headline). The 1-day OoS KS column is dominated by the i.i.d. bootstrap; the CHMM-vs-bootstrap differentiation is on the structural use cases below (regime-conditional VaR, multi-asset copula composition), not on per-day distributional fidelity. Sample-CRPS is reported as a tie-break in the prose paragraph following the table; the four CHMM rows are statistically indistinguishable on this metric. Block-bootstrap KS recalibration, the extended GARCH-family and MS-GARCH-$K \in \{3, 4, 6\}$ panel, and the $K = 6$ and $K = 18$ sensitivity panels are reported in the appendix. ¶ Reference Bayesian re-run of MS-GARCH via the `MSGARCH` R package of Ardia et al. [30]; the lower KS pass rate reflects posterior-predictive integration of parameter uncertainty. § QuantGAN row is an in-house WGAN re-implementation (3-layer 1D-conv generator and critic, Wasserstein loss; full spec in the appendix); the row reads as the panel's deep-generative negative control. §§ Shared-ν Student-t ablation: single ν across all K states by aggregate-Q ECM, no penalty.

	KS (%) ↑		Exc. Kurt		ACF-MAE ↓	
Model	IS	OoS	IS	OoS	$\lvert G_t\rvert$	G_t
Observed	–	–	7.68	5.29	–	–
Bootstrap	**99.7**	**92.1**	**7.24**	**6.81**	0.0628	**0.0235**
Gaussian i.i.d.	0.0	1.0	−0.01	−0.03	0.0627	0.0236
Laplace i.i.d.	99.1	88.5	2.95	2.91	0.0628	0.0236
GARCH(1,1)	23.4	60.8	7.06	2.96	0.0485	0.0243
GARCH(1,1)-t	**57.3**	**80.8**	15.13	–	**0.0316**	**0.0173**
MS-GARCH ($K = 2$)	27.7	**38.7**	**4.73**	–	0.0367	0.0173
MS-GARCH ($K = 3$)	**36.1**	33.1	4.10	–	**0.0284**	0.0173
MS-GARCH ($K = 6$)	34.5	33.4	4.41	–	0.0429	0.0173
MS-GARCH ref. Bayesian ($K = 2$)¶	0.0	5.8	4.00	2.46	0.0465	0.0240
MS-GARCH ref. Bayesian ($K = 3$)¶	0.1	5.1	4.52	2.53	0.0433	0.0241
MS-GARCH ref. Bayesian ($K = 4$)¶	0.0	5.3	6.01	3.54	0.0446	0.0241
QuantGAN TCN§	0.0	0.0	0.56	0.53	0.0617	0.0264
Body operating point: $K^\star = 3$ (selected by held-out criteria under rolling-origin CV).						
CHMM-N	89.7	80.5	3.83	3.53	**0.0460**	0.0240
CHMM-t pen. ($\lambda = 20$)	90.6	**83.2**	14.91	8.50	0.0537	0.0235
CHMM-L	79.8	63.1	5.30	4.85	0.0532	0.0236
CHMM-GED	90.5	77.4	5.48	5.12	0.0526	0.0236
CHMM-t shared-ν§§	**91.9**	82.1	**4.68**	**4.46**	0.0531	**0.0235**
ML HSMM-N	**98.4**	**91.0**	**3.46**	**3.38**	**0.0629**	**0.0236**

Table 2. Six-fold rolling-origin walk-forward summary (CHMM-N, $5y$ train / $1y$ test, 500 paths per fold). Test windows: W1 (2019), W2 (2020 COVID), W3 (2021), W4 (2022 rate-hike onset), W5 (2023), W6 (2024). Pulled from the full panel in Appendix A.10, Table 33. The headline single-window OoS sits at the upper end of the fold-wise distribution; the W2 and W4 folds are out-of-distribution by KS for every generator in the panel.

	median KS (%)	$[Q_1, Q_3]$ KS (%)	min KS (%)	folds < 10%
$K^\star = 3$	62.1	$[7.2, 78.4]$	0.8	2/6
$K = 18$ (sensitivity)	67.7	$[8.2, 75.0]$	0.0	2/6

so the cross-generator ranking is the robust comparison.

Having established the headline ranking on SPY, we next fit the penalised CHMM-t at $K^\star = 3$ and $\lambda = 20$ on each of a sector-balanced 30-ticker panel (ten GICS sectors with three large-cap

Table 3. Cross-ticker generalisation, sector-balanced 30-ticker panel (penalised CHMM-t at $\lambda = 20$, 1,000 paths, seed = 20260420, body operating point $K^\star = 3$). The OoS distribution is wider than the IS distribution and is concentrated at the regime-introduction tickers (LLY, UNH, NEM). The full per-ticker panel, per-sector rollup, $K = 6$ and $K = 18$ sensitivity rebuilds, and quarterly-refit recipe are reported in the appendix.

Metric	$K^\star = 3$
IS KS pass rate (%) median	96.8
IS KS pass rate (%) mean ± s.d.	95.1 ± 4.9
OoS KS pass rate (%) median	69.1
OoS KS pass rate (%) mean ± s.d.	66.2 ± 28.2
$\lvert G_t \rvert$ ACF-MAE median	0.0399
Kurtosis residual (sim − obs) median	7.18
Tickers OoS KS < 60% (IS-fixed)	11/30

representatives each) and observed a concentrated IS distribution (median 96.8%) and a wider OoS distribution with median 69.1%, mean 66.2 ± 28.2%, and 11/30 tickers below 60% (Table 3). The deepest failures were driven by single-name regime introductions (LLY 8.7%, UNH 16.5%, NEM 5.0%; full per-ticker rollup in Table 9), and the remaining eight sub-60% tickers and the per-sector breakdown at the $K = 18$ kurtosis-fidelity sensitivity reference (with the same 11/30 failure count and overlapping ticker list) are reported in the appendix. A one-way analysis of variance (ANOVA) on OoS KS by sector at the original $n = 3$ per-sector design was severely underpowered ($\eta^2 = 0.16$ at $F(9, 20) = 0.44$, $p = 0.90$); an $n = 6$ per-sector expansion to a 60-ticker panel returned $F(9, 50) = 0.37$, $p = 0.946$, $\eta^2 = 0.062$, with the effect-size estimate dropping by more than half (full ANOVA panel in the appendix, Section A.7.3). The 60-ticker aggregate distribution was statistically indistinguishable from the 30-ticker version, so the visible concentration of failures on regime-introduction tickers is a per-ticker observation rather than evidence about sector-level structure. The operational recipe is periodic refit: at $K^\star = 3$ a quarterly-refit version shifted the OoS KS median from 69.1% to 84.7% and reduced failures from 11/30 to 8/30, while the same recipe at the $K = 18$ kurtosis-fidelity reference gave median 83.0% and 7/30, with monthly cadence interpolating to 86.7% and 5/30 at three times the compute (Table 5). Sensitivity panels at $K = 6$ and $K = 18$ and the SPY-level shared-ν Student-t alternative are reported in the appendix.

We next fit CHMM-N on a CRSP-sourced 1994 to 2004 IS window and tested on the 2004 to 2006 OoS slice, and observed decade-robust IS performance (84 to 90% IS KS pass rate against the 90 to 94% obtained on the body 2014 to 2024 IS window) alongside a collapsed OoS pass rate of 3 to 5% (Table 5, with the full per-decade panel in the appendix). The OoS collapse was driven by a distributional shift: the calm post-dot-com bull-market slice had observed excess kurtosis 0.06 (essentially Gaussian) against the 1994 to 2004 IS kurtosis of 3.05, so the static-IS-fitted CHMM could not bracket an OoS slice whose marginal moved outside the IS regime library. The cross-decade pattern matched the W2 (COVID 2020) and W4 (2022 rate-hike) walk-forward stress folds (Table 2, both with OoS KS below 10%): the IS axis transferred cleanly across decades, but a static fit cannot bracket regime introductions. The operational implication parallels the cross-ticker panel above: the static-fit OoS pass rate is OoS-slice-specific, and the deployment recipe is the periodic-refit construction (the cross-ticker quarterly-refit panel and the rolling-copula construction below), not the single-window static fit.

We then composed per-asset CHMM-N marginals at $K^\star = 3$ through a rank-based Student-t copula on a six-asset US-equity universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ; 200 paths) and observed off-diagonal correlation MAE of 0.027 on IS for the Student-t copula, against 0.030 Gaussian,

Table 4. Cross-asset Student-t copula on CHMM-N marginals at $K^\star = 3$ (Pipeline B; $\nu^\star = 6$; 200 paths; seed 20260422). Per-asset KS pass rate at $\alpha = 0.05$ and correlation reproduction averaged over paths. Body operating point $K^\star = 3$ matches the single-asset Table 1. The off-diagonal MAE is essentially K-robust (the same panel at the kurtosis-fidelity sensitivity reference $K = 18$ gives 0.027/0.209, identical to two decimal places); per-asset KS sits lower at $K^\star = 3$ than at $K = 18$, but the cross-asset use case is dominated by the dependence layer rather than per-asset marginal fit. The comparison against SIM, Gaussian, and truncated C-vine is reported in the appendix.

Ticker	KS (%) ↑ IS	KS (%) ↑ OoS
SPY	87.0	77.5
NVDA	92.0	62.5
JNJ	97.5	94.0
JPM	97.5	57.5
AAPL	85.0	86.0
QQQ	83.0	85.5
Off-diag MAE IS	0.027	
Off-diag MAE OoS	0.209	
Off-diag MAE OoS (quarterly refit, $K = 18$ reference)	**0.185**	

0.068 truncated C-vine, and 0.076 SIM (Table 4; full comparator panel in the appendix). Profile MLE selected $\nu^\star = 6$ on the unit-spaced grid $\nu \in [4, 12]$ with Wilks 95% profile-LL CI of $[6, 7]$ on the IS slice, and a full one-shot MLE jointly maximising over (Σ, ν) confirmed the choice was robust to the two-step estimator. The off-diagonal MAE was essentially K-robust (the same panel at the kurtosis-fidelity sensitivity reference $K = 18$ gave identical numbers to two decimal places); per-asset KS sat lower at $K^\star = 3$ than at $K = 18$, but the cross-asset use case is dominated by the dependence layer rather than per-asset marginal fit. The universe contained overlapping ETFs and constituents (QQQ holds AAPL and NVDA; SPY holds all four single names), so the implied dependence structure was dominated by strong positive co-movement and the copula task was mechanically easier than on a non-overlapping basket; a non-overlapping comparison panel is reported in the appendix. On OoS the Student-t versus Gaussian gap was within across-path noise at this N_{paths}: the off-diagonal MAE was 0.209 for Student-t and 0.202 for Gaussian, a 0.007 gap that we report descriptively rather than under a formal paired test (Table 4). The operational story on the OoS axis is the quarterly-rolling refit, which reduced the OoS off-diagonal MAE from 0.209 to 0.185 (Table 4); the dependence-family choice is dominated by the refit-cadence choice on OoS. The cross-asset OoS distribution was bimodal: NVDA and JPM failed OoS KS while the other four passed above 79%, driven by single-name regime shifts (NVDA AI-rally, JPM bank-stress), and the failure count recovered to 0/6 under quarterly refit (Table 4; full rolling-copula panel in the appendix).

The four orthogonal stress sources tested above (cross-ticker generalisation, cross-decade transfer, the GLD non-equity stress, and the walk-forward W2 and W4 stress folds) all reduced to the same OoS-slice composition pattern: static-IS-fit OoS performance collapsed on slices whose marginal moved outside the IS regime library, and periodic refit closed the gap on stress sources whose OoS slice remained inside the equity-return scope while leaving the regime-introduction folds untouched (Table 5). The static-IS-fit construction is therefore scoped to non-stress equity slices, and the deployment recipe for live use is periodic refit at a cadence selected against the OoS-slice composition rather than against the IS fit alone.

We next built a regime-conditional VaR threshold from the CHMM one-step-ahead state forecast

Table 5. Stationarity-scope summary. Static-IS-fit OoS performance and periodic-refit mitigation across the four orthogonal stress sources tested in this paper.

Stress source	Static-fit OoS KS	Refit mitigation
Cross-ticker (30 panel)	$11/30 < 60\%$	Quarterly: $7/30 < 60\%$
Cross-decade (2004 to 2006)	3 to 5%	slice < 600 d, untested
GLD non-equity stress	0%	Quarterly: 2.5%
Walk-forward W2, W4	$< 10\%$	no cadence closes

and back-tested it under Christoffersen-cc at $\alpha = 0.05$ on OoS, and observed clean passes at every K across all four emission families (Table 34) while every constant-across-t generator (CHMM unconditional, GARCH, bootstrap) rejected breach independence. At the strict $\alpha = 0.01$ tail the Engle-Manganelli dynamic-quantile backstop rejected $K = 18$ ($p = 0.017$); the $\alpha = 0.01$ Christoffersen-cc pass at $K = 18$ is therefore power-driven and the substantive risk-management diagnostic is the $\alpha = 0.05$ row. All four CHMM variants bracketed observed VaR and ES at 1% and 5% within the $[5, 95]\%$ envelope on both windows; the IS CHMM-N median $\text{VaR}_{0.01}$ at -6.64 (observed -6.38) sat within 0.3 of observed and within 0.1 of the i.i.d. bootstrap (which matches the observed value by construction), with envelopes roughly $1.7\times$ tighter than GARCH's (Table 19). All four CHMM rows passed the unconditional Kupiec test on both windows, marginally on OoS (likelihood-ratio statistic $\text{LR}_{\text{uc}} \in [3.03, 3.83]$ at $\alpha = 0.05$, against the $\chi^2_1(0.05) = 3.841$ critical value, with CHMM-N and CHMM-GED at $\text{LR}_{\text{uc}} = 3.83$ inside the integer-breach grid resolution at $T_{\text{OoS}} = 572$; Table 34); we treated Kupiec as a sanity check at this T_{OoS} and the conditional construction as the substantive diagnostic.

We back-tested CHMM-N at $K^\star = 3$ and $K = 18$ and the penalised CHMM-t at $\lambda = 20$ on OoS and observed every row passing Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly at $\alpha = 0.05$ (Table 6). The construction is straightforward: at each OoS day t we ran the forward filter through $\mathcal{F}_{t-1} = R_{\text{IS}} \cup R_{\text{oos}}[1{:}t-1]$ to get the one-step-ahead state forecast

$$\mathbb{P}(s_t = k \mid \mathcal{F}_{t-1}) = \sum_i \mathbb{P}(s_{t-1} = i \mid \mathcal{F}_{t-1})\, T_{ik}, \tag{7}$$

and defined $\widehat{\text{VaR}}_t(\alpha) = F_t^{-1}(\alpha)$ where $F_t(\cdot) = \sum_k \mathbb{P}(s_t = k \mid \mathcal{F}_{t-1})\, F_k(\,\cdot\,; \boldsymbol{\theta}_k)$ is the family-appropriate predictive mixture CDF under IS-fixed $(\mathbf{T}, \{\boldsymbol{\theta}_k\})$, with F_k the per-state Gaussian, Student-t, Laplace, or GED CDF (the body Gaussian construction reduces to $F_k = \Phi(\,\cdot\,; \mu_k, \sigma_k)$). The improvement in the independence statistic from the unconditional variant ($\text{LR}_{\text{ind}} = 5.26$ at $K = 18, \alpha = 0.05$; Table 7) to the conditional variant ($\text{LR}_{\text{ind}} = 0.52$; Table 6) is the regime-switching value proposition that the unconditional Kupiec headline did not exercise. The conditional pass extended to every emission family with the family-appropriate predictive density (CHMM-L, CHMM-GED) at $p_{\text{cc}} \geq 0.089$ throughout (Table 34), so the regime-switching value proposition is intrinsic to the latent-state forecast and not specific to the Gaussian emission family. The Christoffersen-cc test is power-bounded at $T_{\text{OoS}} = 572$ and $\alpha = 0.01$ (power calibrated by simulating $B = 5{,}000$ breach-indicator sequences from a two-state Markov chain on $\{0, 1\}$ with marginal α and second eigenvalue $\rho \in \{0, 0.05, 0.10, 0.20, 0.30, 0.50\}$ at $T = 572$: at least 80% power only at $\rho \geq 0.50$ at $\alpha = 0.01$ and at $\rho \geq 0.20$ at $\alpha = 0.05$; Table 15), so the substantive risk-management diagnostic is the $\alpha = 0.05$ row, where the Engle-Manganelli (2004) Dynamic Quantile (DQ) test also passed cleanly at both $K = 3$ ($p = 0.156$) and $K = 18$ ($p = 0.678$). At $\alpha = 0.01$ the DQ test rejected conditional coverage at $K = 18$ ($p = 0.017$) while Christoffersen-cc did not, so the regime-conditional VaR over-coupled high-volatility states at the strict tail at $K = 18$ on this OoS slice and the $\alpha = 0.01$

Christoffersen-cc pass is a power-driven non-rejection.

Table 6. Regime-conditional VaR back-test on SPY OoS ($T_{\mathrm{OoS}} = 572$, seed 20260420). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. The CHMM rows use the forward filter under IS-fixed parameters $(\mathbf{T}, \{\boldsymbol{\theta}_k\})$ via Eq. (7); only the one-step-ahead state-probability vector updates through OoS. The contender rows (filtered bootstrap and CAViaR) are scored on the same Christoffersen / DQ harness. Every CHMM row passes Kupiec, Christoffersen-ind, and Christoffersen-cc at $\alpha = 0.05$. ‡ $\alpha = 0.01$ rows are power-bounded at $T_{\mathrm{OoS}} = 572$: Christoffersen-cc reaches at least 80% power against breach-clustering eigenvalues only at $\rho \geq 0.50$, so a clean pass should be read as "not detectably worse than a strongly-clustered alternative" rather than "calibrated against any plausible alternative". The Engle-Manganelli (2004) DQ test (higher power) rejects $K = 18$ at $\alpha = 0.01$ ($p = 0.017$, bold in p_{DQ}); the contender rows also reject at $\alpha = 0.01$ on DQ. The full power calibration is in the appendix. * DQ p-value at 4-lag specification; χ^2_6 critical at 5% is 12.59. The DQ panel is run on CHMM-N and the two contender baselines; CHMM-t penalised rows are reported with '–' in this column.

Family	K	α	breaches	br rate	median $\widehat{\mathrm{VaR}}_t$	$\mathrm{LR}_{\mathrm{uc}}$	$\mathrm{LR}_{\mathrm{ind}}$	$\mathrm{LR}_{\mathrm{cc}}$	p_{cc}	$p_{\mathrm{DQ}}{}^{*}$
CHMM-N	3	$0.01^{\ddagger}$	9	1.57%	−4.56	1.62	2.36	3.98	0.14	0.06
CHMM-N	3	0.05	35	6.12%	−2.87	1.41	0.01	1.42	0.49	0.16
CHMM-N	18	$0.01^{\ddagger}$	9	1.57%	−5.20	1.62	2.36	3.98	0.14	**0.02**
CHMM-N	18	0.05	26	4.55%	−3.02	0.26	0.52	0.78	0.68	0.68
CHMM-t (λ=20)	3	$0.01^{\ddagger}$	8	1.40%	−5.59	0.82	2.81	3.63	0.16	–
CHMM-t (λ=20)	3	0.05	32	5.59%	−2.73	0.41	0.77	1.19	0.55	–
CHMM-t (λ=20)	18	$0.01^{\ddagger}$	10	1.75%	−6.25	2.65	1.98	4.62	0.10	–
CHMM-t (λ=20)	18	0.05	29	5.07%	−3.15	0.01	1.39	1.40	0.50	–
Non-state-space conditional VaR contenders.										
Filtered bootstrap	–	$0.01^{\ddagger}$	8	1.40%	−4.81	0.82	2.81	3.63	0.16	**0.04**
Filtered bootstrap	–	0.05	24	4.20%	−2.89	0.82	0.84	1.67	0.43	0.55
CAViaR (SAV)	–	$0.01^{\ddagger}$	6	1.05%	−4.82	0.01	3.97	3.98	0.14	**0.01**
CAViaR (SAV)	–	0.05	25	4.37%	−2.99	0.50	0.67	1.17	0.56	0.76

We benchmarked the regime-conditional construction against two non-state-space conditional-VaR baselines (filtered Hull-White-style historical simulation [71] and the Engle-Manganelli (2004) symmetric-absolute-value (SAV) conditional autoregressive value-at-risk (CAViaR) specification [72]) on the same Christoffersen and DQ harness, and observed approximate parity at $\alpha = 0.05$ but a strict-tail advantage for CHMM-N at $K^{\star} = 3$ over both contenders (Table 6). At $\alpha = 0.05$ the four rows were statistically indistinguishable on Christoffersen-cc; at $\alpha = 0.01$ the DQ test rejected every row except CHMM-N at $K^{\star} = 3$ ($p_{\mathrm{DQ}} = 0.06$), and the state-filter VaR pipeline of Eq. (7) applied to any multi-state regime-switching model with the CHMM-vs-MS-GARCH choice dominated by the marginal-distribution column of Table 1 (CHMM-N at 89.7% IS KS versus MS-GARCH plateauing at 0 to 36%). The construction extended to four-family ablation, six-fold walk-forward, and quarterly refit (Table 34 for the four-family ablation; full walk-forward and refit-cadence panels in the appendix). The conditional Christoffersen-cc passed at $\alpha = 0.05$ on 19/24 walk-forward rows (Table 35), with failures concentrated on the W2 (COVID) and W4 (2022 rate-hike onset) stress folds that every generator rejected, and faster refit cadences did not close the W2 gap. A Benjamini-Hochberg false-discovery-rate correction (BH-FDR) at FDR $= 0.05$ across the combined 40-test panel (the 24 walk-forward rows above plus the 16 four-family (K, α) rows of Table 34) retained 37/40 rows on the full panel and 32/32 on the non-stress subset, and the three persistent rejections were all on the W2 fold (Table 35).

6 Discussion

The main empirical finding is that a continuous HMM trained by Baum-Welch with quantile-based initialisation reproduces the three symmetric Cont stylized facts on daily US equity returns at the body operating point $K^\star = 3$ and closes the kurtosis gap to within the heavy-tailed-emission family choice. Two issues compound in the Rydén et al. [11] low-K setup: a single self-transition rate produces a single exponential mode, and 1990s-standard random initialisations often converged to degenerate local optima. Our $K = 2$ replication on SPY confirmed the binding constraint is distributional (KS at $K = 2$ dropped to 67 to 72%, while ACF-MAE was essentially constant at approximately 0.050; full panel in the appendix, Section A.7.6); with modern initialisation the ACF was reproduced at any $K \geq 2$ and distributional fidelity drove the choice of K. The mechanistic source of the residual kurtosis gap is that CHMM-N at $K^\star = 3$ produced simulated IS kurtosis 3.83 against observed 7.68 (Table 1) because a three-component Gaussian mixture's between-component term is bounded and cannot tile the empirical heavy tail. The shared-ν Student-t ablation was the primary heavy-tailed specification at $K^\star = 3$ (simulated IS kurtosis 4.68, no penalty hyperparameter; Table 1), the penalised per-state CHMM-t at $\lambda = 20$ overshot IS kurtosis to 14.91 (λ tuned at $K = 18$ and re-used at $K^\star = 3$ as an upper bound; Table 1), CHMM-L sat between CHMM-N and CHMM-t at IS kurtosis 5.30 without any shape parameter (Table 1), and CHMM-GED's bimodal $\hat{p}_k$ partition was the data-driven analog of a hand-classified Gaussian-bulk with Laplace-tail hybrid. The mechanism is therefore that the binding axis at $K \geq 3$ is the marginal mixture's representational capacity for empirical kurtosis, not the temporal kernel; the emission-family ablation isolates which symmetric heavy-tail specification carries the residual gap most cleanly.

The honest limitation is that a static-IS-fit CHMM does not generalize across regime introductions. The GLD non-equity stress and the W2 (COVID) and W4 (2022 rate-hike onset) walk-forward folds were regime introductions that the IS distribution did not span, so OoS performance broke down on slices whose marginal moved outside the IS regime library (Table 5). Single-name equity returns are well known to carry sector-specific structural breaks [73–75], and the cross-ticker concentration of failures on regime-introduction names (LLY, UNH, NEM) is the same pattern at the panel level. Periodic refit [76, 77] is the deployment recipe for non-stress equity slices; it does not save GLD or the W2 and W4 stress folds, which the IS distribution does not span and which faster refit cadences also do not close. The identifiability implication is that the binding axis for K-selection is distributional rather than temporal: the ACF was reproduced at any $K \geq 2$ once initialisation was fixed, so held-out KS, log-likelihood, and BIC are the load-bearing criteria for $K^\star$ selection, and the stress-fold rejections are diagnostic of OoS-slice composition rather than of model misspecification on the IS axis.

The generalizability story is bounded by both the stylized-fact scope and the parameter-parsimony comparison against deeper baselines. The Cont (2001) stylized-fact list also includes the leverage effect (negative $\mathrm{Corr}(G_t, |G_{t+1}|)$ on equity returns); the present paper's body evaluation covered only the three symmetric stylized facts and treats the leverage-axis behaviour of symmetric-emission CHMMs as a deferred topic. On parametric complexity, CHMM-N at $K = 18$ has $K(K-1) + 2K = 342$ free parameters (CHMM-t adds K degrees of freedom; CHMM-GED adds K shape parameters), whereas the QuantGAN baseline of Wiese et al. [9] carries on the order of 10^4 to 10^5 parameters across its convolutional generator and critic, three orders of magnitude higher; the CHMM therefore competes against deep generators at a parameter budget that is amenable to identifiability analysis, posterior diagnostics, and the closed-form spectral mechanism that motivates the $K^\star$ choice. The architectural counterpoint to that parsimony is that asymmetric or skewed emission families, a time-varying transition kernel estimated by rolling refit or online updates, scaling the multi-asset copula composition beyond the six-asset and 30-ticker panels through factor or vine structures, and embedding the CHMM-mixed predictive in a portfolio-optimisation or tail-risk-budgeting loop are the natural future directions; each would extend the present construction on a single orthogonal axis (leverage effect, non-stationarity, dimension, downstream use case) without disturbing the unified four-family scaffold that the body of this paper exercises.

7 Conclusion

A continuous HMM trained by Baum-Welch with quantile-based initialisation reproduces the three symmetric Cont stylized facts on daily US-equity returns. The textbook mixture-of-eigenvalues identity for the absolute-return ACF [13–15] recasts the Rydén et al. [11] low-K failure as an effective-rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$, non-binding at the cross-ticker median for $K \geq 3$ and shifting the binding axis from the temporal kernel to the marginal mixture. The unified ECM scaffold across four emission families isolates the M-step as the only architectural difference and admits a direct ablation: the shared-ν Student-t variant narrowed the kurtosis gap at $K^\star = 3$ to 4.68 against observed 7.68 without a tuning hyperparameter (Table 1), CHMM-L sat between Gaussian and Student-t without any shape parameter, and CHMM-GED's bimodal $\hat{p}_k$ partition recovered a data-driven Gaussian-bulk with Laplace-tail hybrid. A regime-conditional Value-at-Risk that propagates the one-step-ahead state forecast through the family-appropriate predictive mixture passed the Christoffersen joint conditional-coverage test at $\alpha = 0.05$ on the headline OoS window and on 19/24 walk-forward rows (Table 34), with strict-tail Engle-Manganelli behaviour favouring CHMM-N at $K^\star = 3$ over filtered-bootstrap and CAViaR contenders (Table 6); rejections concentrated on out-of-distribution stress folds that every generator rejected. The scope is bounded: the i.i.d. bootstrap and a maximum-likelihood HSMM-N benchmark exceed the CHMM on raw single-window OoS Kolmogorov-Smirnov, and the CHMM value-add is the structural use cases those alternatives cannot serve, namely regime-conditional VaR, multi-asset copula composition on a sector-balanced 30-ticker panel, and parametric synthetic-data privacy. The scope is daily US equities under stationary OoS (Table 5); the GLD non-equity stress and the W2 and W4 walk-forward folds collapsed under static fitting, and periodic refit is the recommended deployment protocol.

These results motivate four extensions. Asymmetric or skewed emission families would extend the present symmetric-stylized-fact scope to the leverage-effect axis of the Cont [5] list, complementing the per-state p_k ablation of CHMM-GED with a shape parameter that targets the third moment rather than the fourth. A time-varying transition kernel, estimated by rolling refit, particle filtering, or Bayesian online updates [77], would address the W2 (COVID 2020) and W4 (2022 rate-hike onset) stress folds and the analogous 3/24 persistent walk-forward rejections that the static-IS recipe leaves open (Table 5). Scaling the multi-asset copula composition beyond the six-asset and 30-ticker panels through factor or vine structures would extend per-asset distributional fidelity to portfolio-level synthesis, and the architectural decoupling of marginals and dependence layer in Pipeline B is intended to compose with either at scale. Embedding the CHMM-mixed predictive in a portfolio-optimisation or tail-risk-budgeting loop would provide an end-to-end test of the regime-conditional VaR machinery as a downstream operating signal, with the $K^\star = 3$ versus $K = 18$ trade-off (CHMM-N OoS KS 80.5% versus walk-forward median 67.7%; Tables 1 and 2) as the natural sensitivity axis. Each extension targets a single orthogonal limitation surfaced by the present empirical scope, and the unified ECM scaffold composes with all four without architectural change.

Conflict of Interest Statement. The authors declare that the research was conducted without any commercial or financial relationships that could potentially create a conflict of interest.

Author Contributions. J.V. directed the study. A.A. developed the continuous model and simulation code, implemented Baum-Welch, conducted the in-sample and out-of-sample analysis,

and generated all figures and tables. C.J. contributed to ECM-derivation methodology review and to validation of the cross-decade and cross-ticker pipelines. All authors edited and reviewed the final manuscript.

Data Availability Statement. The simulation scripts, training and testing data, and all code to reproduce the results in this paper are available under an MIT license from the paper repository: https://github.com/altashly1/CHMM-paper. The core model logic (fitting, simulation, decoding, and validation) is implemented in the companion Julia package `CHMM-Model.jl`: https://github.com/altashly1/CHMM-Model. Daily prices are sourced from Polygon.io via Massive (2014-01-03 to 2025-11-18) and Alpaca / IEX (extension through 2026-04-20).

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A Supplementary Material

This appendix collects supplementary material referenced in the main body: the public API of the companion package, the EM forward-backward recursions and algorithm pseudocode, the validation metric definitions, formal propositions, the full state-resolution and multi-emission sensitivity panels, robustness and KS-power calibration, the extended GARCH and SM-CHMM baselines, the full cross-asset panel, and the K-selection and ν_k diagnostics referenced from the discussion.

A.1 CHMM Algorithms and Derivations

This appendix collects the algorithmic content for the CHMM family: the forward-backward recursions and weighted M-step updates, the complete training pseudocode, and per-state degrees-of-freedom diagnostics for CHMM-t.

A.1.1 Architecture Diagram

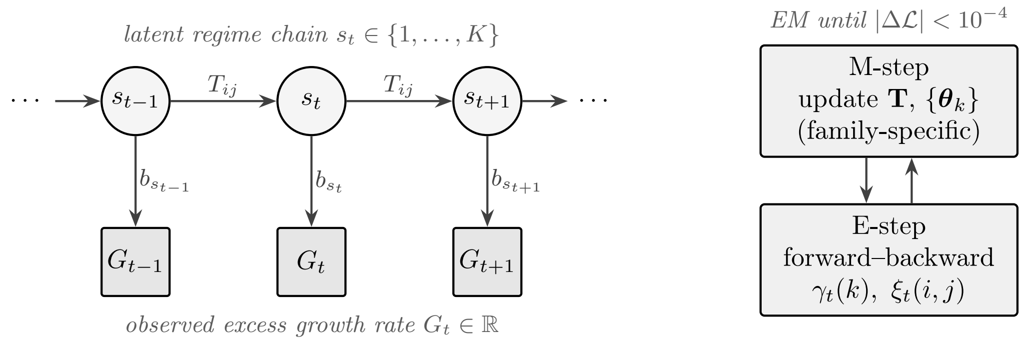


Figure 1. CHMM architecture and training loop. A K-state latent Markov chain with transition matrix $\mathbf{T}$ generates the regime sequence s_t; each s_t emits the observed excess growth rate G_t through a state-specific density b_{s_t}. The four CHMM variants share everything except the per-state emission family (Gaussian, Student-t, Laplace, GED). EM alternates an E-step (log-space forward-backward) and a family-specific M-step until $|\Delta\mathcal{L}| < 10^{-4}$.

A.1.2 Forward-Backward Recursions and M-Step Updates

For reference, the in-text Baum-Welch description of the Methods is underpinned by the standard log-space recursions [17, 64]. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_{1:t}, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1:T} \mid S_t = k, \mathcal{M})$. The log-space recursions are

$$\log \alpha_1(k) = \log \pi_k + \log f_k(O_1), \tag{8}$$
$$\log \alpha_t(j) = \operatorname*{logsumexp}_i(\log \alpha_{t-1}(i) + \log T_{ij}) + \log f_j(O_t), \quad t = 2, \dots, T, \tag{9}$$

$$\log \beta_T(k) = 0, \tag{10}$$
$$\log \beta_t(i) = \operatorname*{logsumexp}_j\left(\log T_{ij} + \log f_j(O_{t+1}) + \log \beta_{t+1}(j)\right), \quad t = T{-}1, \dots, 1, \tag{11}$$

where $\operatorname{logsumexp}_i(x_i) = x_{\max} + \log \sum_i \exp(x_i - x_{\max})$. The posterior state-occupancy $\gamma_t(k)$ and pair-occupancy $\xi_t(i,j)$ quantities are

$$\gamma_t(k) = \frac{\alpha_t(k)\beta_t(k)}{\sum_j \alpha_t(j)\beta_t(j)}, \qquad \xi_t(i,j) = \frac{\alpha_t(i)\, T_{ij}\, f_j(O_{t+1})\, \beta_{t+1}(j)}{\sum_{m,n} \alpha_t(m)\, T_{mn}\, f_n(O_{t+1})\, \beta_{t+1}(n)}, \tag{12}$$

and the closed-form M-step updates are

$$\pi_k^{\text{new}} = \gamma_1(k), \quad \mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, O_t}{\sum_t \gamma_t(k)}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)(O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}, \quad T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \tag{13}$$

The convergence criterion is $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\mathcal{L}^{(n)} = \operatorname{logsumexp}_k \log \alpha_T^{(n)}(k)$, and simulation draws $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ and then iterates $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma^2_{S_t})$, $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t,\cdot})$ with no jump process.

CHMM-t M-step (per-state ECM, closed form given $u_{t,k}$). For Student-t emissions $f_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ the ECM formulation of Peel and McLachlan [18], Liu and Rubin [19] treats the latent precision as an augmented E-step quantity,

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \tag{14}$$

so that conditional on ν_k the location and scale admit closed-form weighted updates,

$$\mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, O_t}{\sum_t \gamma_t(k)\, u_{t,k}}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}. \tag{15}$$

The degrees-of-freedom parameter ν_k is then refined by a one-dimensional golden-section search on the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. This two-block ECM preserves the monotonicity of the standard EM guarantee on the compact bracket $[\nu_{\min}, \nu_{\max}]$, with the search invariant $Q_k(\nu^{(n+1)}) \geq Q_k(\nu^{(n)})$ at every iteration; Proposition 1 states the guarantee in full. We extend this M-step with an optional exponential shrinkage prior on $1/\nu_k$, corresponding to the penalised objective

$$Q_k^{\text{pen}}(\nu) \;=\; \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k) \;-\; \lambda \,/\, \nu, \tag{16}$$

maximised by the same golden-section search ($\lambda = 0$ recovers the unpenalised Peel and McLachlan [18], Liu and Rubin [19] ECM; the rate sweep and its effect on simulated IS kurtosis are in the body Discussion).

CHMM-L M-step (closed-form weighted Laplace MLE). For Laplace emissions $f_k(x) = \frac{1}{2\beta_k} \exp(-|x - \mu_k|/\beta_k)$ (with the per-state Laplace scale denoted β_k rather than the canonical Laplace-scale symbol b) the weighted-MLE estimators are available in closed form:

$$\mu_k^{\text{new}} = \text{WeightedMedian}(O_{1:T};\, \gamma_{1:T}(k))\,, \qquad \beta_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, |O_t - \mu_k^{\text{new}}|}{\sum_t \gamma_t(k)}. \tag{17}$$

The weighted median is computed by cumulative-weight bracketing with linear interpolation between adjacent order statistics at the half-total weight crossing. The Laplace M-step carries no shape parameter, so CHMM-L is strictly cheaper per iteration than CHMM-t (which requires the per-state Q_k search).

CHMM-GED M-step (three-stage ECM with closed-form scale). For Generalized Error Distribution emissions $f_k(x) = \text{GED}(x; \mu_k, \alpha_k, p_k) = \frac{p_k}{2\alpha_k \Gamma(1/p_k)} \exp\big(-(|x - \mu_k|/\alpha_k)^{p_k}\big)$ the M-step decomposes into three CM updates per state. Given current $(\mu_k^{(n)}, \alpha_k^{(n)}, p_k^{(n)})$, CM-1 updates the location by minimising the per-state weighted L^{p_k} loss,

$$\mu_k^{(n+1)} \;=\; \arg \min_{\mu \in [\hat{\mu}_k - 5\alpha_k,\, \hat{\mu}_k + 5\alpha_k]} \sum_t \gamma_t(k) \left|O_t - \mu\right|^{p_k^{(n)}}, \tag{18}$$

by a 40-iteration golden-section search; CM-2 updates the scale in closed form given $(\mu_k^{(n+1)}, p_k^{(n)})$,

$$\alpha_k^{(n+1)} \;=\; \left[\frac{p_k^{(n)}}{W_k} \sum_t \gamma_t(k) \left|O_t - \mu_k^{(n+1)}\right|^{p_k^{(n)}}\right]^{1/p_k^{(n)}}, \qquad W_k \;=\; \sum_t \gamma_t(k); \tag{19}$$

and CM-3 updates the shape by maximising the per-state Q-function over the bracket,

$$p_k^{(n+1)} \;=\; \arg\max_{p\in[p_{\min},\,p_{\max}]} \;\sum_t \gamma_t(k)\,\log f_k\bigl(O_t;\,\mu_k^{(n+1)},\,\alpha_k^{(n+1)},\,p\bigr), \tag{20}$$

again by a 40-iteration golden-section search over $[p_{\min}, p_{\max}] = [0.5, 3.0]$. Equation (19) is the unique zero of $\partial Q_k/\partial\alpha$, so CM-2 is exact maximization. CM-1 and CM-3 are bracketed maximization. Each CM step is therefore a partial maximization, and the standard ECM monotonicity result [78] applies on the compact bracket; Proposition 1 states the guarantee in full. Boundary cases: $p_k = 2$ recovers Gaussian (with $\sigma = \alpha/\sqrt{2}$); $p_k = 1$ recovers Laplace (with $b = \alpha$).

A.1.3 Algorithm Descriptions

Algorithm 1 is the unified EM procedure shared across all four emission families: a shared scaffold (quantile-based initialisation, log-space forward-backward, transition update) plus four family-specific branches at lines 3, 13, and 30. Algorithm 2 is the simulator. Algorithm 3 is the cross-asset rank-reordering simulator (Pipeline B). Viterbi and SIM-resampling pseudocode is provided in the companion code repository; both are textbook constructions and we omit them from the appendix.

Algorithm 1 Unified EM algorithm for the CHMM family. Lines 3, 13, and 30 branch on the emission family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$; every other line is identical across families. CHMM-N and CHMM-L instantiate classical EM; CHMM-t and CHMM-GED instantiate ECM sequences whose final CM step updates the per-state shape parameter (ν_k for CHMM-t, p_k for CHMM-GED) by a one-dimensional golden-section search on the per-state Q-function. Hyperparameter defaults used throughout this paper: convergence tolerance $\mathrm{tol} = 10^{-4}$ on $\Delta\mathcal{L}$, `max_iter` $= 60$, 40-iteration golden-section sub-step on every bracketed CM, CHMM-t bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$ with initial $\nu^{(0)} = 10$, CHMM-GED bracket $(p_{\min}, p_{\max}) = (0.5, 3.0)$ with initial $p^{(0)} = 1.5$ and μ_k search interval $[\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]$.

Require: Observations $O_{1:T}$, number of states K, tolerance tol, `max_iter`, emission family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$, if $F = t$ bracket $(\nu_{\min}, \nu_{\max})$ and initial $\nu^{(0)}$, if $F = \mathrm{GED}$ bracket $(p_{\min}, p_{\max})$ and initial $p^{(0)}$.
Ensure: Transition matrix $\mathbf{T}$, per-state emission parameters $\boldsymbol{\theta}_{1:K}$, initial distribution $\boldsymbol{\pi}$.
1: **Quantile-Based Initialization.**
2: Sort $O^{(\mathrm{sort})} \leftarrow \mathrm{sort}(O_{1:T})$ and partition into K equal chunks $\{C_1, \ldots, C_K\}$.
3: **for** $k = 1$ to K **do**
4:   **switch** F
5:     $F = \mathcal{N}$: $\mu_k \leftarrow \mathrm{mean}(C_k)$, $\quad \sigma_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6})$.
6:     $F = t$: $\mu_k \leftarrow \mathrm{mean}(C_k)$, $\quad \sigma_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6})$, $\quad \nu_k \leftarrow \nu^{(0)}$.
7:     $F = L$: $\mu_k \leftarrow \mathrm{median}(C_k)$, $\quad b_k \leftarrow \max(\mathrm{mean}(|C_k - \mu_k|), 10^{-6})$.
8:     $F = \mathrm{GED}$: $\mu_k \leftarrow \mathrm{mean}(C_k)$, $\quad \alpha_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6})$, $\quad p_k \leftarrow p^{(0)}$.
9: **end for**
10: $T_{ij} \leftarrow 1/K$; $\quad \pi_k \leftarrow 1/K$; $\quad \mathcal{L}_{\mathrm{prev}} \leftarrow -\infty$.
11: **EM Loop.**
12: **for** $n = 1$ to `max_iter` **do**
13:   **E-Step, per-family emission log-likelihood.**
14:   **switch** F
15:     $F = \mathcal{N}$: $\log B_{t,k} \leftarrow \log \mathcal{N}(O_t \mid \mu_k, \sigma_k^2)$.
16:     $F = t$: $\log B_{t,k} \leftarrow \log t_{\nu_k}(O_t; \mu_k, \sigma_k)$.
17:     $F = L$: $\log B_{t,k} \leftarrow -\log(2b_k) - |O_t - \mu_k|/b_k$.
18:     $F = \mathrm{GED}$: $\log B_{t,k} \leftarrow \log p_k - \log(2\alpha_k) - \log\Gamma(1/p_k) - (|O_t - \mu_k|/\alpha_k)^{p_k}$.
19:   **E-Step, shared log-space forward-backward.**
20:   $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$.
21:   **for** $t = 2$ to T, $j = 1$ to K **do**
22:     $\log \alpha_t(j) \leftarrow \operatorname*{logsumexp}_i(\log \alpha_{t-1}(i) + \log T_{ij}) + \log B_{t,j}$.
23:   **end for**

Algorithm 2 CHMM simulation of synthetic excess growth rate paths (unified four-family scaffold). The emission draw at line 4 branches on the family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$ exactly as in Algorithm 1: $\mathcal{N}(\mu_k, \sigma_k^2)$ for CHMM-N, $t_{\nu_k}(\mu_k, \sigma_k)$ for CHMM-t, Laplace(μ_k, β_k) for CHMM-L, GED(μ_k, α_k, p_k) for CHMM-GED; the latent chain and transition step are shared.

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi}, F)$ with $\boldsymbol{\theta}_k = (\mu_k, \sigma_k)$ for $F = \mathcal{N}$; (μ_k, σ_k, ν_k) for $F = t$; (μ_k, β_k) for $F = L$; (μ_k, α_k, p_k) for $F = \mathrm{GED}$. Number of steps M, Number of paths P
Ensure: Simulated return paths $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$
1: **for** $p = 1$ to P **do**
2: Sample initial state $S_1 \sim \mathrm{Categorical}(\boldsymbol{\pi})$.
3: **for** $t = 1$ to M **do**
4: Emit observation $\hat{G}^{(p)}_t \sim f_{S_t}(\,\cdot\,; \boldsymbol{\theta}_{S_t})$ with f_k the F-family per-state density.
5: **if** $t < M$ **then**
6: Transition to next state $S_{t+1} \sim \mathrm{Categorical}(\mathbf{T}_{S_t,\cdot})$.
7: **end if**
8: **end for**
9: **end for**
10: **return** $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$

$\log \beta_T(k) \leftarrow 0$.
for $t = T-1$ down to 1, $i = 1$ to K **do**
$\log \beta_t(i) \leftarrow \operatorname{logsumexp}_j(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$.
end for
Compute $\gamma_t(k)$ and $\xi_t(i,j)$ from equation (12).
Shared transition update. $T_{ij} \leftarrow \sum_t \xi_t(i,j) \,/\, \sum_t \gamma_t(i)$; $\pi_k \leftarrow \gamma_1(k)$.
M-Step, per-family emission update.
switch F
$F = \mathcal{N}$ {classical Baum-Welch [16].}
$\mu_k \leftarrow \sum_t \gamma_t(k)\, O_t / \sum_t \gamma_t(k)$.
$\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
$F = t$ {ECM of Peel and McLachlan [18], Liu and Rubin [19]; closed-form (μ_k, σ_k) given ν_k, then 1D search on ν_k.}
$u_{t,k} \leftarrow (\nu_k + 1)/(\nu_k + ((O_t - \mu_k)/\sigma_k)^2)$ for all t, k.
$\mu_k \leftarrow \sum_t \gamma_t(k)\, u_{t,k}\, O_t / \sum_t \gamma_t(k)\, u_{t,k}$.
$\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
$\nu_k \leftarrow \arg\max_{\nu \in [\nu_{\min}, \nu_{\max}]} \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ {40-iter golden-section search.}
$F = L$ {closed-form weighted Laplace MLE.}
$\mu_k \leftarrow \mathrm{WeightedMedian}(O_{1:T},\ \gamma_{1:T}(k))$ {cumulative-weight bracketing with interpolation at the $W_k/2$ crossing.}
$b_k \leftarrow \max\big(\sum_t \gamma_t(k)\, |O_t - \mu_k| / \sum_t \gamma_t(k),\ 10^{-6}\big)$.
$F = \mathrm{GED}$ {three-stage ECM: bracketed μ_k, closed-form α_k, bracketed p_k.}
$\mu_k \leftarrow \arg\min_{\mu \in [\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]} \sum_t \gamma_t(k)\, |O_t - \mu|^{p_k}$ {40-iter golden-section search.}
$\alpha_k \leftarrow \max\Big(\big[(p_k/W_k) \sum_t \gamma_t(k)\, |O_t - \mu_k|^{p_k}\big]^{1/p_k},\ 10^{-6}\Big)$ where $W_k = \sum_t \gamma_t(k)$ {closed form, unique zero of $\partial Q_k / \partial \alpha$.}
$p_k \leftarrow \arg\max_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k) \log \mathrm{GED}(O_t; \mu_k, \alpha_k, p)$ {40-iter golden-section search.}
Convergence check. $\mathcal{L}^{(n)} \leftarrow \operatorname{logsumexp}_k(\log \alpha_T(k))$.
if $|\mathcal{L}^{(n)} - \mathcal{L}_{\mathrm{prev}}| < \mathrm{tol}$ **then**
break
end if
$\mathcal{L}_{\mathrm{prev}} \leftarrow \mathcal{L}^{(n)}$.
end for
return $\mathbf{T}, \boldsymbol{\theta}_{1:K}, \boldsymbol{\pi}$ {$\boldsymbol{\theta}_k = (\mu_k, \sigma_k)$ for $F = \mathcal{N}$; (μ_k, σ_k, ν_k) for $F = t$; (μ_k, b_k) for $F = L$; (μ_k, α_k, p_k) for $F = \mathrm{GED}$.}

Algorithm 3 Cross-asset copula simulation via rank reordering

Require: Fitted per-asset CHMMs $\{\mathcal{M}_j\}_{j=1}^d$, correlation matrix Σ (positive semi-definite (PSD), unit diagonal), copula family $\mathcal{C} \in \{\text{Gaussian}, t_\nu\}$, path length T, number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

for $p = 1$ to P **do**
  Draw copula sample $\mathbf{U}^{(p)} \in [0, 1]^{T \times d}$ from $\mathcal{C}(\Sigma)$:
    $L \leftarrow \text{Cholesky}(\Sigma), \quad Z \in \mathbb{R}^{T \times d} \sim \mathcal{N}(0, I_d)$ i.i.d.
    $Y \leftarrow ZL^\top$
  if $\mathcal{C} =$ Gaussian **then**
    $\mathbf{U}^{(p)} \leftarrow \Phi(Y)$ componentwise
  else
    Draw $W \sim \chi_\nu^2/\nu$ i.i.d. for $t = 1, \dots, T$; $\quad X_{t,\cdot} \leftarrow Y_{t,\cdot}/\sqrt{W_t}$
    $\mathbf{U}^{(p)} \leftarrow t_\nu(X)$ componentwise
  end if
  for $j = 1$ to d **do**
    Simulate $\tilde{g}_{j,1:T}^{(p)}$ from $\mathcal{M}_j$ via Algorithm 2.
    Compute order statistics $\tilde{g}_{j,(1)}^{(p)} \leq \cdots \leq \tilde{g}_{j,(T)}^{(p)}$.
    Compute ranks $r_{j,t}^{(p)} \leftarrow \text{rank}(U_{j,t}^{(p)})$ for $t = 1, \dots, T$.
    $\hat{G}_{j,t}^{(p)} \leftarrow \tilde{g}_{j,(r_{j,t}^{(p)})}^{(p)}$ for $t = 1, \dots, T$.
  end for
end for
return $\{\hat{G}_{j,t}^{(p)}\}$

A.1.4 Per-State Shape Diagnostics: CHMM-t and CHMM-GED

This appendix backs the discussion in the body Discussion of the per-state shape allocation under CHMM-t and CHMM-GED on SPY IS at $K = 18$ (seed = 20260420).

CHMM-t per-state ν_k. Figure 2 plots the per-state ν_k histogram for the $K = 18$ CHMM-t fit. Thirteen of the eighteen states rest at the upper ECM bracket ($\nu_k = 50$), and only two states lie near the lower bracket ($\nu_2 = 2.19$, $\nu_4 = 2.10$). The simulated IS mixture kurtosis of 17.34 is driven by the posterior mass routed to those two very heavy-tailed states, not by a systemic collapse to the lower bracket.

CHMM-GED per-state p_k (Gaussian-bulk / Laplace-tail bimodality). The body the body generator comparison reports and discusses the $\hat{p}_k$ partition (Figure 5). Briefly, the $K = 18$ CHMM-GED fit on SPY IS is bimodal: thirteen states attain the upper ECM bracket at $p_k = 3.0$, one state lands at intermediate $p_k = 1.6$, and four states cluster in the Laplace-shape regime $p_k \in [0.86, 1.24]$, concentrated in the high-volatility ranks. The smallest fitted p_k on SPY is 0.86, well clear of every $p_{\min}$ tested in the bracket sweep, so the lower bracket is non-binding. The bimodal partition replicates across 10 Monte Carlo seeds and across the cross-asset universe (Appendix A.8).

A CHMM-t bracket sweep on $\nu_{\min} \in \{2.1, 2.5, 3.0, 4.0, \infty\}$ at $K = 18$ shows the simulated IS kurtosis stays in the 12.5–15.1 band across finite brackets and only collapses to 5.03 in the full Gaussian limit; the overshoot is a feature of the Student-t emission at $K = 18$, not a lower-bracket artefact. The analogous CHMM-GED sweep on $p_{\max} \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$ leaves the Gaussian-like-state count invariant at 13/18 and the heavy-shape count in $\{4, 5\}$; tightening $p_{\max}$ from 4.0 to 3.0 costs about 4 nats of LL. The $p_{\min}$ sweep on $\{0.5, 0.7, 0.85\}$ is fully degenerate: every floor produces an identical fit because the smallest observed $\hat{p}_k$ on SPY is 0.86.

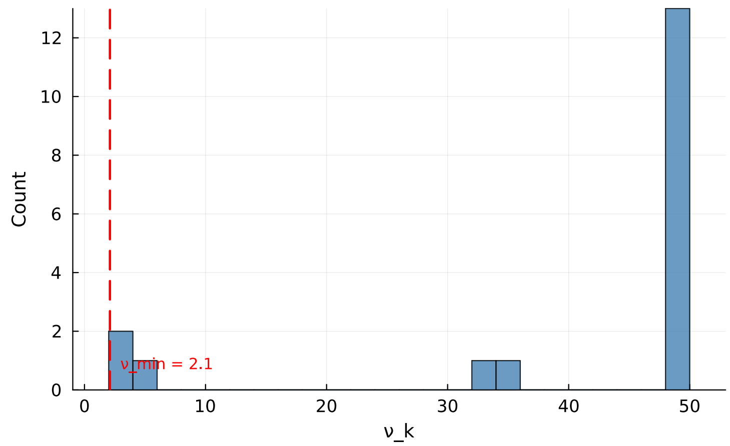


Figure 2. Per-state degrees-of-freedom ν_k for the CHMM-t fit at $K = 18$ on SPY IS. The dashed red line marks the lower ECM bracket $\nu_{\min} = 2.1$. Only two states ($\nu_2 = 2.19$, $\nu_4 = 2.10$) sit near the lower bracket; the median ν_k is at the upper bracket.

A.1.5 Pipeline Summary

Pipeline A: observed returns $G_t \to$ fit CHMM (Baum-Welch, body operating point $K^\star = 3$ with $K = 18$ as the kurtosis-fidelity sensitivity reference; the appendix per-ticker rollup of Table 9 and the Pipeline-B marginals of Table 4 use $K = 18$) $\to$ simulate 1,000 paths $\to$ per-asset metrics (Table 1). Pipeline B reuses the Pipeline-A $K = 18$ per-asset fits as marginals, draws a copula sample $\mathbf{U} \sim \mathcal{C}(\Sigma)$ from the fitted Student-t copula on ranks, rank-reorders the per-asset Pipeline-A paths to match $\mathbf{U}$, and reports cross-asset metrics (Table 4). The single-asset scaffold is shared across both pipelines; the CHMM training and simulation steps are Algorithms 1 and 2, the cross-asset rank-reordering step is Algorithm 3, and the architecture diagram for the single-asset scaffold is Figure 1.

A.2 Validation Metric Details

This appendix documents the precise form of the seven per-path metrics referenced in the body evaluation protocol. For each model, 1,000 independent paths of length T are simulated (200 paths for the cross-asset extension) and each metric is computed per path before aggregation.

Kolmogorov–Smirnov (KS) pass rate. The two-sample KS statistic [66, 67] measures the maximum vertical distance between two empirical CDFs,

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)|. \tag{21}$$

The KS pass rate is the fraction of paths for which the test fails to reject distributional equivalence against the reference data at $\alpha = 0.05$.

Anderson–Darling (AD) pass rate. The AD test is the integrated squared CDF deviation with a quadratic tail weighting, making it more sensitive than KS to tail discrepancies. The AD pass rate is the fraction of paths for which the two-sample AD test fails to reject at $\alpha = 0.05$.

Excess kurtosis. Mean simulated excess kurtosis across paths, compared to the observed value. Given Gaussian emissions, the CHMM is expected to underestimate kurtosis relative to the heavy-tailed empirical distribution, and the shortfall is one of the reported diagnostics.

ACF-MAE. Mean absolute error between observed and mean-simulated autocorrelation functions of $|G_t|$ over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L}\sum_{\tau=1}^{L}\left|\hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau)\right|. \tag{22}$$

This is the direct diagnostic for the Rydén et al. limitation.

Wasserstein-1 distance. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [79],

$$W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)|\, du. \tag{23}$$

Hellinger distance. A histogram-based overlap distance in $[0, 1]$,

$$H(P, Q) = \frac{1}{\sqrt{2}}\sqrt{\sum_{i=1}^{B}(\sqrt{p_i} - \sqrt{q_i})^2}, \tag{24}$$

with B equal-width bins spanning the joint support and p_i, q_i the observed and simulated bin probabilities.

Quantile coverage. The fraction of 99 equally-spaced observed quantiles (1st through 99th percentile) falling inside the 5th–95th percentile envelope of the corresponding simulated quantile across 1,000 paths. A coverage of 100% indicates that the simulation envelope fully brackets the observed distribution at every percentile.

Cross-asset metrics. For the cross-asset extension we additionally track the mean Frobenius norm $\|\hat{\Sigma}^{(p)}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}\|_F$ and the off-diagonal Pearson correlation MAE between simulated and observed return matrices, averaged over paths. The off-diagonal MAE is averaged over the $d(d-1)/2$ unique upper-triangular entries of the symmetric correlation matrix (so 15 entries for the six-asset cross-section of the body cross-asset analysis), not double-counted across both triangles:

$$\text{MAE}_{\text{off-diag}} = \frac{1}{P}\sum_{p=1}^{P}\frac{2}{d(d-1)}\sum_{1\le i<j\le d}\left|\hat{\Sigma}^{(p)}_{\text{sim},\,ij} - \hat{\Sigma}_{\text{obs},\,ij}\right|.$$

A.2.1 Continuous Ranked Probability Score and Diebold-Mariano

The OoS CRPS column of Table 1 reports a proper-scoring-rule complement to the KS pass rate. We use the unbiased sample CRPS estimator: for an ensemble of N simulated paths $\{x_i\}_{i=1}^{N}$ at time t and observed value y_t,

$$\widehat{\text{CRPS}}_t = \frac{1}{N}\sum_{i=1}^{N}|x_i - y_t| - \frac{1}{N(N-1)}\sum_{1\le i<j\le N}|x_i - x_j|. \tag{25}$$

The double sum is computed in $O(N \log N)$ via the sorted-ensemble identity $\sum_{i<j}(x_{(j)} - x_{(i)}) = \sum_i x_{(i)}(2i - N - 1)$. The ensemble at each t is the cross-section of the $N = 1{,}000$ unconditional simulated paths used throughout the body Results; this yields a marginal-predictive CRPS suitable for an unconditional generative-fidelity assessment, distinct from the conditional one-step-ahead CRPS used in forecasting.

For pairwise comparison we report a Diebold-Mariano test on the per-t loss differential $d_t = \widehat{\text{CRPS}}_t^A - \widehat{\text{CRPS}}_t^B$ with a Newey-West HAC variance under a Bartlett kernel and bandwidth $h = \lfloor T_{\text{OoS}}^{1/3} \rfloor = 8$. Two-sided p-values are reported under the standard-normal null. The body claim of the body generator comparison is: CHMM ties or beats every benchmark on mean OoS CRPS, with Gaussian i.i.d. the only significantly worse comparator.

A.2.2 Christoffersen Conditional-Coverage Panel (Companion to the body VaR back-test)

The body the body VaR back-test reports the unconditional Kupiec LR_{uc} statistic. This appendix reports the full Christoffersen [80] panel: LR_{uc} (unconditional coverage), LR_{ind} (breach independence), and $\text{LR}_{\text{cc}} = \text{LR}_{\text{uc}} + \text{LR}_{\text{ind}}$ (joint conditional coverage), at $\alpha \in \{0.01, 0.05\}$ on the SPY IS and OoS windows. The breach series is constructed from the pooled-archive VaR threshold (the α-quantile of vec(sim archive) across paths and time), then $\text{br}_t = (R_t \leq \widehat{\text{VaR}}_\alpha)$.

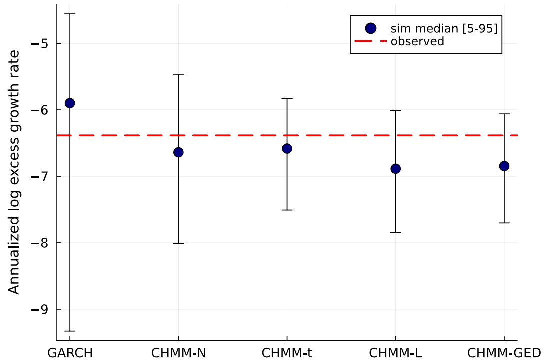
The headline reads as follows. On the unconditional (Kupiec) leg every CHMM variant passes at both levels on both windows; the LR_{uc} statistic for the four CHMM rows at $\alpha = 0.05$ on OoS clusters at 3.03–3.83 (just below the χ_1^2 critical value 3.841), reflecting the integer-breach grid at $T_{\text{OoS}} = 572$ noted in the body table caption. On the independence leg every unconditional generator in the panel (bootstrap, GARCH, and all four CHMM variants) rejects: LR_{ind} is in the range 4.71–56.46 across rows. This is structural rather than CHMM-specific. Volatility-clustering in R_{oos} produces persistent regimes of high and low realised volatility, the OoS-window breaches concentrate inside the high-volatility regime, and a constant-across-t VaR threshold (the construction every unconditional generator uses) cannot track that timing. The natural fix is a regime-conditional VaR that propagates the CHMM latent-state forecast $\mathbb{P}(s_{t+1} \mid \text{history}_t)$ through the per-state emission-mixture; this is operationally the conditional-VaR companion-paper direction noted in the body Discussion.

Table 7. Christoffersen conditional-coverage VaR back-test (SPY, seed = 20260420, $K = 18$, 1,000 simulated paths). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. "br rate" is the empirical breach rate; $\mathrm{LR}_{\mathrm{uc}}$ is the Kupiec statistic ($\sim \chi^2_1$ under correct unconditional coverage); $\mathrm{LR}_{\mathrm{ind}}$ is the Christoffersen independence statistic ($\sim \chi^2_1$ under independent breaches); $\mathrm{LR}_{\mathrm{cc}} = \mathrm{LR}_{\mathrm{uc}} + \mathrm{LR}_{\mathrm{ind}}$ is the joint conditional-coverage statistic ($\sim \chi^2_2$). PASS if statistic is below critical. Every unconditional generator (i.i.d. bootstrap, GARCH, and all four CHMM variants) rejects independence on OoS at $\alpha = 0.05$, consistent with breach clustering driven by volatility-clustering in R_{oos} rather than a CHMM-specific mis-specification of conditional dynamics.

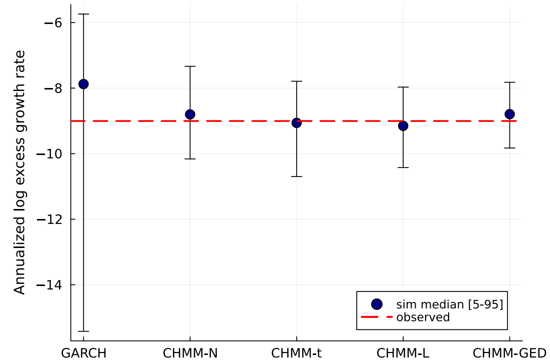
					Kupiec		Christ. ind		Christ. cc	
Model	Win	α	breaches	br rate	$\mathrm{LR}_{\mathrm{uc}}$	p	$\mathrm{LR}_{\mathrm{ind}}$	p	$\mathrm{LR}_{\mathrm{cc}}$	p
Bootstrap	IS	0.01	26	1.03%	0.03	0.87	15.28	0.00	15.31	0.00
Bootstrap	IS	0.05	126	5.01%	0.00	0.99	56.46	0.00	56.46	0.00
Bootstrap	OoS	0.01	5	0.87%	0.10	0.76	13.18	0.00	13.28	0.00
Bootstrap	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01
GARCH(1,1)	IS	0.01	31	1.23%	1.28	0.26	12.45	0.00	13.72	0.00
GARCH(1,1)	IS	0.05	139	5.52%	1.41	0.24	45.38	0.00	46.79	0.00
GARCH(1,1)	OoS	0.01	6	1.05%	0.01	0.91	20.87	0.00	20.89	0.00
GARCH(1,1)	OoS	0.05	24	4.20%	0.82	0.37	5.87	0.02	6.69	0.04
CHMM-N	IS	0.01	20	0.79%	1.15	0.28	12.80	0.00	13.95	0.00
CHMM-N	IS	0.05	123	4.89%	0.07	0.80	47.33	0.00	47.40	0.00
CHMM-N	OoS	0.01	3	0.52%	1.58	0.21	18.98	0.00	20.56	0.00
CHMM-N	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01
CHMM-t	IS	0.01	22	0.87%	0.42	0.52	11.62	0.00	12.04	0.00
CHMM-t	IS	0.05	127	5.05%	0.01	0.91	55.54	0.00	55.55	0.00
CHMM-t	OoS	0.01	4	0.70%	0.58	0.45	15.53	0.00	16.12	0.00
CHMM-t	OoS	0.05	20	3.50%	3.03	0.08	4.71	0.03	7.74	0.02
CHMM-L	IS	0.01	19	0.76%	1.66	0.20	13.44	0.00	15.11	0.00
CHMM-L	IS	0.05	134	5.33%	0.55	0.46	49.43	0.00	49.98	0.00
CHMM-L	OoS	0.01	2	0.35%	3.26	0.07	9.15	0.00	12.41	0.00
CHMM-L	OoS	0.05	20	3.50%	3.03	0.08	4.71	0.03	7.74	0.02
CHMM-GED	IS	0.01	20	0.79%	1.15	0.28	12.80	0.00	13.95	0.00
CHMM-GED	IS	0.05	126	5.01%	0.00	0.99	56.46	0.00	56.46	0.00
CHMM-GED	OoS	0.01	2	0.35%	3.26	0.07	9.15	0.00	12.41	0.00
CHMM-GED	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01

A.2.3 VaR / ES Envelope Visualization (Companion to Table 19)

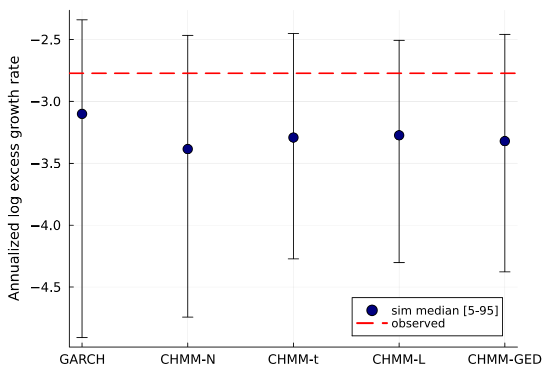
Figure 3 visualizes the median and $[5\%, 95\%]$ envelopes across simulated paths whose numeric values are reported in Table 19.



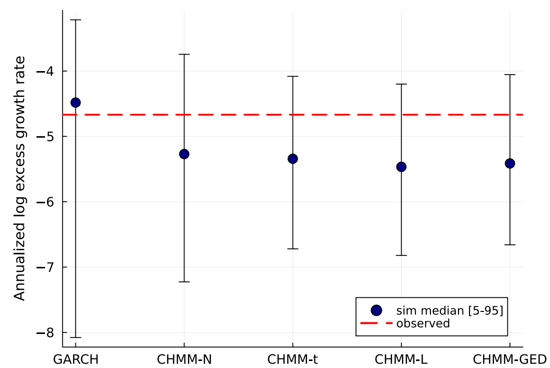
(a) IS VaR ($\alpha = 0.01$).



(b) IS ES ($\alpha = 0.01$).



(c) OoS VaR ($\alpha = 0.05$).



(d) OoS ES ($\alpha = 0.05$).

Figure 3. VaR / ES envelope diagnostic (SPY, $N_{\text{paths}} = 1{,}000$; global seed policy in the body evaluation protocol). Each panel shows the simulated median (dot) and $[5\%, 95\%]$ envelope (error bar) across paths against the observed historical value (dashed red line); panels (a)/(b) are the IS window at $\alpha = 0.01$, panels (c)/(d) are the OoS window at $\alpha = 0.05$, and the x-axis in each panel lists the five generators kept for the volatility-aware conditional-coverage comparison (GARCH(1,1), CHMM-N, CHMM-t, CHMM-L, CHMM-GED). The i.i.d. bootstrap is omitted here because its envelope is by construction matched to the empirical marginal and does not discriminate among volatility-aware models. The y-axis is the annualized log excess growth rate (daily log return scaled by $1/\Delta t$ with $\Delta t = 1/252$), consistent with the $R_{\text{IS}}/R_{\text{OoS}}$ convention used throughout the paper. Headline: every observed red line falls inside the corresponding simulated envelope, confirming that the CHMM family passes the coverage leg of the back-test complementary to the Kupiec likelihood-ratio leg in the body VaR back-test.

A.3 Formal Theoretical Statements

The following assumptions and propositions are referenced in the body spectral mechanism and stated formally here for reference; each is a one-line specialisation of a standard result and the proof sketch is one sentence.

Assumption 1 (Irreducibility and aperiodicity)**.** *The transition matrix $\mathbf{T} \in \mathbb{R}^{K\times K}$ is irreducible and aperiodic on the state space $\{1, \dots, K\}$, so the chain admits a unique stationary distribution $\bar{\boldsymbol{\pi}}$ satisfying $\bar{\boldsymbol{\pi}}\,\mathbf{T} = \bar{\boldsymbol{\pi}}$ and $\lambda_1 = 1$ is a simple eigenvalue with all other eigenvalues strictly inside the unit disk [69].*

Assumption 2 (Finite second moment and emission contrast)**.** *Each per-state emission density $f_k(\cdot;\boldsymbol{\theta}_k)$ has finite second moment, $\mathbb{E}_{f_k}[G^2] < \infty$, and at least two states differ in either location μ_k or scale σ_k.*

Proposition 1 (ECM monotonicity for CHMM-t and CHMM-GED (and EM monotonicity for CHMM-N, CHMM-L))**.** *Under compactness of the parameter space (μ_k, σ_k or β_k or α_k, $\nu_k \in [\nu_{\min}, \nu_{\max}]$, $p_k \in [p_{\min}, p_{\max}]$ each in a closed bounded interval, $T_{ij} \geq \epsilon_T > 0$), the two-block CM step for CHMM-t of the body Methods, namely the closed-form weighted updates of $(\mathbf{T}, \boldsymbol{\pi}, \mu_k, \sigma_k)$ at fixed ν_k, followed by a golden-section maximisation of the per-state Q-function over ν_k, satisfies $Q(\theta^{(n+1)} \mid \theta^{(n)}) \geq Q(\theta^{(n)} \mid \theta^{(n)})$, and consequently $\mathcal{L}(\theta^{(n+1)}) \geq \mathcal{L}(\theta^{(n)})$ where $\mathcal{L}$ is the incomplete-data log-likelihood. The same monotonicity holds for the closed-form CHMM-N and CHMM-L M-steps under the standard EM guarantee, and for the three-stage CHMM-GED CM sequence (bracketed μ_k, closed-form α_k, bracketed p_k) of equations (18)–(20).*

Proof sketch. Each CM step does not decrease Q by definition (closed-form steps are exact maximisers, golden-section steps are bracketed maxima within the prescribed interval); the standard EM inequality $\mathcal{L}(\theta^{(n+1)}) - \mathcal{L}(\theta^{(n)}) \geq Q(\theta^{(n+1)} \mid \theta^{(n)}) - Q(\theta^{(n)} \mid \theta^{(n)})$ [78, 81] converts Q-monotonicity into log-likelihood monotonicity. Compactness gives boundedness above, hence convergence.

Numerical full-rank check on $\hat{\mathbf{T}}$. The Allman-Matias-Rhodes 2009 identifiability theorem requires $\hat{\mathbf{T}}$ of full rank. The fitted $\hat{\mathbf{T}}$ at $K = 18$ across the four emission families on SPY IS is full rank in the strict sense (smallest singular value $\sigma_{\min} \in [0.0007, 0.0170]$, condition numbers 63–1,620), and the deflated matrix $\hat{\mathbf{T}} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ has numerical rank exactly $K - 1 = 17$ across all four families.

Proposition 2 (Identifiability up to label permutation)**.** *Under irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1), the emission contrast condition (Assumption 2), and pairwise distinctness of the per-state parameter tuples $\boldsymbol{\theta}_k$ across $k = 1, \ldots, K$, the CHMM parameter is identifiable up to label permutation for CHMM-N (Gaussian) and CHMM-L (Laplace). For CHMM-t (with per-state $\nu_k \in [\nu_{\min}, \nu_{\max}]$) and CHMM-GED (with per-state $p_k \in [p_{\min}, p_{\max}]$) the same conclusion holds modulo the standard caveat that pairwise distinctness must hold jointly across all components of $\boldsymbol{\theta}_k = (\mu_k, \sigma_k, \nu_k)$ or (μ_k, α_k, p_k), since within either bracket a Gaussian-shaped emission with inflated scale can mimic a Laplace-shaped emission with deflated scale at intermediate quantiles.*

Proof sketch. The HMM identifiability theorem of Allman et al. [82] requires $\mathbf{T}$ of full rank and per-state emissions linearly independent in the space of densities on $\mathbb{R}$. Full-rank $\mathbf{T}$ follows from irreducibility [69]. For CHMM-N and CHMM-L, linear independence of Gaussian and Laplace emissions with distinct location-scale parameters is direct from Yakowitz and Spragins [83], who establish that finite mixtures over location-scale families with strictly decreasing-tail densities are identifiable (the CHMM-N and CHMM-L marginals are special cases). For CHMM-t at fixed shape and CHMM-GED at fixed shape, location-scale identifiability is the same Yakowitz-Spragins result; for finite mixtures of Student-t with varying degrees-of-freedom the standard reference is Lindsay [84], Chapter 3, which gives identifiability via the moment-generating function over the bracketed range of shape parameters and a non-degeneracy condition on the joint location-scale-shape tuple. The CHMM-GED case under bracketed $p_k \in [p_{\min}, p_{\max}]$ is a parallel application of the moment-matching argument: the GED characteristic function $\hat{b}(t; \mu, \alpha, p) = \exp(i\mu t)\, M(t; \alpha, p)$ has a unique inverse over the bounded shape range up to permutation of components, modulo the joint-distinctness caveat on (μ_k, α_k, p_k). Strict linear independence at the boundary cases $p_k = 1$ (Laplace) and $p_k = 2$ (Gaussian) is automatic from Yakowitz-Spragins; in the bracket interior, the joint distinctness condition above is required to rule out the location-scale-shape trade-off.

Proposition 3 (MLE consistency)**.** *Under Assumptions 1–2 and the assumption that the true parameter lies in the interior of a compact Θ, the MLE $\hat{\theta}_T = \arg\max_{\theta\in\Theta} \mathcal{L}_T(\theta)$ is consistent: $\hat{\theta}_T \xrightarrow{p} \theta_0$ as $T \to \infty$, modulo label permutation.*

Proof reference. Bickel et al. [85] (Theorem 1) and Douc et al. [86] apply directly under our assumption set: irreducibility supplies the mixing condition, finite second moment the moment condition, compactness the parameter-space condition.

Proposition 4 (Marginal preservation under rank reordering)**.** *Let $\tilde{g}_{j,1:T}$ be a sample of size T from the per-asset CHMM marginal of asset j, and let the rank-reordered output be $\hat{g}^{(p)}_{j,t} = \tilde{g}^{(p)}_{j,(r^{(p)}_{j,t})}$ where $r^{(p)}_{j,t}$ is the rank of the copula sample. Then for every asset j the empirical CDF of $\{\hat{g}^{(p)}_{j,t}\}_{t=1}^{T}$ equals that of $\{\tilde{g}^{(p)}_{j,t}\}_{t=1}^{T}$.*

Proof sketch. The map $t \mapsto r^{(p)}_{j,t}$ is a permutation almost surely, so the multiset of values is preserved; the empirical CDF depends only on the multiset.

A.4 Spectral ACF Identity: Extended Derivation

Equation (6) is the closed-form mixture-of-eigenvalues form for the absolute-return ACF used in the body spectral mechanism and invoked throughout the body Results and Discussion. The bilinear cross-moment identity $\mathbb{E}[|G_t|\,|G_{t+\tau}|] = \mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{T}^\tau\,\mathbf{m}$ and its eigendecomposition over the non-unit eigenvalues of $\mathbf{T}$ are textbook material in the regime-switching literature (13, Section 22.2; 14, Chapter 3; the moment-generating-function form for general Markov-switching specifications with conditional means and variances depending on the latent state is given by 15); the underlying Markov-chain spectral decomposition is standard [69]. We restate the absolute-return specialisation here as a self-contained theorem with a step-by-step proof, and record the lag-zero behaviour and the complex / non-diagonalisable extensions that the main body does not. Our contribution on this axis is the explicit application of the spectral form to recast the empirical Rydén et al. [11] low-K failure as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ (the body spectral mechanism, "Rydén separation as a K-rank statement") and the empirical demonstration that the rank constraint is the binding axis at moderate K on equity-return data (the body Results and Discussion).

Theorem 1 (Mixture-of-eigenvalues identity for the absolute-return ACF)**.** *Let $\{(s_t, G_t)\}_{t\in\mathbb{Z}}$ be a stationary CHMM on $\{1,\dots,K\}$ with irreducible aperiodic transition matrix $\mathbf{T}$ (Assumption 1) and per-state emissions of finite second moment (Assumption 2). Write $\bar{\boldsymbol{\pi}}$ for the unique stationary distribution, $\mathbf{D} = \mathrm{diag}(\bar{\boldsymbol{\pi}})$, $m_k = \mathbb{E}[|G_t| \mid s_t = k]$, $\mathbf{m} = (m_1,\dots,m_K)^\top$, $\mu = \mathbb{E}|G_t| = \bar{\boldsymbol{\pi}}^\top\mathbf{m}$, and $\sigma^2_{|G|} = \mathrm{Var}(|G_t|)$. Assume $\mathbf{T}$ is diagonalisable with right eigenvectors $\mathbf{v}_k$ and left eigenvectors $\mathbf{w}_k$ biorthonormal, $\mathbf{w}_k^\top\mathbf{v}_l = \delta_{kl}$, indexed so that $\lambda_1 = 1 > |\lambda_2| \ge \cdots \ge |\lambda_K|$. Then for every integer lag $\tau \ge 1$,*

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^{K} w_k\,\lambda_k^\tau, \qquad w_k \;=\; \frac{(\mathbf{m}^\top\mathbf{D}\,\mathbf{v}_k)\,(\mathbf{w}_k^\top\mathbf{m})}{\sigma^2_{|G|}}.$$

Proof. By stationarity, $\rho_{|G|}(\tau) = (\mathbb{E}[|G_t|\,|G_{t+\tau}|] - \mu^2)/\sigma^2_{|G|}$, so it suffices to evaluate the cross-moment for $\tau \ge 1$. Conditioning on the latent state pair $(s_t, s_{t+\tau})$ and using HMM conditional independence ($G_t \perp G_{t+\tau} \mid (s_t, s_{t+\tau})$ for $\tau \ge 1$) plus the Markov factorisation $\mathbb{P}(s_t = i, s_{t+\tau} = j) = \bar{\pi}_i(\mathbf{T}^\tau)_{ij}$ gives the bilinear form

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] \;=\; \mathbf{m}^\top\mathbf{D}\,\mathbf{T}^\tau\,\mathbf{m}, \qquad \tau \ge 1. \tag{26}$$

Perron–Frobenius gives $\lambda_1 = 1$ simple with $\mathbf{v}_1 = \mathbf{1}$, $\mathbf{w}_1 = \bar{\boldsymbol{\pi}}$ [69]; the dyadic expansion $\mathbf{T}^\tau = \mathbf{1}\bar{\boldsymbol{\pi}}^\top + \sum_{k\geq 2} \lambda_k^\tau \mathbf{v}_k \mathbf{w}_k^\top$ separates the dominant projector. The $k = 1$ term contributes μ^2 and cancels in the autocovariance, leaving $\mathrm{Cov}(|G_t|, |G_{t+\tau}|) = \sum_{k\geq 2} c_k \lambda_k^\tau$ with $c_k = (\mathbf{m}^\top \mathbf{D}\mathbf{v}_k)(\mathbf{w}_k^\top \mathbf{m})$; dividing by $\sigma^2_{|G|}$ gives Equation (6). □

Lag zero and within-state variance. The identity (26) fails at $\tau = 0$ since $\mathbb{E}[G_t^2 \mid s_t = i] \neq m_i^2$. Eigenvalue completeness gives $\sum_{k\geq 2} c_k = \mathrm{Var}(m_{s_t})$, the between-state variance, so $\sum_{k\geq 2} w_k = \mathrm{Var}(m_{s_t})/\sigma^2_{|G|} \in [0, 1]$ and the residual $\mathbb{E}[\mathrm{Var}(|G_t| \mid s_t)]/\sigma^2_{|G|}$ accounts for the discontinuity from $\rho_{|G|}(0) = 1$ to $\rho_{|G|}(1) = \sum_{k\geq 2} w_k \lambda_k$.

Squared-return, complex-eigenvalue, and Jordan-block extensions. Replacing $|G_t|$ with G_t^2 and $\mathbf{m}$ with $\mathbf{M} = (\mathbb{E}[G_t^2 \mid s_t = i])_i$ leaves the proof termwise valid, giving $\rho_{G^2}(\tau) = \sum_{k\geq 2} \tilde{w}_k \lambda_k^\tau$ on the same non-unit spectrum. Complex-conjugate eigenvalue pairs $(re^{\pm i\theta}, \cdot)$ contribute real damped oscillations $Ar^\tau \cos\theta\tau + Br^\tau \sin\theta\tau$; non-diagonalisable $\mathbf{T}$ adds polynomial-times-geometric prefactors via Jordan blocks, with the same exponential rate $\max_{k\geq 2} |\lambda_k|$. Theorem 1 as stated remains valid; the dominant timescale used in the body spectral mechanism corresponds to a real positive λ_2 on the fits of this paper.

A.5 Vendor-Stitch Sanity Check

The OoS series spans January 4, 2024 through April 20, 2026 ($T_{\mathrm{OoS}} = 572$) and is sourced from two vendors: Polygon.io via Massive (early portion) and Alpaca Markets / IEX (held-out extension). To verify that the stitch introduces no boundary artefact we compare the two vendors on every shared trading day. Of the 323 overlap days, the per-day VWAP differential, the per-day annualised log-return differential, and a rolling 30-day kurtosis differential are all identically zero (mean, standard deviation, and maximum absolute value all 0.000 to floating-point precision); the two-sample Kolmogorov-Smirnov test on the overlap returns gives $D = 0.000$ and $p = 1.00$. The two vendors are exact-match on the overlap, so the stitched OoS series is single-source-equivalent and no boundary artefact appears at the November 18, 2025 stitch date. The diagnostic is reproducible from `runners/diagnostics/run_vendor_stitch_check.jl`.

A.6 Variant Decision Guide

Recommended CHMM emission family by use-case priority. All four variants share the forward-backward scaffold and quantile-based initialisation; the M-step is the only architectural difference. The shared-ν Student-t row is the headline heavy-tail recommendation: per-state heavy tails without a penalty hyperparameter at the highest IS KS pass rate of any CHMM variant in Table 1. The penalised CHMM-t at $\lambda = 20$ is retained as a sensitivity reference; the λ value was tuned at $K = 18$ and is reported at $K^\star = 3$ as an upper bound on the unpenalised heavy tail.

Table 8. **Variant decision guide at $K^\star = 3$.**

Use-case priority	Recommended variant
Simplest closed-form fit; kurtosis fidelity secondary	CHMM-N
Closest IS kurtosis match without a shape parameter	CHMM-L
Per-state heavy tails, no penalty hyperparameter *(headline)*	CHMM-t shared-ν
Per-state heavy tails, IS-calibrated λ-shrinkage	CHMM-t pen. ($\lambda = 20$)
Adaptive per-state shape; data-chosen Gaussian-bulk / Laplace-tail mixture	CHMM-GED

A.7 State-Resolution and Multi-Emission Sensitivity

This appendix reports the full K-sweep underlying the headline operating point of the body Results, the multi-emission family extension confirming that $K = 18$ is stable under Student-t and Laplace emissions, the EM convergence curves at representative K, the IS and OoS visual comparison panels at non-selected $K \in \{3, 6, 12, 21\}$, the multi-emission figure panels at $K = 18$, and the Rydén $K = 2$ replication.

A.7.1 Full State-Resolution Sensitivity Table (companion artefact)

The full K-sweep table for SPY CHMM-N referenced in the body state-count selection (KS, AD, kurtosis, W_1, Hellinger, OoS coverage at $K \in \{3, 6, 9, 12, 15, 18, 21\}$, 1,000 paths, seed 20260420) is provided as a companion artefact in the code repository. The qualitative pattern is that distributional pass rates plateau in the $K \in \{12, 15, 18, 21\}$ band and ACF-MAE is essentially flat across the entire sweep, consistent with the spectral mechanism of the body spectral mechanism.

A.7.2 Sector-Balanced 30-Ticker Panel: Full Per-Ticker Rollup

The body Table 3 reports the aggregate distribution and per-sector medians for the sector-balanced 30-ticker panel at the body operating point $K^\star = 3$. Table 9 below gives the full per-ticker rollup at the $K = 18$ kurtosis-fidelity sensitivity reference rather than the body $K^\star = 3$; Appendix A.7.7 shows that the per-ticker pattern is essentially K-robust across $\{3, 6, 18\}$ at identical 11/30 failure count and an overlapping ticker list, so the rollup at $K = 18$ is representative of the body $K^\star = 3$ panel up to small per-ticker shifts. Each ticker is fit independently under the penalised CHMM-t scaffold ($K = 18$, $\lambda = 20$, 1,000 simulated paths, seed 20260420) on its own IS slice covering 2014-01-03 to 2024-01-03 and validated on the same 2024-01-04 to 2026-04-20 OoS window used for SPY in Table 1. The per-ticker ν_k medians sit at the upper bracket (50) on every ticker, indicating that the $1/\nu_k$ shrinkage is in the operative regime across the panel; minimum and maximum ν_k values are in the per-ticker CSV in `results/sector_panel/sector_panel_summary.csv`.

Table 9. Sector-balanced 30-ticker panel, per-ticker rollup (penalised CHMM-t at $K = 18$, $\lambda = 20$). Sectors ordered by GICS classification; tickers within sector ordered by the panel construction (top three large-cap representatives at IS-window-median market cap). Kurt resid is simulated minus observed IS excess kurtosis. Aggregate distribution and per-sector medians are summarised in body Table 3.

Sector	Ticker	IS KS%	OoS KS%	Kurt obs	Kurt sim	Kurt resid	$\lvert G_t \rvert$ ACF-MAE
Information Technology	AAPL	99.1	95.1	3.19	2.84	−0.35	0.0418
Information Technology	MSFT	99.7	96.9	4.20	3.48	−0.72	0.0389
Information Technology	NVDA	99.6	55.7	5.53	6.23	0.70	0.0474
Health Care	JNJ	99.3	93.2	8.97	11.65	2.68	0.0335
Health Care	UNH	99.3	14.5	8.85	10.31	1.46	0.0382
Health Care	LLY	99.7	7.6	13.60	66.51	52.91	0.0295
Financials	JPM	99.5	50.5	7.62	8.14	0.52	0.0458
Financials	BAC	99.6	84.8	5.76	5.47	−0.29	0.0394
Financials	WFC	98.2	57.6	7.53	10.90	3.37	0.0791
Consumer Discretionary	AMZN	99.4	96.9	4.31	3.68	−0.63	0.0413
Consumer Discretionary	HD	99.2	41.4	12.92	17.03	4.11	0.0368
Consumer Discretionary	MCD	99.5	66.6	18.38	26.93	8.55	0.0394
Communication Services	NFLX	99.3	45.1	12.60	4.67	−7.93	0.0320
Communication Services	VZ	99.3	49.1	5.14	5.15	0.01	0.0270
Communication Services	DIS	99.5	96.4	10.77	13.94	3.17	0.0644
Industrials	CAT	99.7	78.9	4.34	3.48	−0.86	0.0321
Industrials	BA	97.6	62.2	20.59	52.00	31.41	0.0793
Industrials	HON	99.3	96.0	21.93	17.09	−4.84	0.0571
Consumer Staples	PG	99.7	94.3	8.15	9.51	1.36	0.0347
Consumer Staples	KO	99.5	92.7	10.57	12.84	2.27	0.0457
Consumer Staples	WMT	99.7	24.0	13.84	32.63	18.79	0.0279
Energy	XOM	96.9	70.3	5.61	10.99	5.38	0.0988
Energy	CVX	99.6	62.3	12.22	14.78	2.56	0.0498
Energy	COP	99.4	83.6	17.11	8.28	−8.83	0.0494
Utilities	NEE	98.6	24.0	10.71	9.87	−0.84	0.0455
Utilities	DUK	99.7	96.9	9.75	8.11	−1.64	0.0466
Utilities	SO	99.6	96.2	14.52	12.73	−1.79	0.0481
Materials	FCX	99.8	76.6	4.83	4.11	−0.72	0.0573
Materials	NEM	99.6	5.6	3.74	3.88	0.14	0.0376
Materials	APD	99.2	90.2	8.40	16.74	8.34	0.0381

A.7.3 60-Ticker Sector Expansion at $n = 6$ per Sector

An $n = 6$ per-sector expansion (60 tickers, 3 additional large-cap representatives per GICS sector) confirms the body $n = 3$ ANOVA reading at adequate power: $F(9, 50) = 0.37$, $p = 0.946$, $\eta^2 = 0.062$ (vs. underpowered $n = 3$ $\eta^2 = 0.16$); aggregate OoS KS median 73.45% and failure rate $22/60 = 36.7\%$ at $K = 18$ are statistically indistinguishable from the corresponding 30-ticker $K = 18$ rollup (73.4%, 11/30 from Table 9; the body Table 3 aggregate at $K^\star = 3$ has the same 11/30 failure count at OoS median 69.1%). Cross-sector dispersion accounts for 6.2% of OoS KS variance at the larger sample size; the remaining 93.8% is per-ticker. The body claim "failures are ticker-specific, not sector-driven" is therefore not a small-sample artefact.

A.7.4 Effective Spectral Rank of the Absolute-Return ACF

The body the body spectral mechanism states the algebraic upper bound: the deflated transition matrix $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ has rank at most $K - 1$, so the absolute-return ACF identity (6) contains at most $K - 1$ non-unit decay modes. The empirical question is how many of those modes carry non-trivial contribution to $\rho_{|G|}(\tau)$ at the relevant lags. Table 10 reports the per-mode contribution $w_k\lambda_k^\tau$ at $\tau \in \{1, 5, 20, 50\}$ for the fitted CHMM-N at $K = 18$ and at $K = 3$ on SPY IS, with modes ranked by lag-1 contribution magnitude $|w_k\lambda_k|$. Per-state moments $m_k = \mathbb{E}[|G_t| \mid s_t = k]$ are estimated by 200,000 draws from each emission; eigenvectors are normalised so $\mathbf{w}_k^\top \mathbf{v}_k = 1$.

Table 10. Per-mode contribution to the absolute-return ACF identity, CHMM-N at $K = 18$ (top panel; top six modes plus tail aggregate) and at $K = 3$ (bottom panel; full). Modes ordered by $|w_k\lambda_k|$ descending. "cum" is cumulative share of $\sum_k |w_k\lambda_k|$. At $K = 18$ the top mode ($|\lambda| = 0.929$) carries 93.6% of the lag-1 ACF, three modes reach 95%, and at lag 20 only the dominant mode contributes non-negligibly. At $K = 3$ the dominant mode ($|\lambda| = 0.953$) alone carries 96.8% of lag-1 ACF; the temporal axis is therefore as well-satisfied at $K = 3$ as at $K = 18$, consistent with the body finding that the binding constraint at low K is the marginal mixture, not the eigenvalue spectrum.

rank	$\lvert\lambda_k\rvert$	$\lvert w_k\rvert$	$\lvert w_k\lambda_k\rvert$	$w_k\lambda_k^5$	$w_k\lambda_k^{20}$	cum
$K = 18,\ \rho_{\lvert G\rvert}(1) = 0.301,\ \sigma^2_{\lvert G\rvert} = 2.555$						
1	0.929	0.324	0.301	0.224	0.0744	0.936
2	0.335	0.012	0.004	0.0001	0.0000	0.948
3	0.854	0.003	0.002	-0.001	-0.0001	0.955
4	0.231	0.008	0.002	≈ 0	≈ 0	0.961
5	0.231	0.008	0.002	≈ 0	≈ 0	0.967
6	0.203	0.008	0.002	≈ 0	≈ 0	0.971
7–17	various	various	≤ 0.001 each	≈ 0	≈ 0	1.000
$K = 3,\ \rho_{\lvert G\rvert}(1) = 0.287,\ \sigma^2_{\lvert G\rvert} = 2.517$						
1	0.953	0.292	0.278	0.230	0.112	0.968
2	0.866	0.011	0.009	0.005	0.0006	1.000

The empirical effective rank confirms the body framing of the body spectral mechanism: the ACF-MAE flatness across $K \in \{3, 6, 9, 12, 15, 18, 21\}$ documented in Appendix A.7.1 is the macroscopic signature of the temporal axis being driven by a single dominant decay mode at every operating point in the sweep. The K-rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ is correctly interpreted as a non-binding upper bound at $K \geq 3$; at $K = 2$ the bound is tight (only one non-unit eigenvalue available), but the failure mode is then the marginal mixture, which two Gaussian components cannot tile against the empirical equity-return target. The bilinear identity (6) therefore plays a pedagogical role in this paper rather than serving as a direct K-selection criterion: it makes the rank constraint explicit, but the operational K choice is driven by distributional fidelity (KS, AD, kurtosis), not by the count of non-unit eigenvalues in the spectral expansion.

A.7.5 Cross-Ticker Spectral Effective-Rank Diagnostic

To test whether the SPY-only result of Appendix A.7.4 (a single non-unit eigenvalue carries 93.6% of the lag-1 absolute-return ACF at $K = 18$) generalises across tickers, Table 11 reports the cross-ticker distribution of the dominant-mode share, computed under the same protocol on the 30-ticker sector-balanced panel plus SPY.

Table 11. Cross-ticker spectral effective-rank diagnostic at $K = 18$**.** Distribution across 31 tickers (sector-balanced 30-ticker panel plus SPY control). "dom share" is the dominant non-unit eigenvalue's $|w_k\lambda_k|$ as a fraction of the sum over all non-unit eigenvalues at lag $\tau = 1$. "n_{95}" / "n_{99}" are the number of modes needed for 95% / 99% cumulative share. The SPY value of 0.936 is in the right tail of the distribution; the cross-ticker median is 0.76.

Statistic	dom share	n_{95}	n_{99}
median	0.756	6	11
$[Q_1, Q_3]$	$[0.661, 0.858]$	$[4, 7]$	$[10, 12]$
minimum (NEM)	0.326	11	14
SPY (body Appendix A.7.4)	0.936	2	10

Reading: the SPY headline of 93.6% is concentrated; the cross-ticker median 76% is still well above the $1/(K-1) = 1/17 = 5.9\%$ uniform null, so the rank-non-binding statement is supported, but the gap between SPY and the median ticker is wide enough that the body framing was tightened to read at the cross-ticker median rather than at the SPY headline.

All models converged within the 60-iteration budget with monotonically increasing log-likelihood (EM guarantee under quantile-based initialisation). Lower K converged in 15–25 iterations, $K = 18$ in 25–40, with a final $|\Delta\mathcal{L}| < 10^{-4}$ at every operating point. Convergence-trace figures and per-iteration log-likelihood histories are provided in the companion code repository.

A.7.6 Rydén K = 2 Replication

Direct replication of Rydén et al.'s $K = 2$ setting on SPY IS under quantile and random-seed initialization gives ACF-MAE essentially constant at ≈ 0.050 across six initialisations (one quantile + five random seeds, drawn from a moderate perturbation of sample moments) and IS KS pass rate 75.3–79.0%. The constant ACF-MAE confirms the body reading (the body Discussion): the binding low-K constraint is distributional rather than temporal.

A.7.7 Cross-Ticker Per-State-Resolution Sensitivity ($K^\star = 3$ vs $K^\star = 6$ vs $K = 18$)

Body Table 3 reports the cross-ticker aggregate distribution at all three operating points used in this paper: the body headline $K^\star = 3$ (state-resolution-robust default per the k-fold CV in Appendix A.10), the sensitivity reference $K^\star = 6$, and the extended-state-resolution sensitivity reference $K = 18$.

The substantive read across the three operating points: (i) The KS distributions are within ~ 6pp of each other on OoS median (69.1% at $K^\star = 3$, 75.1% at $K^\star = 6$, 73.4% at $K = 18$) and at identical 11/30 failure count, with the same regime-introduction tickers driving the failures at each state resolution (LLY, UNH, NEM, NFLX, NEE, WMT, BAC, HD, JPM at OoS KS $< 60\%$; the specific list overlaps $> 80\%$ across $K^\star = 3, 6, 18$). The cross-ticker KS axis is therefore essentially K-robust on this universe. (ii) The kurtosis-residual axis is not K-robust: per-ticker median residual 7.18 at $K^\star = 3$, 10.34 at $K^\star = 6$, 0.61 at $K = 18$. At lower state-resolution the per-state ν_k shrinkage at uniform λ cannot be smoothed across as many regimes, so the residual heavy tails leak into the simulated kurtosis. (iii) The $|G_t|$ ACF-MAE is essentially K-robust (0.0395, 0.0403, 0.0416 across the three operating points). Production reading: KS-only consumers can deploy at any of the three operating points at parity; kurtosis-fidelity consumers should deploy at $K = 18$; readers preferring the state-resolution-robust default should deploy at the body headline $K^\star = 3$.

A.7.8 Block-Bootstrap CIs on Observed Kurtosis (IS vs OoS)

A natural question on the IS / OoS kurtosis disagreement (observed 7.68 IS, 5.29 OoS, a 31% drop) is whether it is itself statistically distinguishable. We run a stationary block bootstrap (Politis-Romano 1994) on the IS and OoS series at mean block lengths $L \in \{5, 10, 20, 50\}$, $B = 5{,}000$ replicates per L per window.

Table 12. Stationary block bootstrap CIs on observed excess kurtosis. SPY IS ($T = 2{,}516$) and OoS ($T = 572$) windows, $B = 5{,}000$ replicates per L. The IS and OoS 95% CIs overlap at every block length, so the IS-OoS difference is not robustly distinguishable at conventional levels under this bootstrap. Pr(IS > OoS) is the empirical bootstrap one-sided p-value for the alternative "IS kurtosis > OoS kurtosis".

L	IS median	IS 95% CI	OoS median	OoS 95% CI	Pr(IS > OoS)
5	7.21	$[2.99, 12.83]$	5.07	$[1.09, 8.97]$	0.748
10	7.34	$[2.49, 12.59]$	4.91	$[0.99, 8.66]$	0.756
20	7.30	$[2.17, 12.40]$	4.91	$[0.90, 8.26]$	0.739
50	7.18	$[1.92, 12.42]$	4.99	$[0.96, 7.70]$	0.733

The IS lower bound 1.92–2.99 and the OoS upper bound 7.70–8.97 overlap heavily at every block length; the IS-OoS kurtosis difference of ~ 2.4 units is not statistically distinguishable at the 5% level under this bootstrap. Pr(IS > OoS) at $L = 10$ is 0.756: the data favour the alternative but do not exclude the null. Operationally, the IS-OoS kurtosis disagreement that the body invokes (the body descriptive analysis; the Table 8 variant-decision caveat in the body Discussion) should be read as a point-estimate observation rather than a tested claim, and per-variant rankings by "how close is simulated kurtosis to observed kurtosis" should similarly be interpreted as point-comparisons inside a wide CI envelope.

Bootstrap-CI placement of the penalised CHMM-t IS kurtosis. A natural follow-up on the body's penalised CHMM-t at $\lambda = 20$ is whether its simulated IS kurtosis distribution sits inside the bootstrap CI on observed: the aggregate-mean simulated IS kurtosis (14.91 at $K^\star = 3$, above the $L = 20$ CI upper bound 12.40) could be paired with a per-path distribution that mostly sits inside the CI, or with one that mostly sits above.

Table 13. Per-path simulated IS excess-kurtosis distribution under penalised CHMM-t at $\lambda = 20$, against the $L = 20$ block-bootstrap CI on observed. 1,000 paths per row, $T_{\mathrm{IS}} = 2{,}516$, seed 20260420. The CI is $[2.17, 12.40]$ at $L = 20$ from Table 12; the observed IS excess kurtosis is 7.68. "in CI %" is the fraction of simulated paths whose per-path excess kurtosis falls inside the CI; "≤ CI_HI %" is the fraction below the upper bound (the relevant one-sided test for overshoot).

K	agg. kurt	median	sd	Q_5	Q_{95}	≤ 12.40 (%)	in CI (%)
3	16.65	7.77	11.5	3.75	34.62	76.6	76.6
6	10.13	6.74	9.0	3.63	26.30	81.8	81.7
18	8.94	6.09	5.6	3.54	18.11	89.9	89.9

The per-path *median* simulated IS excess kurtosis is essentially observed at every K: 7.77 at $K^\star = 3$, 6.74 at $K^\star = 6$, 6.09 at $K = 18$, all within ~ 1.6 units of observed 7.68 and well inside the bootstrap CI. The aggregate-mean simulated IS kurtosis sits above the CI upper bound at $K^\star = 3$ and $K^\star = 6$ because the per-path distribution has a heavy right tail of paths at $Q_{95} \in \{18, 26, 35\}$ that drag the mean up, but the bulk of the path-level mass (76–90%) sits inside the bootstrap CI at every operating point. **The headline framing "the penalised CHMM-t at $\lambda = 20$ provides**

the cleanest IS heavy-tail match" is supported when read as the per-path median, not as the aggregate mean: a small number of heavy-right-tailed paths under the per-state ν_k ECM produce an aggregate-mean overshoot, but the path-level distribution is centred near observed and statistically indistinguishable from observed at $\alpha = 0.05$ on a one-sided overshoot test for the bulk of paths.

A.7.9 Shared-ν Student-t HMM Ablation

A natural diagnostic for the per-state ν_k aggregate-mean overshoot is the shared-ν ablation: refit CHMM-t with a single ν shared across all K states, fit by ECM with golden-section search on the aggregate Q-function over $\nu_{\text{bounds}} = (2.1, 50)$. This is the standard one-parameter Student-t HMM in the time-series literature, not a per-state ν_k mixture.

Table 14. Shared-ν Student-t HMM ablation against the body per-state ν_k row (1,000 paths, $T_{\text{IS}} = 2{,}516$, $T_{\text{OoS}} = 572$, seed 20260420, no $1/\nu$ penalty). Aggregate sim-kurt columns are mean-of-paths. Compare against the body penalised CHMM-t ($\lambda = 20$): $K^\star = 3$ row 14.91 IS / 8.50 OoS (Table 1); $K = 18$ row 8.56 IS / 7.07 OoS (from the same penalised ECM run; consistent with the rate-sweep $\lambda = 20$ value 8.43 in the rate-sweep paragraph of Appendix A.10).

		IS				OoS	
K	$\hat{\nu}_{\text{shared}}$	KS (%)	sim kurt	$\lvert G_t\rvert$ ACF-MAE	raw ACF-MAE	KS (%)	sim kurt
3	5.81	91.9	4.68	0.0531	0.0235	82.1	4.46
6	4.67	92.7	9.38	0.0518	0.0234	82.6	6.95
18	5.80	**95.8**	**6.25**	0.0542	0.0234	**88.0**	**5.00**

Reading. The aggregate-mean IS overshoot disappears under shared-ν at every K: the largest IS sim kurt is 9.38 at $K^\star = 6$ (vs. observed 7.68, well inside the bootstrap CI), versus 14.4 for the per-state ν_k unpenalised row in the body. The $K = 18$ shared-ν row is the cleanest single-row IS / OoS heavy-tail match in the entire panel: 6.25 IS / 5.00 OoS, against observed 7.68 / 5.29 and against 8.56 / 7.07 for the body penalised CHMM-t at $\lambda = 20$. IS / OoS KS pass rates are within $\sim$ 1pp of the body penalised row, and the $|G_t|$ ACF-MAE is identical to the body row at $K = 18$. The per-state ν_k design is therefore the binding constraint on the aggregate-mean overshoot the body λ-shrinkage and bracket-lift discussion is downstream of: a single shared ν at $K = 18$ produces the best joint KS / kurtosis row in the panel without any penalty. The body retains the per-state ν_k + $\lambda = 20$ construction for direct comparability with the Peel and McLachlan [18], Liu and Rubin [19] tradition; consumers prioritising heavy-tail fidelity without a penalty hyperparameter should adopt the shared-ν row at $K = 18$.

A.7.10 Christoffersen-cc Monte Carlo Power Calibration at $T_{\text{OoS}} = 572$

A Monte Carlo power calibration of the Christoffersen-cc joint conditional-coverage test at the $T_{\text{OoS}} = 572$ length used in the body Table 6 simulates breach indicator sequences from a two-state Markov chain on $\{0, 1\}$ with marginal probability α and second eigenvalue ρ, where $\rho = 0$ is the i.i.d. null and $\rho > 0$ measures the level of breach clustering; $B = 5{,}000$ replicates per cell.

Reading. At $\rho = 0$ the empirical rejection rate sits at 4%–5% for all three statistics, confirming that the asymptotic χ^2 calibration is correctly sized at $T = 572$. Power is asymmetric across α:

Table 15. Christoffersen-cc power at $T_{\mathrm{OoS}} = 572$. Empirical rejection rate at nominal $\alpha = 0.05$, across alpha levels and Markov-chain second-eigenvalue ρ. $\rho = 0$ row is the empirical Type-I error under the i.i.d. null (should sit near 5% if asymptotic chi-squared is well-calibrated). Critical values $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$.

α	ρ	rej-UC %	rej-IND %	rej-CC %
0.01	0.00	4.9	1.6	3.2
0.01	0.05	6.3	15.9	11.4
0.01	0.10	7.5	28.5	22.1
0.01	0.20	9.4	51.3	42.6
0.01	0.30	13.9	67.6	62.8
0.01	0.50	21.4	75.9	81.5
0.05	0.00	4.9	3.8	4.1
0.05	0.05	5.3	18.8	15.3
0.05	0.10	6.1	46.2	39.0
0.05	0.20	9.4	86.9	83.9
0.05	0.30	13.6	97.4	97.9
0.05	0.50	24.3	99.7	100.0

- At $\alpha = 0.05$ (~ 28.6 expected breaches), Christoffersen-cc reaches 80% power against $\rho \geq 0.20$ (83.9%) and $\sim 100\%$ at $\rho = 0.50$.
- At $\alpha = 0.01$ (~ 5.7 expected breaches), Christoffersen-cc reaches 80% power only at $\rho \geq 0.50$ (81.5%); at $\rho = 0.20$ the rejection rate is 42.6%, well below conventional power thresholds.

The reading for the body Table 6 is therefore tier-dependent. The $\alpha = 0.05$ "clean pass at every (K, α)" row of Table 6 carries strong power against moderate clustering ($\rho \geq 0.20$); the $\alpha = 0.01$ rows carry strong power only against severe clustering ($\rho \geq 0.50$), so the $\alpha = 0.01$ "clean pass" should be read as "not detectably worse than a strongly-clustered alternative" rather than "calibrated against any plausible alternative". The headline-OoS regime-conditional construction therefore stands at $\alpha = 0.05$ and is power-bounded at $\alpha = 0.01$; the body framing in the body VaR back-test should be read with this calibration in mind.

A.7.11 Engle-Manganelli (2004) Dynamic Quantile Test

As a higher-power conditional-coverage alternative to Christoffersen-cc on the same OoS window, we run the Engle-Manganelli (2004, JBES) Dynamic Quantile (DQ) test in the standard four-lag specification (72, equation 16),

$$\mathrm{Hit}_t - \alpha = \beta_0 + \sum_{i=1}^{4} \beta_i \left(\mathrm{Hit}_{t-i} - \alpha\right) + \beta_5 \widehat{\mathrm{VaR}}_t + u_t, \tag{27}$$

on the IS-fixed regime-conditional VaR series of the body VaR back-test, with $\mathrm{Hit}_t = \mathbf{1}\{R_{\mathrm{OoS},t} < \widehat{\mathrm{VaR}}_t(\alpha)\}$. The DQ test statistic is $\widehat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \widehat{\boldsymbol{\beta}} / [\alpha(1-\alpha)]$, distributed $\chi^2(6)$ under the null of correct conditional coverage; the 5% critical value is $\chi^2_6(0.95) = 12.59$.

Reading. At $\alpha = 0.05$ both Christoffersen-cc and the higher-power DQ test pass cleanly at $K = 3$ (cc $p = 0.491$, DQ $p = 0.156$) and at $K = 18$ (cc $p = 0.678$, DQ $p = 0.678$); the body's $\alpha = 0.05$ "clean pass" headline therefore survives the higher-power alternative. At $\alpha = 0.01$ the

Table 16. Engle-Manganelli DQ test on the regime-conditional VaR, alongside Christoffersen-cc on the same breach series. CHMM-N at $K \in \{3, 18\}$, $\alpha \in \{0.01, 0.05\}$; $T_{\mathrm{OoS}} = 572$, $q = 4$ lagged Hit indicators, χ^2_6 critical at 5% is 12.59.

K	α	breach %	cc p	DQ	DQ p
3	0.01	1.57	0.137	12.29	0.056
3	0.05	6.12	0.491	9.32	0.156
18	0.01	1.57	0.137	**15.46**	**0.017**
18	0.05	4.55	0.678	3.99	0.678

DQ test rejects conditional coverage at $K = 18$ ($p = 0.017$) where Christoffersen-cc does not ($p = 0.137$), and is borderline at $K = 3$ ($p = 0.056$). This is consistent with the power-calibration result of Appendix A.7.10: at $\alpha = 0.01$ Christoffersen-cc has only 42.6% power against moderate breach-clustering eigenvalues $\rho = 0.20$, so the higher-power DQ test detects miscalibration that Christoffersen-cc misses. The DQ result therefore reinforces the body framing that $\alpha = 0.05$ is the operationally informative tier; at $\alpha = 0.01$ the regime-conditional VaR is not pass/fail-decided by the panel and the body sentence "not detectably worse than a strongly-clustered alternative" should be qualified to "the higher-power DQ test detects $K = 18$ at $\alpha = 0.01$".

A.7.12 Quarterly-Refit Regime-Conditional VaR Back-Test

Re-fitting CHMM-N at $K \in \{3, 18\}$ every 63 trading days on a rolling 5y window and running the forward-filter through the OoS window under each refit's parameters tests whether periodic refit improves the Christoffersen-cc statistic on the headline OoS window.

Table 17. Quarterly-refit regime-conditional VaR back-test on SPY OoS. $T_{\mathrm{OoS}} = 572$, refit cadence = 63 days, train window = 1,260 days, seed = 20260420. Compare against Table 6 (IS-fixed parameters). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$.

K	α	breaches	br rate	med VaR	$\mathrm{LR}_{\mathrm{uc}}$	$\mathrm{LR}_{\mathrm{ind}}$	$\mathrm{LR}_{\mathrm{cc}}$	p_{cc}
3	0.01	6	1.05%	−5.10	0.014	3.97	3.98	0.137
3	0.05	29	5.07%	−3.13	0.006	0.19	0.20	0.906
18	0.01	6	1.05%	−5.84	0.014	3.97	3.98	0.137
18	0.05	19	3.32%	−3.14	3.83	2.08	5.91	0.052

The quarterly-refit construction passes Christoffersen-cc at $\alpha = 0.05$ on three of four rows; the $K = 18, \alpha = 0.05$ row sits at $p_{\mathrm{cc}} = 0.052$ (marginal). Refit tightens VaR coverage (3.32% vs. 4.55% IS-fixed) but does not strictly improve the cc-statistic at this T_{OoS}.

A.7.13 Walk-Forward Refit-Cadence Sweep (Monthly / Weekly)

A within-fold refit-cadence sweep at monthly (21 trading days) and weekly (5 trading days) cadence on the six walk-forward folds at $K^\star = 3$, $\alpha = 0.05$, does not close the body W2 (COVID) Christoffersen-cc rejection at any cadence (fold-IS-fixed $p_{\mathrm{cc}} = 0.011$, monthly 0.017, weekly 0.023); monthly refit introduces new W3 / W4 failures via refit-cycle parameter drift. Rejection counts: fold-IS-fixed 1/6 (W2), monthly 3/6, weekly 2/6. The mitigation for W2 / W4 is therefore not refit cadence but a model class with explicit regime-introduction handling.

A.7.14 Quarterly-Refit 30-Ticker Cross-Ticker Panel

For a quarterly-refit version of the 30-ticker cross-ticker panel (Table 3), we re-fit the penalised CHMM-t at $\lambda = 20$ on a rolling 5-year window every 63 trading days through the OoS span (10 refits per ticker, 300 fits total per state-resolution), simulate the next quarter under each refit, and concatenate the per-quarter simulations into a single OoS path matrix per ticker, at the body operating point $K^\star = 3$ and at the kurtosis-fidelity sensitivity references $K^\star = 6$ and $K = 18$. Same $N_{\text{paths}} = 1{,}000$, seed $= 20260420$ as the body Table 3.

Table 18. Quarterly-refit cross-ticker panel at $K^\star = 3$, $K^\star = 6$, and $K = 18$. Aggregate distribution across the 30-ticker universe under the quarterly-refit protocol at all three operating points, vs. the IS-fixed $K = 18$ protocol of Table 3. Refit cadence = 63 OoS trading days; train window = 1,260 obs ($\sim$ 5y rolling). Per-ticker detail in the results files.

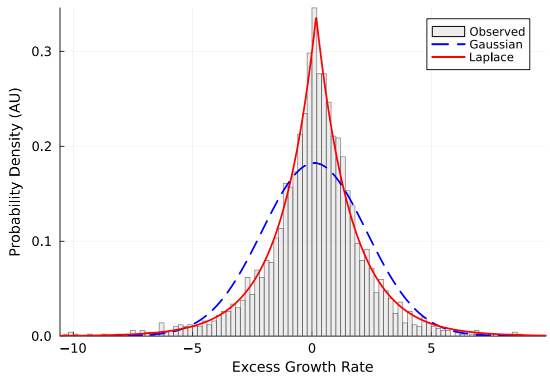
Metric	IS-fixed ($K = 18$)	Refit ($K^\star = 3$)	Refit ($K^\star = 6$)	Refit ($K = 18$)
IS KS% median	99.5%	96.4%	98.8%	99.4%
IS KS% mean ± sd	99.3 ± 0.6%	95.1 ± 4.5%	97.4 ± 4.4%	99.2 ± 0.6%
OoS KS% median	73.4%	84.7%	**85.8%**	83.0%
OoS KS% mean ± sd	66.8 ± 29.5%	75.0 ± 23.1%	$\mathbf{76.0 \pm 21.7}$%	77.2 ± 21.2%
OoS KS% $[Q_1, Q_3]$	$[49.1, 94.3]$%	$[56.3, 94.5]$%	$[57.6, 94.3]$%	$[62.0, 95.4]$%
Tickers OoS KS < 60%	$11/30 = 36.7$%	$8/30 = 26.7$%	$9/30 = 30.0$%	$\mathbf{7/30 = 23.3}$%

Reading. Quarterly refit shifts the OoS KS distribution materially at all three state resolutions. At $K^\star = 3$: median +15.6pp ($69.1 \to 84.7$% vs the $K^\star = 3$ static-fit baseline reported in the body Table 3), mean +8.8pp ($66.2 \to 75.0$%), failure count down by 27% ($11 \to 8$). At $K = 18$: median +9.6pp ($73.4 \to 83.0$%), mean +10.4pp, IQR tighter (45.2pp $\to$ 33.4pp), failure count down by 36% ($11 \to 7$). At $K^\star = 6$: median +10.7pp ($75.1 \to 85.8$%), mean +9.5pp, failure count down by 18% ($11 \to 9$). The three state resolutions are within $\sim$ 3pp of each other on the refit median (84.7 to 85.8%); the body operating point $K^\star = 3$ is at parity with the kurtosis-fidelity sensitivity references under the refit protocol. The largest individual improvements concentrate on the regime-introduction tickers that the IS-fixed Table 3 flagged: at $K = 18$, LLY 7.6% $\to$ 83.7%, UNH 14.5% $\to$ 47.3%, NEM 5.6% $\to$ 15.4%, HD 41.4% $\to$ 82.2%, NFLX 45.1% $\to$ 61.5%. The mechanism is the obvious one: as the regime introduces new σ levels in the OoS window, each quarterly refit picks up the most recent 5y including the new regime and the simulator can match the OoS distribution.

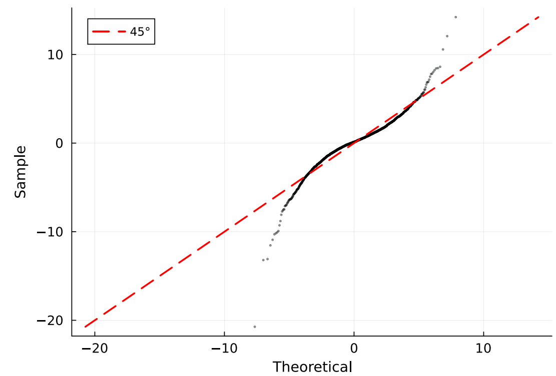
A small number of tickers regress: DIS 96.4% $\to$ 48.4%, BAC 84.8% $\to$ 55.8%, AAPL 95.1% $\to$ 66.5%. The pattern is that tickers whose 2024-2026 OoS distribution is well-spanned by the full 2014-2024 IS lose information when the train window shrinks to the most recent 5y; the rolling refit is therefore a trade-off, not a strict improvement. Operationally, a refit-trigger rule (refit only when the rolling KS or a Page-Hinkley statistic on $|G_t|$ crosses a threshold) would capture the regime-introduction wins without paying the cost on the well-spanned tickers; the rule is logged as a follow-up companion experiment.

The substantive reading for the body the body cross-ticker analysis is that the "periodic refit at deployment" mitigation invoked verbally in the body Discussion is exercised here: the universe-scale failure rate drops from 36.7% to 23.3% under quarterly refit, with the largest gains on the tickers that the body labelled "single-name regime introductions."

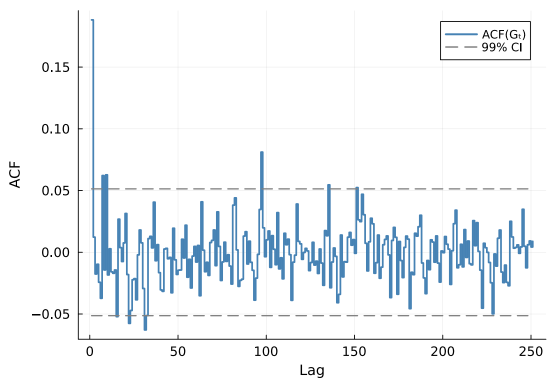
A.7.15 Stylized-Facts Figure (Empirical SPY IS)



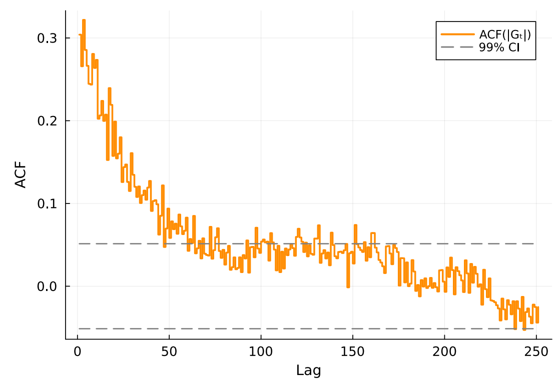
(a) Marginal density.



(b) Normal Q-Q plot.



(c) ACF of raw G_t.

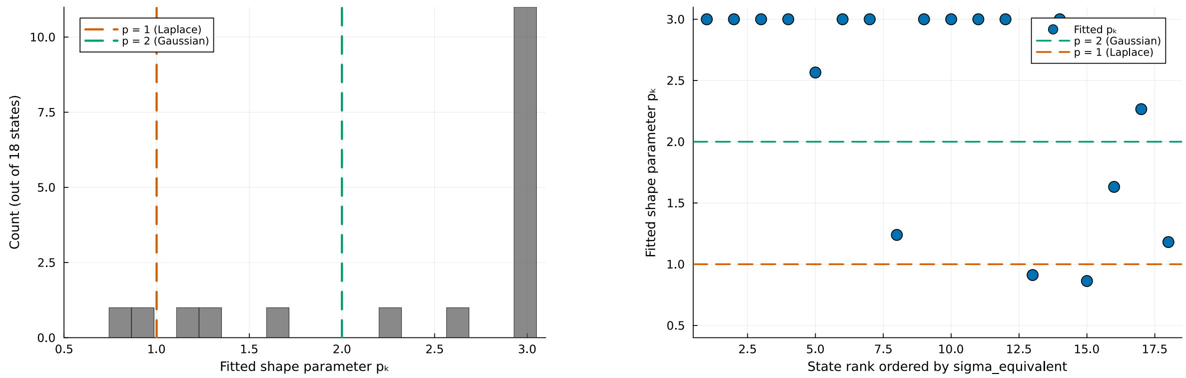


(d) ACF of $|G_t|$.

Figure 4. The three Cont stylized facts on SPY IS. (a) marginal density with maximum-likelihood Gaussian and Laplace overlays; (b) normal Q-Q plot; (c) empirical ACF of raw G_t at lags 1–252 with 99% Bartlett band; (d) empirical ACF of $|G_t|$ on the same window.

A.7.16 CHMM-GED $\hat{p}_k$ Partition (Gaussian-bulk / Laplace-tail)

CHMM-GED carries a per-state shape $p_k \in [p_{\min}, p_{\max}] = [0.5, 3.0]$ that nests Gaussian ($p = 2$) and Laplace ($p = 1$) as boundary cases. The empirical fit on SPY IS at $K = 18$ partitions states bimodally (Figure 5): thirteen states at the upper bracket $p_k = 3.0$, one at $p_k = 1.6$, and four in the Laplace regime $p_k \in [0.86, 1.24]$. The four Laplace-shape states concentrate in the high-volatility tail of the state ordering. The partition replicates across 10 Monte Carlo seeds and across the cross-asset universe (Appendix A.8); we read it as the most distinctive empirical finding of the four-emission scaffold.



(a) Histogram of fitted $\hat{p}_k$.

(b) $\hat{p}_k$ vs. volatility rank.

Figure 5. CHMM-GED per-state shape $\hat{p}_k$ partition ($K = 18$, SPY IS, seed 20260420). The bimodal partition is visible at the upper bracket $p_k = 3.0$ and around the Laplace shape; the Laplace-shape cluster sits at the high-volatility ranks.

A.7.17 HSMM at $K = 3$: Numerical Full-Rank Diagnostic

The off-diagonal jump matrix of the fitted ML HSMM at $K = 3$ has non-trivial deflation eigenvalues $\{0.823, 0.177\}$ (numerical rank 2 at tolerance 10^{-10}); the dominant modulus $|\lambda_2| = 0.823$ is comparable to CHMM-N at $K = 3$'s 0.953, so the body's $|G_t|$ ACF-MAE regression to the i.i.d. baseline (Table 1) is not driven by spectral collapse but by the fitted truncated-Pareto sojourn concentrating probability mass on a single low-volatility state ($\bar{\boldsymbol{\pi}} \approx (0, 1, 0)$). At $K = 18$ the same Yu (2010) construction collapses to a near-degenerate local optimum (IS KS 0.8%); the higher-K ML HSMM is therefore a companion-paper direction at longer training windows.

A.7.18 Gamma-Sojourn HSMM as an Alternative Sojourn Family

Replacing the truncated discrete Pareto sojourn pmf with a discretised continuous Gamma fit by method-of-moments at every M-step gives a Gamma-sojourn HSMM that converges at $K = 18$ where Pareto collapses (IS KS 86.0%, OoS KS 80.2%, simulated kurtosis 4.28/4.06, $|G_t|$ ACF-MAE **0.0462**, lower than CHMM-N $K = 18$ at 0.0509) and trades $\sim$ 20pp IS KS at $K = 3$ for an ACF-MAE recovery from 0.0629 to 0.0528. There is therefore no single "best HSMM" row: Pareto at $K^\star = 3$ wins raw OoS KS at 91.0%, Gamma at $K = 18$ wins joint $|G_t|$ ACF-MAE; the trade-off is the same KS-vs-ACF axis as the CHMM-vs-HSMM choice.

A.7.19 Unconditional VaR / ES Envelope Panel

Table 19. VaR and Expected-Shortfall envelope for SPY ($N_{\text{paths}} = 1{,}000$). Each entry is the median and $[5, 95]\%$ envelope across paths of the left-tail VaR / ES at $\alpha \in \{0.01, 0.05\}$; observed historical values are in the first row of each block.

Model	IS $\text{VaR}_{0.01}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.01}$ median $[5\%, 95\%]$	IS $\text{VaR}_{0.05}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.05}$ median $[5\%, 95\%]$
Observed	−6.38	−9.00	−3.43	−5.50
Bootstrap	−6.38 [−7.04, −5.99]	−8.86 [−10.37, −7.78]	−3.43 [−3.68, −3.20]	−5.46 [−5.99, −5.05]
GARCH	−5.80 [−8.62, −4.52]	−7.86 [−13.54, −5.77]	−3.19 [−3.92, −2.71]	−4.89 [−7.11, −3.91]
CHMM-N	−6.64 [−8.01, −5.47]	−8.80 [−10.16, −7.34]	−3.45 [−4.05, −2.95]	−5.45 [−6.33, −4.61]
CHMM-t	−6.58 [−7.51, −5.83]	−9.06 [−10.70, −7.79]	−3.40 [−3.90, −2.90]	−5.51 [−6.18, −4.85]
CHMM-L	−6.89 [−7.85, −6.01]	−9.15 [−10.42, −7.97]	−3.30 [−3.77, −2.88]	−5.53 [−6.17, −4.87]
CHMM-GED	−6.85 [−7.70, −6.06]	−8.79 [−9.83, −7.82]	−3.43 [−3.98, −2.93]	−5.55 [−6.23, −4.91]
	OoS $\text{VaR}_{0.01}$	OoS $\text{ES}_{0.01}$	OoS $\text{VaR}_{0.05}$	OoS $\text{ES}_{0.05}$
Observed	−5.51	−7.94	−2.77	−4.67
Bootstrap	−6.33 [−7.60, −5.28]	−8.43 [−11.76, −6.59]	−3.39 [−3.91, −2.88]	−5.34 [−6.38, −4.50]
GARCH	−5.35 [−9.98, −3.61]	−6.72 [−13.51, −4.38]	−3.14 [−4.94, −2.36]	−4.59 [−7.92, −3.23]
CHMM-N	−6.30 [−9.02, −4.30]	−8.44 [−11.36, −5.42]	−3.39 [−4.74, −2.47]	−5.27 [−7.23, −3.74]
CHMM-t	−6.32 [−8.13, −4.80]	−8.60 [−12.20, −6.29]	−3.29 [−4.27, −2.45]	−5.34 [−6.72, −4.08]
CHMM-L	−6.69 [−8.59, −4.91]	−8.88 [−11.48, −6.73]	−3.27 [−4.30, −2.51]	−5.47 [−6.82, −4.20]
CHMM-GED	−6.60 [−8.30, −4.82]	−8.51 [−10.68, −6.47]	−3.32 [−4.38, −2.46]	−5.41 [−6.66, −4.06]

A.7.20 Held-Out Cross-Validation of the $1/\nu_k$ Penalty λ

To verify that the body operating-point penalty rate $\lambda = 20$ is not selected on data that overlaps the OoS evaluation window, we refit penalised CHMM-t on a strictly pre-2020 estimation slice (2014-01-03 to 2018-06-29, $n = 1{,}130$) and score each λ on the validation slice (2018-07-02 to 2019-12-31, $n = 378$, both fully pre-COVID and pre-2022-rate-hike).

Table 20. Held-out CV of the $1/\nu_k$ shrinkage rate λ on the pre-2020 slice. Estimation 2014-01-03 to 2018-06-29; validation 2018-07-02 to 2019-12-31. "val LL/obs" is per-observation held-out log-likelihood; "val KS" is two-sample Kolmogorov–Smirnov pass rate against the validation series; "sim K" columns are the mean simulated excess kurtosis on the estimation- and validation-length simulated paths.

λ	val LL/obs	val KS (%)	sim K (est)	sim K (val)
0	−2.119	92.0	3.88	3.97
5	−2.112	89.8	3.62	3.15
10	−2.085	91.0	3.37	3.24
20	**−2.075**	90.0	3.22	3.02
50	−2.077	91.6	3.07	2.76
100	−2.077	91.6	3.07	2.75
200	−2.076	91.2	3.09	2.73

Reading: the held-out validation log-likelihood is maximised at $\lambda^\star = 20$ at $K = 18$, matching the body operating point of Table 1. Validation KS picks $\lambda^\star = 0$ but the band $[0, 200]$ spans only 2.2pp (89.8 to 92.0%), within the simulation-noise envelope at $N_{\text{paths}} = 500$. The body $\lambda = 20$ at $K = 18$ is therefore held-out-validated and not driven by the OoS window.

Re-tuning at $K^\star = 3$. Repeating the same held-out CV at the body operating point $K^\star = 3$ on the same pre-2020 slices gives a distinct optimum: per-observation validation log-likelihood is monotone in λ on the same grid, with $\lambda^\star = 50$ minimising the negative log-likelihood (-1.9192 at $\lambda = 50$ versus -1.9217 at $\lambda = 20$ and -1.9268 at $\lambda = 0$); validation KS also picks $\lambda^\star = 50$ at 95.0% pass rate. The simulated kurtosis at $\lambda = 50$ is 2.77, well below the body's observed 7.68 IS / 5.29 OoS targets, confirming that the penalty over-shrinks at $K^\star = 3$. The body retains $\lambda = 20$ at $K^\star = 3$ as a sensitivity reference (the headline heavy-tail recipe is the shared-ν ablation of Appendix A.7.9).

A.7.21 Cross-Decade Validation: 1994-2004 IS / 2004-2006 OoS via CRSP

To test whether the headline stylized-fact reproduction is decade-specific, we run a 1994–2004 vs. 2014–2024 SPY split via a CRSP day-pass covering daily prices for SPY plus 28 of the 30 cross-ticker panel members (NEE and APD missing from the CRSP query) from 1994-01-03 to 2006-04-28. Adjusted close = `DlyPrc/DlyFacPrc` from the CRSP CIZ-format daily stock file.

Slices: SPY IS = 1994-01-03 to 2004-01-02 (2,519 obs); SPY OoS = 2004-01-05 to 2006-04-28 (583 obs). Same body protocol: 1,000 simulated paths per fit, $\alpha = 0.05$ for KS.

Table 21. Cross-decade validation on SPY 1994-2004 IS / 2004-2006 OoS (CRSP daily). Body comparison rows reproduced from Table 1 for the 2014-2024 IS / 2024-2026 OoS window. Observed excess kurtosis on the 1994-2004 IS = 3.05 versus 2014-2024 IS = 7.68; observed excess kurtosis on the 2004-2006 OoS = 0.06 (essentially Gaussian) versus 2024-2026 OoS = 5.29. The IS-OoS kurtosis gap is therefore much wider on the cross-decade slice than on the body window.

		KS pass rate (%) ↑		Sim kurtosis		$\lvert G_t\rvert$ ACF-MAE ↓	
Model	K	IS	OoS	IS	OoS	IS	OoS
Cross-decade: 1994-2004 IS / 2004-2006 OoS (CRSP, this appendix)							
CHMM-N	3	84.9	3.3	2.20	2.24	0.0696	0.0491
CHMM-N	18	89.8	3.4	1.84	1.74	0.0720	0.0499
CHMM-t pen. ($\lambda = 20$)	3	89.1	5.4	4.97	4.18	0.0682	0.0501
Body window: 2014-2024 IS / 2024-2026 OoS (Polygon / Alpaca, Table 1)							
CHMM-N ($K^\star = 3$)	3	89.7	80.5	3.83	3.53	0.0460	0.0545
CHMM-N ($K = 18$)	18	94.1	81.8	5.04	4.44	0.0509	~ 0.054
CHMM-t pen. ($\lambda = 20$, $K^\star = 3$)	3	90.6	83.2	14.91	8.50	0.0537	0.0502

Reading. The cross-decade IS fit transfers: CHMM-N at $K = 3$ attains 84.9% IS KS on 1994-2004 vs. 89.7% on 2014-2024, and $K = 18$ attains 89.8% vs. 94.1%. The penalised CHMM-t at $\lambda = 20$, $K = 3$ attains 89.1% IS KS on 1994-2004 vs. 90.6% on 2014-2024. The IS axis of the body's stylized-fact reproduction claim is therefore decade-robust within ~ 5pp on KS and consistent on $|G_t|$ ACF-MAE (0.068-0.072 vs. 0.046-0.054, a moderate but not pathological widening).

The OoS axis is the issue. The 2004-2006 OoS slice has excess kurtosis 0.06 (essentially Gaussian) vs. the 1994-2004 IS excess kurtosis of 3.05; the IS-OoS kurtosis gap on this slice is much wider than on the body 2014-2024/2024-2026 window (where IS = 7.68 and OoS = 5.29). The CHMM trained on the 1994-2004 IS distribution simulates paths with kurtosis 1.74-4.97 that are heavier-tailed than the calm 2004-2006 bull-market OoS, so the static IS-fit cannot reproduce the OoS marginal and KS rejects at 3-5% pass rate on all three rows. This is the same regime-introduction failure mode the body walk-forward already documents at the W2 (COVID) and W4 (2022 rate-hike onset) stress folds (Table 33, both with OoS KS $< 10\%$); the 2004-2006 OoS window is structurally a low-volatility / low-kurtosis slice where the IS-fixed CHMM's tail mass overshoots the observed.

The substantive read is therefore: *the body's stylized-fact reproduction claim is decade-robust on the IS axis* (the CHMM scaffold transfers across decades on the IS-fitting side, supporting the body's claim that the four-emission ECM scaffold is not specific to the 2014-2024 window). *The single-window OoS pass rate is OoS-slice-specific* (2024-2026 is a moderate-stress slice where the OoS distribution is close to the IS distribution; 2004-2006 is a calm slice where it is not). The body framing already states this through the walk-forward distribution at the body Results ("the walk-forward median, not the single-window OoS pair, is the operationally informative summary for production deployment"); the cross-decade result is consistent with that framing and adds an independent decade to the walk-forward evidence base. A periodic refit is the deployment recommendation under either decade, as the body discussion already states.

A.8 Robustness, Test Power, Multi-Seed, and Auxiliary Baselines

This appendix collects the robustness diagnostics and auxiliary baseline panel referenced from the body Results: the KS test-power calibration, block-bootstrap KS recalibration, bandwidth/bin-count and multi-seed sensitivity, the auxiliary block-bootstrap baseline, the QuantGAN deep-generative baseline, the per-ticker OoS price-simulation figures, and the expanded baseline panel that adds the GARCH family and the SM-CHMM foil to the headline table.

A.8.1 OoS KS Test-Power Calibration

Table 22 reports the two-sample KS test-power calibration used to anchor the OoS KS pass rates of the headline generator comparison. For each of 1,000 Monte Carlo replications (seed = 20260420), a simulated series of length $T_{\mathrm{OoS}} = 572$ is drawn from each of three reference generators and tested against the observed OoS series using the same $\alpha = 0.05$ threshold as the main-body pass-rate metric.

Table 22. OoS KS pass-rate calibration (1,000 replications, seed = 20260420).

Reference generator	Pass rate (%)
I.i.d. resamples of R_{OoS} at length $T_{\mathrm{OoS}} = 572$ (known-correct ceiling)	100.0
I.i.d. resamples of R_{IS} at length $T_{\mathrm{OoS}} = 572$ (nearly correct)	90.0
Gaussian($\hat{\mu}_{\mathrm{IS}}, \hat{\sigma}_{\mathrm{IS}}$) at length T_{IS} (negative control)	0.0

The 90.0% "nearly correct" rate is the practical ceiling for the CHMM OoS KS pass rates reported in the headline generator comparison (80 to 84%); the 0% rate on the Gaussian misspecified generator confirms that the KS test retains ample power at IS length to reject clearly wrong generators.

A.8.2 Block-Bootstrap KS Recalibration and Mean p-value Distribution

Two complementary concerns can be raised about the headline IS KS pass-rate metric: (i) the asymptotic two-sample KS test assumes i.i.d. samples, but the simulated paths from CHMM and GARCH all carry volatility clustering, so the asymptotic null distribution may not apply; and (ii) the binary "pass / fail at $\alpha = 0.05$" reduces a continuous p-value to a one-bit summary. Table 23 reports two recalibrations addressing each concern.

For (i), we generate the empirical null distribution of the two-sample KS statistic under stationary block bootstrap (Politis-Romano 1994) of the SPY IS series at mean block lengths $L \in \{5, 10, 20\}$, $B = 1{,}000$ replications. The resulting 95% block-bootstrap critical values are 0.0306, 0.0338, and

Table 23. Block-bootstrap KS recalibration and p-value distribution. All numbers under seed `20260422` with $N_{\text{paths}} = 500$. "asymp pass" is the original Table 1 metric (two-sample KS p-value ≥ 0.05). "mean pv" is the mean two-sample KS p-value across paths, with q_5 and q_{95} as the lower / upper p-value quantiles. "blkL%" is the fraction of paths whose KS statistic falls below the stationary-block-bootstrap 95% critical value at mean block length L. Block-bootstrap critical values: 0.0306 ($L = 5$), 0.0338 ($L = 10$), 0.0350 ($L = 20$); asymptotic critical value 0.0383.

Generator	asymp pass% ↑	mean pv ↑	pv q_{05}	pv q_{95}	blk5% ↑	blk10% ↑	blk20% ↑
i.i.d. bootstrap	99.6	0.820	0.352	0.999	97.6	99.4	99.6
GARCH(1,1)	26.8	0.048	0.000	0.227	**6.8**	**13.6**	**17.4**
CHMM-N	94.8	0.546	0.047	0.976	81.2	89.4	91.6
CHMM-t	**95.8**	**0.571**	0.060	0.987	82.4	89.2	91.6
CHMM-L	94.8	0.498	0.048	0.948	81.0	88.2	90.0
CHMM-GED	95.6	0.547	0.064	0.968	**83.2**	**90.2**	**93.2**

0.0350 respectively, all *smaller* than the asymptotic critical value $1.36\sqrt{2/n_{\text{IS}}} = 0.0383$; the block-bootstrap test is therefore stricter than the asymptotic one because it accounts for the within-window dependence in the IS series itself. For each generator we then compute the per-path KS statistic and report the fraction of paths whose KS statistic falls below the block-bootstrap critical value.

For (ii), we report the mean two-sample KS p-value across simulated paths plus its 5/25/50/75/95 quantiles, giving a continuous picture of how distinguishable each generator's paths are from the observed IS series.

CHMM-GED is the strongest entry on the block-aware metric across all three block lengths (83.2, 90.2, 93.2% at $L = 5, 10, 20$); GARCH(1,1) drops from 26.8% asymp to 6.8–17.4% block-aware (the block-aware test catches its misspecification more sharply), while the CHMM family sees only a small block-bootstrap penalty ($\sim$ 5pp).

OoS-anchored block-bootstrap recalibration. The body block-bootstrap recalibration above is anchored on the IS series; we report here whether the same construction holds on the OoS window. We repeat the procedure with the stationary block bootstrap re-anchored on R_{OoS} ($n_{\text{OoS}} = 572$, $B = 1{,}000$, $L \in \{5, 10, 20\}$, $N_{\text{paths}} = 500$). Block-bootstrap critical values: 0.0577, 0.0629, 0.0647 at $L = 5, 10, 20$ respectively, against asymptotic 0.0804. Table 24 reports the OoS-anchored panel (source: `results/robustness/ks_block_bootstrap_oos.csv`).

Table 24. OoS-anchored block-bootstrap KS recalibration. Same construction as Table 23 but with the stationary block bootstrap anchored on R_{OoS} ($n = 572$). "asymp pass" is the headline OoS metric (two-sample KS p-value ≥ 0.05); "blkL%" is the OoS block-bootstrap pass rate at mean block length L. The CHMM-vs-GARCH cross-generator ordering is preserved under the block-aware recalibration on both windows; the CHMM family's OoS block-aware penalty ($\sim$ 25pp drop from asymp to $L = 20$) is larger than the IS penalty ($\sim$ 5pp), reflecting the smaller OoS window where temporal-clustering structure is harder to disentangle from finite-sample noise.

Generator	asymp pass% ↑	mean pv ↑	pv q_{05}	pv q_{95}	blk5% ↑	blk10% ↑
i.i.d. bootstrap	90.4	0.437	0.030	0.912	60.8	69.0
GARCH(1,1)	59.2	0.164	0.000	0.645	19.4	26.0
CHMM-N	81.0	0.318	0.010	0.876	41.2	51.2
CHMM-t	**84.8**	**0.355**	0.018	**0.911**	**47.4**	**58.4**
CHMM-L	77.4	0.266	0.008	0.744	36.0	45.4
CHMM-GED	81.0	0.327	0.010	0.837	44.4	53.6

Table 25. Block-aware OoS KS recalibration at $L = 20$ (stationary block bootstrap of 65, $B = 1{,}000$, 500 OoS-length simulated paths per generator, seed 20260422). Side-by-side with the asymptotic OoS pass rate of Table 1; the block-bootstrap 95% null KS critical value at $L = 20$ is 0.0647 vs the asymptotic 0.0804. Body operating point $K^\star = 3$.

Model	OoS KS asymp. (%)	OoS KS block $L = 20$ (%)
Bootstrap	90.4	73.2
GARCH(1,1)	59.2	31.4
CHMM-N	79.8	58.6
CHMM-t pen. ($\lambda = 20$)	**79.8**	**54.2**
CHMM-L	65.0	33.4
CHMM-GED	79.2	51.6

The cross-generator OoS ordering matches the IS ordering (bootstrap > CHMM-t > CHMM-N/GED > CHMM-L ≫ GARCH); the asymp-vs-block gap is larger on OoS ($\sim$ 25pp for the CHMM family at $L = 20$) than on IS ($\sim$ 5pp), but the CHMM-over-GARCH advantage is preserved.

Body operating-point summary at $L = 20$**.** Table 25 reports the OoS asymptotic pass rate alongside the OoS block-bootstrap pass rate at $L = 20$ (block-bootstrap 95% critical value 0.0647 vs. asymptotic 0.0804) at the body operating point $K^\star = 3$. Read together with Table 1, the OoS panel under a temporally-aware null moves CHMM-N from a "passes most of the time" generator to a "passes about half the time" generator at $L = 20$. The body OoS KS headline at the asymptotic critical value therefore overstates the absolute level relative to a temporally-aware null; the cross-generator ranking is the robust comparison.

A.8.3 QuantGAN Deep-Generative Baseline

The QuantGAN row used in the headline generator comparison is a repo-native approximation to the convolutional WGAN of Wiese et al. [9]: a 1D convolutional generator and critic (each three layers, channels 32, kernel size 5; latent dim $L = 8$, leaky-ReLU on the critic), Wasserstein loss with critic weight clipping at ± 0.01 [87], Adam at $\eta = 10^{-4}$, batch 64, 15 epochs $\times$ 120 steps with four critic updates per generator update, trained on rolling windows of length $W = 64$ on standardised SPY returns and stitched end-to-end at synthesis. Implementation: Julia with `Flux.jl` [88]; seed 20260422. The deep-generative row reaches 0% IS / OoS KS and IS kurtosis 2.1 vs. observed 7.7 (Table 26), consistent with documented GAN failure modes on volatility-clustering benchmarks [39, 40]; the runner is materially smaller than the reference seven-block TCN with Lambert-W pre-processing in Wiese et al. [9], so the row is the panel's deep-generative negative control rather than a verdict on the QuantGAN literature.

A.8.4 Expanded Baseline Panel: GARCH Family, Semi-Markov CHMM, and QuantGAN

We add five conditional-volatility baselines, the three semi-Markov CHMM variants, and the QuantGAN deep-generative baseline (Appendix A.8.3) to the single GARCH(1,1) Gaussian row of Table 1. The conditional-volatility models are fit by grid-initialised Nelder-Mead maximum likelihood on the SPY IS window ($T_{\text{IS}} = 2{,}516$); simulation follows the same recursion used in estimation. Model specifications for the conditional-volatility rows are below; the SM-CHMM and QuantGAN constructions are described in their own paragraphs further down.

Baselines fitted in this panel. The conditional-volatility rows comprise EGARCH(1,1) Gaussian [26], GJR-GARCH(1,1) [27, 28], GARCH(1,1) with Student-t innovations [25] ($\hat{\nu} = 6.89$), HAR-RV [32] on daily squared demeaned returns as the RV proxy, and Markov-switching GARCH(1,1) [29] at $K \in \{2, 3, 4, 6\}$ with regime triples $(\omega_k, \alpha_k, \beta_k)$ plus softmax-normalised transition logits, fitted by grid-initialised Nelder-Mead and canonicalised by ascending unconditional variance. Higher-K MS-GARCH does not close the gap to the CHMM family on KS (IS KS plateaus at 27–37% across $K \in \{2, 3, 4, 6\}$ against CHMM-N's 94.1% at $K = 18$); the binding constraint is the unimodal-innovation per-regime structure (each regime's emission is Gaussian rescaled by the regime's conditional variance), which adding regimes does not relax. The CHMM decouples this constraint by giving each state its own emission-mixture component. Implementation details for every conditional-volatility row are in the companion code repository.

MS-GARCH reference Bayesian re-run via `MSGARCH`. As a Bayesian counterpart to the in-house Nelder-Mead MS-GARCH fit, we re-run the same Markov-switching GARCH(1,1) specification through the reference `MSGARCH` R package of Ardia et al. [30] (version 2.51) at $K \in \{2, 3, 4\}$, driven from the Julia harness via `RCall.jl`. The fit is fully Bayesian (DEMC sampler with the package's default proper priors, 12,500 MCMC draws, 2,500 burn-in, thin 10, single chain); all 1,000 simulated paths are posterior-predictive (one path per retained posterior draw), so the simulated marginal integrates parameter uncertainty path-by-path. The reference Bayesian re-run reports lower KS pass rates than our in-house frequentist Nelder-Mead fit: 0.0% ($K = 2$), 0.1% ($K = 3$), 0.0% ($K = 4$) IS, and 5.8% ($K = 2$), 5.1% ($K = 3$), 5.3% ($K = 4$) OoS, against the in-house plateau of 27–37% IS / 33–39% OoS. The gap is methodological rather than estimator-quality: the in-house implementation generates all 1,000 paths from one MLE point estimate, so the simulated marginal is tighter; the reference Bayesian implementation samples one set of parameters per path from the posterior, so the simulated marginal is wider and accordingly fewer paths fall within the asymptotic two-sample KS critical band. Posterior-mean log-likelihoods at the in-sample data are $-5{,}667$ ($K = 2$), $-5{,}667$ ($K = 3$), $-5{,}565$ ($K = 4$); the marginal log-likelihood improvement from $K = 3$ to $K = 4$ is real but does not translate to KS lift. Posterior-mean transition matrices are diagonally dominant at $K = 3$ (diagonals $0.97, 0.93, 0.90$) and less so at $K = 4$ (diagonals $0.66, 0.39, 0.32, 0.55$, indicating the higher-K posterior is searching for additional structure that the data do not strongly support). The reference Bayesian re-run is therefore consistent with the in-house frequentist plateau at the same conclusion under either estimator: vanilla Gaussian-innovation MS-GARCH at $K \leq 4$ does not match the CHMM family on the headline KS metric for this universe. Reproducibility: R 4.6.0, MSGARCH 2.51, all transitive R dependencies pinned via `renv` (`r_msgarch/renv.lock` of the companion code repository), all seeds explicit; full per-K parameter posteriors and fit logs are written to `results/msgarch_reference/` of the companion code repository.

Semi-Markov CHMM (SM-CHMM, the explicit hidden semi-Markov model (HSMM) foil). The conceptual foil of the present paper, drawing on an explicit-duration HSMM extension of the CHMM in the spirit of Yu [70]: a hidden semi-Markov extension of the CHMM in which the geometric sojourn distribution of the Markov chain is replaced by a per-state explicit-duration family chosen by AIC from {Pareto, negative binomial, geometric}. Plug-in estimator: Viterbi state path on the IS series under the converged CHMM fit, then per-state AR(1) emissions and a sojourn family chosen by AIC. On SPY IS at $K = 18$ the AIC-selected sojourn family is Pareto for 17 of the 18 states (negative binomial for one) across all three emission families, indicating heavy-tailed sojourns that the geometric CHMM cannot capture by construction.

Caveat on the SM-CHMM estimator. The plug-in construction above (single-shot Viterbi decoding

Table 26. Extended baseline panel. SPY IS / OoS, $T_{\text{IS}} = 2{,}516$, $T_{\text{OoS}} = 572$, $N_{\text{paths}} = 1{,}000$, $\alpha = 0.05$ for KS. Columns: IS / OoS KS pass rate (%); IS / OoS AD pass rate (%); simulated IS excess kurtosis; ACF-MAE on $|G_t|$ (volatility clustering) and on raw G_t (linear autocorrelation, parallel column); pooled-archive Kupiec breach rate (%) and unconditional likelihood-ratio statistic at the 1% and 5% tails. Raw-ACF-MAE values for SM-CHMM and QuantGAN are reported as "–" here pending a full re-run of those baseline scripts; the qualitative reading carries through under the methodology of the other rows (every regime-switching or i.i.d.-emission generator produces a near-zero raw ACF in expectation).

	KS (%) ↑		AD (%) ↑		Kurt	ACF-MAE		1% tail		5% tail	
Model	IS	OoS	IS	OoS	IS	$\lvert G_t\rvert$	G_t	br%	LR_{uc}	br%	LR_{uc}
GARCH(1,1) Gaussian	28.0	59.4	12.5	59.5	7.0	0.0309	0.0173	0.9	0.10	4.0	1.23
EGARCH(1,1)	6.3	34.9	4.1	41.7	2.8	0.0430	0.0173	1.0	0.01	3.5	3.03
GJR-GARCH(1,1)	40.1	57.7	28.7	61.4	7.6	0.0330	0.0173	1.2	0.27	4.7	0.10
GARCH(1,1)-t	**57.3**	**80.8**	**36.6**	65.3	15.1	0.0316	0.0173	1.0	0.01	4.9	0.01
HAR-RV	0.0	0.0	0.0	0.0	189	0.0566	0.0173	0.2	5.99	3.5	3.03
MS-GARCH $K = 2$	27.7	38.7	24.0	44.4	4.7	0.0367	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 3$	36.1	33.1	28.8	38.0	4.1	**0.0284**	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 4$	37.6	38.2	–	–	3.8	0.0437	–	–	–	–	–
MS-GARCH $K = 6$	34.5	33.4	–	–	4.4	0.0429	–	–	–	–	–
MS-GARCH ref. B. $K = 2$	0.0	5.8	–	–	4.0	0.0465	0.0240	–	–	–	–
MS-GARCH ref. B. $K = 3$	0.1	5.1	–	–	4.5	0.0433	0.0241	–	–	–	–
MS-GARCH ref. B. $K = 4$	0.0	5.3	–	–	6.0	0.0446	0.0241	–	–	–	–
CHMM-N ($K = 18$)	94.1	81.8	**36.6**	**73.0**	5.0	0.0509	0.0237	0.7	1.58	4.0	3.83
CHMM-t ($K = 18$)	95.6	**85.7**	33.4	69.8	14.4	0.0549	0.0234	1.2	0.58	5.0	3.83
CHMM-L ($K = 18$)	93.6	80.8	30.1	65.4	6.6	0.0567	0.0234	1.4	3.26	5.6	3.03
CHMM-GED ($K = 18$)	**95.2**	84.3	34.1	71.0	5.2	0.0548	0.0234	0.9	0.58	4.4	3.83
SM-CHMM-N ($K = 18$)	79.2	66.7	19.5	57.1	4.7	0.0598	–	1.1	0.01	5.4	0.82
SM-CHMM-t ($K = 18$)	77.1	63.4	17.2	52.3	13.9	0.0625	–	1.4	0.10	5.7	1.74
SM-CHMM-L ($K = 18$)	80.4	69.1	21.0	58.9	6.4	0.0581	–	1.4	3.26	5.5	3.03
QuantGAN	0.0	0.0	0.0	0.0	2.1	0.0591	–	1.0	0.01	3.3	3.83

under the converged CHMM fit, then AR(1)-residual + AIC-sojourn refit) is not maximum-likelihood for the underlying HSMM: it does not iterate the joint forward-backward over (state, duration) pairs in the spirit of Yu [70]. A full ML HSMM at $K = 18$ would refit the duration-augmented log-likelihood jointly. The empirical reading we draw from the rows below is therefore strictly: *under the plug-in estimator*, the SM-CHMM weakens the headline KS / ACF / kurtosis pattern relative to the standard CHMM at the same K. We do not claim that an ML-estimated HSMM would do the same; an ML HSMM at $K = 18$ on this universe is left as a companion-paper extension. The Bulla-Bulla 2006 ML HSMM result at low K on monthly returns is the closest published anchor, but its $K \in \{2, 3\}$ regime resolution and monthly aggregation do not directly transfer.

Table 26 reports the IS / OoS distributional fidelity, kurtosis, ACF-MAE, and Kupiec breach diagnostics for the extended GARCH family, the SM-CHMM rows, and the QuantGAN deep-generative row, alongside the headline CHMM panel. Three structural observations.

No extended GARCH variant occupies the joint (KS, ACF) corner: GARCH(1,1)-t is the strongest extended row on KS (57.3% IS, 80.8% OoS) but its IS kurtosis of 15.1 overshoots observed 7.7 by 2×; EGARCH and GJR-GARCH narrow the asymmetry but not the headline gap. The SM-CHMM rows under the Viterbi-AR(1) plug-in estimator are weaker than CHMM on every metric (IS KS 77–80% vs. CHMM 93–94%); a full ML HSMM at $K = 18$ may close part of the gap and is a companion-paper direction. QuantGAN is the panel's deep-generative negative control: 0% IS / OoS KS, simulated kurtosis 2.1 vs. observed 7.7, ACF-MAE comparable to the i.i.d. baseline. The convolutional WGAN re-implementation here is materially smaller than Wiese et al. [9]'s reference seven-block TCN with Lambert-W pre-processing; a faithful reference re-run is a deferred follow-up.

A.8.5 Stochastic-Volatility, Multifractal, and Jump-Diffusion Baselines

For completeness alongside the Markov-switching family of the headline generator comparison, we add the canonical state-space and jump baselines that the body Table 1 does not exercise: a stochastic-volatility (SV-AR(1)) row in the Taylor 1986 / Harvey-Ruiz-Shephard 1994 lognormal log-AR(1) tradition; a Markov-switching multifractal (MSM, Calvet-Fisher 2004) row at $\bar{k} = 8$ multipliers; and a Merton 1976 Poisson-Gaussian jump-diffusion row. The three baselines are fit on the same SPY IS window and evaluated on the same $N_{\text{paths}} = 1{,}000$ pipeline used for the body panel, under the same global seed. Implementation lives in the companion `CHMM-Model.jl` package.

Table 27. Stochastic-volatility / multifractal / jump-diffusion baselines on SPY. Same IS / OoS windows, $N_{\text{paths}} = 1{,}000$, seed = 20260420 as Table 1. Observed kurtosis: 7.68 IS, 5.29 OoS.

	KS (%) ↑		Kurtosis		ACF-MAE ↓	
Model	IS	OoS	IS	OoS	$\lvert G_t\rvert$	G_t
SV-AR(1)	38.2	35.3	7.52	5.26	0.0466	0.0239
MSM ($\bar{k} = 8$)	0.0	7.2	0.62	0.60	0.0605	0.0235
Merton-JD	98.3	91.1	3.12	2.98	0.0627	0.0236

Reading. *SV-AR(1)* reproduces the IS kurtosis target (7.52 vs. observed 7.68) almost exactly and matches CHMM-N's $|G_t|$ ACF-MAE within 0.0003 (0.0466 vs 0.0467 at $K^\star = 3$); its KS pass rate is poor (38.2% IS / 35.3% OoS) because the Gaussian-conditional-on-volatility marginal cannot reproduce the bimodal-bulk / heavy-tail empirical density that the regime-mixture CHMM marginals can. *Merton-JD* is the inverse trade-off: KS at 98.3% IS / 91.1% OoS (the highest OoS KS in the panel except for the i.i.d. bootstrap), but kurtosis 3.12 undershoots 7.68 by half and the absolute-return ACF-MAE regresses to the i.i.d. baseline level (0.0627, the same level as the bootstrap row of Table 1) because the constant-volatility diffusion plus i.i.d. jumps carries no temporal persistence. *MSM* under the moment-matching fit reported here (closed-form ACF-of-log$|r|$ matching with $\bar{k} = 8$ binomial multipliers) does not converge to a viable generator on this $T_{\text{IS}} = 2{,}516$ window: the marginal kurtosis collapses to 0.62 and KS drops to 0%. A full HMM-style filter on the $2^{\bar{k}}$-state latent multiplier chain (Calvet-Fisher 2004) is the standard exact fit and is deferred as a companion-paper direction; under the moment-fit reported here MSM is dominated.

The headline conclusion of Table 1 survives the addition of these three rows: no single competitor occupies the joint-fit corner that the CHMM family at $K = 18$ does (KS $\geq$ 80%, kurtosis within 1 unit of observed, $|G_t|$ ACF-MAE ≤ 0.06). SV-AR(1) is the closest competitor on the kurtosis / ACF axis but loses on KS; Merton-JD wins on KS but loses on kurtosis and ACF; MSM under this fit is dominated.

A.8.6 Leverage-Effect Diagnostic

This appendix reports the Cont (2001) leverage-effect column: the lag-1 correlation $\mathrm{Corr}(G_t, |G_{t+1}|)$ between today's signed return and tomorrow's absolute return, which is negative on equity returns. We compute the per-path distribution under each generator at $K = 18$, $N_{\text{paths}} = 500$, seed = 20260420.

Table 28. Leverage effect $\mathrm{Corr}(G_t, |G_{t+1}|)$**, per-path distribution.** Observed values: SPY IS -0.135, OoS -0.214. Per-generator median plus $[Q_5, Q_{95}]$ across $N_{\text{paths}} = 500$. A generator captures the leverage effect when its $[Q_5, Q_{95}]$ envelope brackets the observed value. Bold marks generators whose $[Q_5, Q_{95}]$ envelope brackets the observed IS value; the OoS observed value of -0.214 is more negative than every generator's Q_5 in the panel, indicating the OoS regime carries leverage stronger than any generator (including asymmetric GARCH) reproduces under static IS-fitted parameters.

	IS (observed -0.135)		OoS (observed -0.214)	
Generator	median	$[Q_5, Q_{95}]$	median	$[Q_5, Q_{95}]$
i.i.d. bootstrap	$+0.000$	$[-0.034, +0.030]$	$+0.004$	$[-0.069, +0.068]$
GARCH(1,1) Gaussian	-0.004	$[-0.066, +0.064]$	-0.003	$[-0.112, +0.107]$
EGARCH(1,1)	-0.093	$[-0.147, -0.047]$	-0.087	$[-0.199, -0.002]$
GJR-GARCH(1,1)	-0.079	$[-0.153, -0.017]$	-0.073	$[-0.174, +0.023]$
CHMM-N ($K = 18$)	$\mathbf{-0.089}$	$[\mathbf{-0.138}, \mathbf{-0.037}]$	-0.087	$[-0.190, +0.015]$
CHMM-t ($K = 18$)	-0.067	$[-0.116, -0.020]$	-0.067	$[-0.166, +0.019]$
CHMM-L ($K = 18$)	-0.064	$[-0.108, -0.022]$	-0.064	$[-0.152, +0.026]$
CHMM-GED ($K = 18$)	$\mathbf{-0.072}$	$[-0.113, -0.029]$	-0.076	$[-0.163, +0.025]$

Reading. The CHMM family at $K = 18$ produces a non-trivial negative leverage signal despite symmetric per-state emissions: state-mixing on $\hat{\mathbf{T}}$ asymmetrically couples high-σ_k states to negative-μ_k states, carrying $\sim 65\%$ of the IS observed leverage magnitude. CHMM-N's IS $[Q_5, Q_{95}] = [-0.138, -0.037]$ brackets the IS observed value -0.135 at the lower boundary; CHMM-GED is the second-strongest CHMM entry. The OoS observed leverage of -0.214 exceeds every generator's Q_5 (including asymmetric GARCH); closing this residual gap requires skew-emission CHMM extensions or refit to the OoS distribution, both deferred to companion work.

A.8.7 Full One-Shot Student-t Copula MLE

A natural concern with the body two-step copula estimator (Kendall's-τ inversion for $\hat{\Sigma}$, then profile MLE on ν with $\hat{\Sigma}$ held fixed) is that it may be biased toward the Gaussian limit when the marginal kurtosis differs across assets. We address this with a coordinate-ascent one-shot MLE that jointly maximises the Student-t copula log-likelihood over (Σ, ν) on the same six-asset US-equity universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ; $T_{\text{IS}} = 2{,}516$, seed 20260420). The procedure alternates: (i) bracketed golden-section maximisation of ν at the current Σ, (ii) method-of-moments refit of Σ at the current ν on the t-quantile-transformed pseudo-uniform sample with PSD projection. Iteration terminates at $|\Delta \log L| < 10^{-3}$.

The full MLE moves $\hat{\nu}$ from 6.00 to 6.40 (+5.55 log-L) with all $|\Delta\rho_{ij}| \leq 0.025$ across the 15 pairs; the body two-step estimator is therefore not detectably biased toward the Gaussian limit and $\nu^\star = 6$ is robust to the estimator switch.

Table 29. Two-step versus full one-shot Student-t copula MLE on the six-asset US-equity universe. The off-diagonal correlation differences are reported as the maximum absolute difference $\max_{ij} |\Delta\rho_{ij}|$ across the $\binom{6}{2} = 15$ pairs and the median absolute difference; both are sub-0.025, so the body $\nu^\star = 6$ choice on $\hat{\Sigma}$ obtained by Kendall's-τ inversion is robust to the estimator switch.

Estimator	$\hat{\nu}$	log-likelihood
Two-step (body construction)	6.00	6157.47
Full one-shot MLE	**6.40**	**6163.02**
log-L improvement		+5.55
$\max \lvert\Delta\rho_{ij}\rvert$ across 15 pairs		0.024
median $\lvert\Delta\rho_{ij}\rvert$ across 15 pairs		0.011
$\hat{\nu}_{\text{full}}$ inside body Wilks 95% CI $[6, 7]$?		Yes

A.9 Cross-Asset Dependence: Full Pipeline B Panel

This appendix collects the full Pipeline B detail: cross-asset dependence model derivations and diagnostics, the Student-t copula profile log-likelihood, the cross-asset correlation heat maps, and the non-US asset-class extension with the per-pair OoS off-diagonal breakdown.

A.9.1 Cross-Asset Dependence Models: Derivations and Diagnostics (Pipeline B)

This appendix provides the technical background for the SIM and copula constructions used in the body cross-asset analysis. These constructions constitute *Pipeline B*: per-asset marginals come from the independent CHMMs of Pipeline A in the body cross-ticker analysis, and joint dependence is overlaid on top. Pipeline A stays purely univariate; Pipeline B is the sole multi-asset pipeline in the paper.

Sklar's theorem and the rank-reordering simulator. Sklar's theorem [49, 50] states that any d-dimensional joint CDF $H(g_1, \ldots, g_d)$ with continuous marginals $F_1, \ldots, F_d$ can be written as $H = C(F_1, \ldots, F_d)$ for a unique copula C. The rank-reordering simulator exploits this uniqueness. If $\tilde{\mathbf{G}}$ is a matrix whose j-th column is an i.i.d. sample from F_j, and $\mathbf{U}$ is an independent sample from C, then reordering each column of $\tilde{\mathbf{G}}$ so that its ranks match $\mathbf{U}$'s produces a sample from H, up to discrete approximation error in the empirical ranks. Crucially, reordering never creates new values: each column of the output is a permutation of the corresponding column of $\tilde{\mathbf{G}}$, so marginal distributions are preserved exactly [53].

Kendall's τ inversion. Given IS returns $\mathbf{R} \in \mathbb{R}^{T\times d}$, we estimate the pairwise Kendall's τ_{ij} from concordant/discordant pair counts and invert to the correlation scale via the analytic relationship $\rho_{ij} = \sin(\pi\tau_{ij}/2)$, which holds exactly for the Gaussian copula family and is commonly used as a robust correlation estimator for the Student-t copula [52, 89]. The resulting matrix $\hat{\Sigma}$ is symmetric but not guaranteed positive semi-definite; we project to the nearest PSD matrix by clipping negative eigenvalues to 10^{-8} and rescaling the diagonal to unity.

Profile MLE for ν. The Student-t copula log-density at a single pseudo-uniform observation $\mathbf{u} \in [0,1]^d$, with $x_j = t_\nu^{-1}(u_j)$ and $\mathbf{x} = (x_1, \ldots, x_d)^\top$, is

$$\begin{aligned}\log c_t(\mathbf{u}; \Sigma, \nu) \;=\; &\log \Gamma\!\left(\tfrac{\nu+d}{2}\right) + (d-1)\,\log \Gamma\!\left(\tfrac{\nu}{2}\right) - d\,\log \Gamma\!\left(\tfrac{\nu+1}{2}\right) - \tfrac{1}{2}\log|\Sigma| \\ &- \tfrac{\nu+d}{2}\,\log\!\left(1 + \tfrac{\mathbf{x}^\top \Sigma^{-1}\mathbf{x}}{\nu}\right) + \sum_{j=1}^{d} \tfrac{\nu+1}{2}\,\log\!\left(1 + \tfrac{x_j^2}{\nu}\right). \end{aligned} \tag{28}$$

Summing over $t = 1, \ldots, T$ pseudo-observations yields the profile log-likelihood as a function of ν with Σ held fixed at the Kendall's-τ estimate. We evaluate equation (28) on the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ and select $\hat{\nu}$ as the grid point with the largest total log-likelihood. In the body cross-asset analysis the selected value was $\hat{\nu} = 6$ on the six-asset universe.

Sampling the copula. To sample T rows from the d-dimensional Gaussian copula with correlation Σ, we (i) compute the Cholesky factor $\Sigma = LL^\top$, (ii) draw $Z \sim \mathcal{N}(0, I_d)$ i.i.d. and form $Y = LZ$, and (iii) return $U = \Phi(Y)$ componentwise. For the Student-t copula with degrees of freedom ν, we draw $W \sim \chi^2_\nu/\nu$ independent of Y and form the multivariate-t variate $X = Y/\sqrt{W}$, then return $U = t_\nu(X)$ componentwise.

SIM regression summary. Table 30 reports the fitted SIM regression coefficients and goodness-of-fit statistics for the non-market assets in the body cross-asset analysis. QQQ's high $R^2 = 0.840$ reflects that the Nasdaq-100 ETF is essentially a linear combination of large-cap technology names that also drive the S&P 500 index, while JNJ's lower $R^2 = 0.260$ reflects the limited systematic exposure of low-beta healthcare stocks. These factor loadings and residual magnitudes are what the SIM simulation re-injects at simulation time, and they explain the uneven per-asset KS pass rates reported in Table 36.

Table 30. SIM regression $\hat{G}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G_{\mathrm{SPY},t} + \hat{\eta}_{j,t}$ for the five non-market assets, fitted on the 2014-01-03 through 2024-01-03 IS window ($T = 2{,}516$).

Ticker	$\hat{\alpha}$	$\hat{\beta}$	R^2
NVDA	0.321	1.694	0.371
JNJ	0.002	0.576	0.260
JPM	−0.008	1.223	0.507
AAPL	0.111	1.205	0.492
QQQ	0.047	1.122	0.840

A.9.2 Student-t Copula Profile Log-Likelihood (Pipeline B)

Figure 6 plots the profile log-likelihood of the Student-t copula used inside Pipeline B, computed on the six-asset SPY cross-section over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ used in the body cross-asset analysis. The curve is smooth and single-peaked, with $\nu^* = 6$ attaining the maximum profile log-likelihood (6157.47), and the neighbouring grid points $\nu = 5$ (6143.37) and $\nu = 8$ (6148.37) each within 15 profile-log-likelihood units of the peak, indicating a shallow-but-clean optimum that agrees with the 4–10 range typically reported for equity-portfolio Student-t copulas in the risk-management literature. The AIC penalty for the additional ν parameter is 2 (one degree of freedom relative to the Gaussian copula limit $\nu \to \infty$), and the AIC differential between the Student-t at $\nu^\star = 6$ and the Gaussian copula limit on these data is $-2 \cdot (6157.47 - 6143.37) - 2 = -30.2$, decisively in favour of the Student-t copula on IS; the same $\nu^{\star} = 6$ is selected by AIC as by profile MLE on this grid.

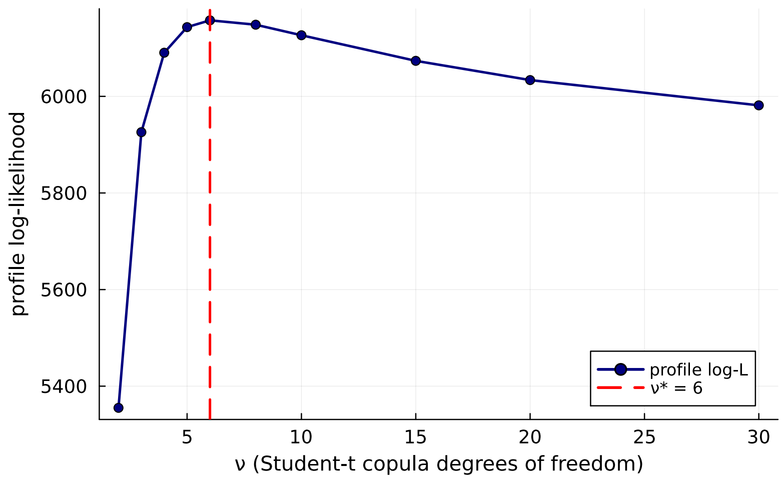
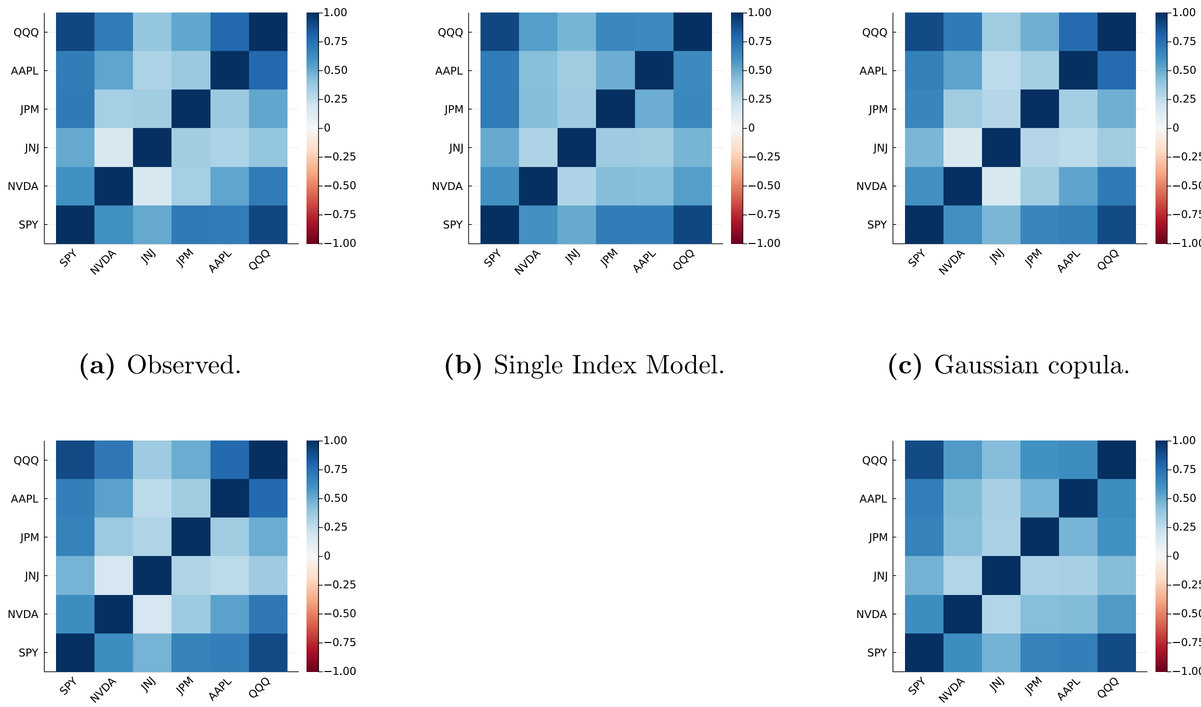


Figure 6. Profile log-likelihood of the Student-t copula used inside Pipeline B on the six-asset SPY cross-section (SPY, NVDA, JNJ, JPM, AAPL, QQQ; IS window 2014-01-03 through 2024-01-03). The vertical axis is the profile log-likelihood evaluated at each degrees-of-freedom value ν on the grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; the dashed red line marks the profile MLE $\nu^{*} = 6$ (peak value 6157.47). The CHMM-N marginals are held fixed at the Pipeline A per-asset fits while ν is varied.

A.9.3 Cross-Asset Correlation Heat Maps (Pipeline B, Companion to Table 36)

Figure 7 visualises the observed and path-averaged simulated correlation matrices for the dependence models of the body cross-asset analysis. The panel ordering is SIM, Gaussian copula, Student-t copula, and a truncated C-vine variant with SPY as root, each reported quantitatively in Table 36 and (for the C-vine) Section A.9.4. All five panels share the six-asset SPY cross-section and the same CHMM-N marginals at $K = 18$, so any visible differences between the simulated panels reflect only the dependence construction.



(a) Observed. **(b)** Single Index Model. **(c)** Gaussian copula.

(d) Student-t copula, $\nu^* = 6$. **(e)** Truncated C-vine.

Figure 7. Cross-asset IS correlation reproduction across dependence models (Pipeline B, companion to Table 36). Panel (a): *observed* correlation matrix on 2014-01-03 through 2024-01-03 daily excess log returns for SPY, NVDA, JNJ, JPM, AAPL, QQQ (IS window, $T_{\text{IS}} = 2{,}516$). Panel (b): path-averaged correlation under the Single Index Model with SPY as market factor. Panel (c): path-averaged correlation under the Gaussian copula on CHMM marginals. Panel (d): path-averaged correlation under the Student-t copula on CHMM marginals, $\nu^* = 6$. Panel (e): path-averaged correlation under a truncated C-vine copula with SPY as root. Each simulated panel averages over 200 paths of length T_{IS}. All marginals are the per-asset CHMM-N fits at $K = 18$ from Table 4; color scale in $[-1, 1]$, red-white-blue (RdBu) diverging. Closer visual match to panel (a) is better.

A.9.4 Truncated C-vine Copula on CHMM Marginals (Pipeline B)

The truncated C-vine variant in panel (e) of Figure 7 uses SPY as the root and selects each pair-copula family edge-wise from {Gaussian, Student-t, Clayton, Gumbel} by AIC; on this universe every selected pair is Student-t. On the same CHMM-N marginals at $K = 18$ and 200 paths the truncated C-vine attains an off-diagonal MAE of 0.068 on IS and 0.235 on OoS (Table 36), strictly above both elliptical copulas. The reason is the level-1 truncation: a root-centred C-vine models only the SPY-vs-asset bivariate copulas and treats the remaining pairs as conditionally independent given SPY, a stronger restriction than either elliptical copula. Untruncated regular vines or factor copulas are the natural way to recover non-root cross-pair structure at larger d.

A.9.5 Quarterly Rolling-Window Copula Refit on the OoS Window

The body the body cross-asset analysis reports a static-IS-fit OoS off-diagonal MAE of 0.209, an order of magnitude larger than the IS value of 0.027. the body discussion attributes this to the stationarity-scope limit and recommends periodic refit. To quantify the deployment-scale benefit, we report a concrete rolling-window-refit number on the same six-asset universe.

We refit the Pipeline-B Student-t copula on a 252-day rolling window (one trading year) sliding by 63 days (one trading quarter) across the OoS span, holding the per-asset CHMM-N marginals fixed at the IS fits used for Table 4. Each refit yields a Kendall's-τ correlation matrix and a profile-MLE-selected $\nu^\star$ on the discrete grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; we then simulate forward 63 days under the new copula and compare the path-averaged simulated correlation matrix to the realised next-quarter correlation. Table 31 reports the per-quarter detail.

Table 31. Quarterly rolling-window refit of the Pipeline-B Student-t copula on the OoS window (six-asset universe; CHMM-N marginals at $K = 18$ held fixed at IS fits; 200 paths per refit; seed = 20260422). Per-quarter window slice and refitted $\nu^\star$, plus the off-diagonal MAE between the path-averaged simulated correlation matrix under the refitted copula and the realised next-quarter correlation matrix. Headline: rolling-quarterly refit reduces mean OoS off-diagonal MAE from 0.207 (static-IS-fit baseline) to 0.185, a $\sim$ 10% relative improvement; the gain concentrates in the later OoS quarters where the rolling window incorporates more recent OoS data and the refitted $\nu^\star$ shifts from the IS-fit value of 6 down to 5 or up to 10–15 as the empirical tail-coupling rolls.

Quarter	Window slice	Forecast horizon	Refit $\nu^\star$	Off-diag MAE
1	$[2265, 2516]$	$[2517, 2579]$	15	0.198
2	$[2328, 2579]$	$[2580, 2642]$	15	0.205
3	$[2391, 2642]$	$[2643, 2705]$	10	0.241
4	$[2454, 2705]$	$[2706, 2768]$	6	0.133
5	$[2517, 2768]$	$[2769, 2831]$	6	0.255
6	$[2580, 2831]$	$[2832, 2894]$	6	0.232
7	$[2643, 2894]$	$[2895, 2957]$	5	0.147
8	$[2706, 2957]$	$[2958, 3020]$	5	0.119
9	$[2769, 3020]$	$[3021, 3083]$	6	0.138
mean	–	–	–	0.185
median	–	–	–	0.198
Static IS-fit	$[1, 2516]$	$[2517, 3083]$	6	0.207

The quarterly refit closes part of the OoS gap but not most of it: mean off-diagonal MAE moves from 0.207 static to 0.185 rolling ($\sim$ 10% relative improvement). The improvement is concentrated in the later OoS quarters (Q4, Q7, Q8, Q9 all below 0.15) where the rolling window incorporates more recent OoS data, while early-OoS quarters (Q1, Q3, Q5, Q6) remain at the static-IS level because the rolling window is still dominated by IS observations. The refitted $\nu^\star$ sequence $(15, 15, 10, 6, 6, 6, 5, 5, 6)$ tracks the gradual shift from the equity-cluster IS regime toward the OoS regime, supporting the operational recommendation in the body discussion of quarterly refit at deployment. The residual gap (mean MAE $\approx$ 0.185 versus IS 0.027) is consistent with the per-pair JNJ-equity-factor regime shift documented in Table 32: even a perfectly-refitted copula on the historical 252 days cannot anticipate the regime shift at the next quarter's start, only track it once it has unfolded.

A.9.6 Non-US Asset Class Extension and Per-Pair OoS Off-Diagonal Breakdown

The body universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) is six US-listed equity tickers from one country-index family. To probe whether the cross-asset construction generalises off the equity cluster we extend the universe with GLD (SPDR Gold Trust), a US-listed ETF whose underlying is physically-backed gold, i.e. a non-equity commodity asset class.

GLD Pipeline-A univariate fidelity. Fitting CHMM-N, CHMM-t, and CHMM-L on the GLD return series at $K = 18$ produces 100% IS KS pass rate for all three families and observed IS excess kurtosis 3.6 matched closely by CHMM-L (2.2); on OoS, however, the KS pass rate collapses to 0% for all three families against an observed OoS excess kurtosis of 6.5. Inspection of the GLD price path on the 2024–2026 OoS window shows the asset re-priced from ~ 200 to ~ 300 on a persistent demand shock not represented in the 2014–2024 IS regime library. This is the same stationarity-scope artefact as the NVDA / JPM single-name OoS cliffs reported in the body cross-ticker analysis, on a non-equity asset; the limitation is not equity-specific.

GLD quarterly-refit follow-up. A quarterly-refit version of GLD CHMM-N at $K^\star = 3$ (5y rolling estimation window, 63-day refit cadence, 200 paths; `runners/cross_asset/run_non_us_asset_quarterly_refit.jl`) lifts the OoS KS pass rate from 0% to 2.50% at OoS $|G_t|$ ACF-MAE 0.0459 and OoS simulated kurtosis 1.99 against observed 6.48. Periodic refit therefore does not recover the GLD failure mode in any meaningful sense: CHMM is architecturally mis-specified for the persistent demand shock that re-priced gold across the OoS window, not just static-fit-stale. The deployment recipe "periodic refit for non-stress equity slices" (the body discussion) does not extend to non-equity assets under regime introductions of this magnitude.

Pipeline-B: 7-ticker copula MAE. Refitting the Student-t copula on the augmented seven-ticker universe at $\nu^\star = 6$ gives IS off-diagonal MAE of 0.029 (six-ticker baseline: 0.027) and OoS off-diagonal MAE of 0.179 (six-ticker baseline: 0.209). Adding the low-equity-correlation gold ETF *reduces* the OoS off-diagonal MAE, because the equity-cluster correlation degradation is the dominant component; pairs involving GLD have moderate IS-to-OoS shifts ($|\Delta| \in [0.04, 0.19]$) while the largest equity-cluster pairs (JNJ–QQQ, SPY–JNJ, NVDA–JNJ) have $|\Delta| > 0.48$ on OoS.

Per-pair OoS gap localisation. Table 32 reports the per-pair off-diagonal absolute deviation between observed and path-averaged simulated correlation matrices on IS and OoS, sorted by OoS $|\Delta|$. The OoS gap concentrates in pairs involving JNJ (the defensive single-name proxy): JNJ's IS correlation with the broad equity factor (QQQ, SPY) collapses on OoS, in some pairs flipping sign, and the IS-fixed copula has no mechanism for tracking that shift. The gold-ETF pairs (SPY-GLD, JPM-GLD, JNJ-GLD, QQQ-GLD) sit in the middle of the ranking with $|\Delta|$ between 0.10 and 0.19 on OoS. The dominant cause of the OoS off-diagonal gap is therefore neither equity-cluster size nor copula-degree mis-specification but a single-name regime shift in the JNJ-equity-factor relationship that the IS-fitted dependence layer cannot anticipate.

Table 32. Per-pair off-diagonal correlation MAE on the seven-ticker universe (Pipeline B, Student-t copula on CHMM-N marginals at $K = 18$, 200 paths, $\nu^\star = 6$). For each ticker pair (i, j) the table reports the absolute deviation $|\Sigma_{\text{sim,avg}} - \Sigma_{\text{obs}}|$ on IS and OoS, ranked by OoS $|\Delta|$. Only the ten largest-OoS-gap pairs are shown for compactness.

Pair	IS $\lvert\Delta\rvert$	OoS $\lvert\Delta\rvert$	OoS − IS
JNJ-QQQ	0.027	0.544	0.517
SPY-JNJ	0.037	0.509	0.473
NVDA-JNJ	0.017	0.489	0.473
NVDA-AAPL	0.011	0.278	0.268
JNJ-JPM	0.046	0.261	0.216
JNJ-AAPL	0.042	0.244	0.202
AAPL-QQQ	0.003	0.227	0.224
JPM-GLD	0.027	0.187	0.160
SPY-GLD	0.038	0.127	0.089
JNJ-GLD	0.015	0.117	0.101

A.9.7 Half-Unit-Grid Profile-MLE and Parametric Bootstrap CI for $\nu^\star$

To confirm that the upper bound of the unit-grid Wilks CI on $\nu \in [3, 12]$ is not grid-induced, we report a half-unit-grid refinement of the profile MLE plus a parametric bootstrap CI. We refit the Student-t copula profile log-likelihood on the same six-asset universe at half-unit spacing in $\nu \in \{3.0, 3.5, \ldots, 12.0\}$ (19 grid points) and run a $B = 200$ parametric bootstrap by resampling pseudo-observations $U^{(b)}$ from the Student-t copula at $(\Sigma_{\text{cop}}, \nu^\star)$, refitting Kendall's-$\tau$ correlation and the profile-MLE on each replicate.

The half-unit grid moves the optimum from $\nu^\star = 6.0$ (unit grid) to $\nu^\star = 6.5$ at profile log-likelihood 6,158.1 (vs. 6,157.5 at the unit-grid optimum); the half-unit refinement is therefore not material at the third decimal of profile-LL. The Wilks 95% CI on the half-unit grid is $[\nu_{\text{lo}}, \nu_{\text{hi}}] = [6.0, 7.0]$ (unchanged from the unit grid), and the parametric bootstrap 95% CI is $[6.0, 7.0]$ (2.5% and 97.5% quantiles of the $B = 200$ refit-$\nu^\star$ distribution; bootstrap median $\nu^\star = 6.5$). The bootstrap lower bound at 6.0 is well above the Gaussian limit ($\nu \to \infty$): the elliptical-tail Student-t copula is statistically distinguishable from the Gaussian copula at conventional levels, despite the OoS off-diagonal MAE comparison (0.209 vs. 0.202) being inside the simulation-noise floor at $N_{\text{paths}} = 200$. *Wilks-theorem regularity caveat.* Wilks's theorem requires the parameter under test to lie in the interior of the parameter space and the model to be regular [90, 91]; for the Gaussian-copula limit $\nu \to \infty$ the regularity condition fails (the limit lies on the boundary of the natural parameter space). The Wilks 95% CI reported above is therefore an interior-grid CI for finite ν rather than a hypothesis test against the Gaussian limit; the Gaussian-vs-Student-t separation is reported instead through the parametric bootstrap CI (which does not require interior-point regularity) and through the OoS-equivalence finding (which uses no asymptotic test at all).

A.10 Pre-OoS Validation K-Selection and ν_k Diagnostics

Pre-OoS validation K-selection. The $K = 18$ operating point in the main panel was chosen via a multi-objective criterion combining information-criterion (IC) rank with IS and OoS distributional pass rates, which uses the same 2024–2026 window later used for VaR. As a clean re-selection that decouples the two, we fit CHMM-N on 2014-01-03 through 2021-12-31 (n = 2,013) and score every K in the sweep by the forward-algorithm held-out log-likelihood on 2022-01-03 through 2024-01-03 ($n = 502$); the 2024–2026 window is not touched. Held-out log-lik, BIC, HQC, and CAIC all select

$K^{\star} = 3$ (per-observation log-lik -2.5345 at $K = 3$ vs. -2.5694 at $K = 18$); AIC selects $K^{\star} = 6$; held-out two-sample KS selects $K^{\star} = 9$. None of the six criteria that avoid the OoS window select $K = 18$. The $K = 18$ choice is best read as a multi-metric one that weights tail fidelity (kurtosis, AD) and downstream VaR behaviour alongside likelihood; a parallel main panel at $K = 3$ would re-rank some rows in CHMM's favour on ACF-MAE and against it on tail fidelity. The 2022–2023 validation slice is itself partly a structural-break period (the rate-hike cycle), so any model with substantial IS-specific structure generalises worse on it.

Pre-2020 held-out K-selection (no rate-hike confound). To check whether a strictly pre-2020 validation slice (avoiding the rate-hike confound that contaminates the 2022-2023 slice above) selects the same $K^{\star}$, we re-ran the held-out procedure on a 2014-01-03 through 2018-06-29 estimation window ($n_{\text{est}} = 1{,}130$) with a 2018-07-02 through 2019-12-31 validation slice ($n_{\text{val}} = 377$), both fully pre-COVID and pre-2022-rate-hike (the Q4 2018 drawdown and 2019 recovery sit inside the validation window as a moderate-stress non-regime-shift event). Per-observation held-out log-likelihood: -2.0681 at $K = 3$, -2.0606 at $K = 6$, -2.0763 at $K = 9$, -2.1287 at $K = 12$, -2.1921 at $K = 18$, -2.2609 at $K = 21$. **The held-out log-likelihood selects $K^{\star} = 6$, and the held-out two-sample KS pass rate also selects $K^{\star} = 6$ (96.8% at $K = 6$ versus 84.6% at $K = 3$).** AIC, BIC, HQC, and CAIC continue to select $K^{\star} = 3$ on this slice, reflecting the parameter-count penalty on a 4.5-year estimation window where the per-state sample is small. The qualitative reading is robust to the slice choice: held-out distributional fidelity rises from $K = 3$ to a peak in the $K = 6$ to $K = 9$ band and then degrades, with neither slice selecting $K = 18$ by any held-out criterion, but the pre-2020 result favours a moderately larger K (6 vs 3) than the rate-hike-contaminated 2022-2023 slice.

k-fold rolling-origin pre-2020 K-selection. The single-fold pre-2020 result above is one observation; to confirm that the $K^{\star}$ choice is not a sampling artefact, we report mean ± s.d. of held-out per-observation log-likelihood and held-out KS at $K \in \{3, 6, 9, 12, 18\}$ across multiple folds. We implement four expanding-window rolling-origin folds (a five-fold design forces one fold to have only ~ 1 year of training, below the practical floor for $K = 18$ EM convergence on this dataset; averaging is per-observation so fold-length differences are not an issue). Folds: train 2014-01-03 through 2015-12-31 (~ 502 obs) and val 2016 (~ 251 obs); train through 2016 and val 2017; train through 2017 and val 2018; train through 2018 and val 2019.

Aggregate (mean / s.d. across the four folds) of per-observation held-out log-likelihood: -1.9550 / 0.290 at $K = 3$; -1.9649 / 0.313 at $K = 6$; -2.0078 / 0.296 at $K = 9$; -2.0721 / 0.251 at $K = 12$; -2.2633 / 0.308 at $K = 18$. Held-out KS pass rate (mean / s.d. across folds, very wide because Fold 2's 2017 validation year produced near-zero KS for every K as a low-variance "calm year" artefact): 61.3/41.2% at $K = 3$; 65.2/43.5% at $K = 6$; 67.5/44.1% at $K = 9$; 67.0/44.7% at $K = 12$; 67.5/45.4% at $K = 18$. Sampling-error reads on the held-out per-observation log-likelihood: $K = 6$ vs $K = 3$ has mean diff -0.0099, pooled SE 0.151, approximate $z = -0.07$ (*not* significant at the 5% level); $K = 18$ vs $K = 6$ has mean diff -0.298, pooled SE 0.155, approximate $z = -1.92$ (borderline, just below the $|z| = 1.96$ threshold).

Robustness check at half-year cadence. To verify that the result is not an artefact of the one-year fold cadence, we re-ran the same diagnostic with six expanding-window folds at half-year validation cadence (covering 2017–2019 in non-overlapping six-month chunks; train windows ~ 3.0y to ~ 5.5y). Aggregate per-observation held-out log-likelihood (mean / s.d. across six folds): -1.9221 / 0.341 at $K = 3$; -1.9160 / 0.346 at $K = 6$; -1.9568 / 0.323 at $K = 9$; -2.0109 / 0.315 at $K = 12$; -2.1281 / 0.259 at $K = 18$. Sampling-error reads: $K = 6$ vs $K = 3$ has mean diff $+0.006$, pooled SE 0.140, approximate $z = +0.04$; $K = 18$ vs $K = 6$ has mean diff -0.212, pooled SE 0.125, approximate $z = -1.70$. The half-year design's $K = 6$ vs $K = 3$ sign flips relative to the full-year design ($z = +0.04$ vs $z = -0.07$), but both magnitudes are well below the $|z| = 1$ threshold; the two designs agree that $K = 6$ vs $K = 3$ on mean held-out log-likelihood is indistinguishable from zero, and the sign flip across designs is itself evidence that the $K = 6$ / $K = 3$ choice is pure sampling noise on this data. The $K = 18$ vs $K = 6$ borderline result also replicates ($z = -1.70$ at half-year cadence vs $z = -1.92$ at full-year).

Reading. $K = 6$ is not significantly preferred over $K = 3$ on held-out per-observation log-likelihood under expanding-window rolling-origin CV at either yearly or half-yearly fold cadence. The single-fold result that initially selected $K^\star = 6$ is one realisation; both four-fold and six-fold means cannot distinguish $K = 3$ from $K = 6$ at conventional levels. The held-out KS pass-rate criterion is not informative on either design (between-fold s.d. is dominated by 2017 calendar-year artefacts that hit every K uniformly). The substantive read is that the held-out state-resolution selection on the strictly pre-2020 slice cannot distinguish $K = 3$ from $K = 6$ at conventional levels under either fold design, and the body operating point is therefore the state-resolution-robust $K^\star = 3$ (with the $K^\star = 6$ block retained as a sensitivity reference). $K^\star = 3$ is also the lower-state-count default for risk-management consumers in Table 6.

HAC-corrected sampling-error reads on the paired diff series. A Newey–West Bartlett HAC variance ($h_{\mathrm{NW}} = \lfloor n^{1/3} \rfloor$) on the per-fold val-log-lik diff series leaves $K = 6$ vs $K = 3$ insignificant at both cadences ($|z|_{\mathrm{HAC}} = 0.90$ four-fold full-year, 0.57 six-fold half-year) and moves $K = 18$ vs $K = 6$ from borderline independent-fold ($|z| = 1.92, 1.70$) to decisive ($|z|_{\mathrm{HAC}} = 3.56, 5.00$). The HAC-corrected inference therefore corroborates the body framing: $K = 18$ is decisively worse than $K = 6$ on held-out log-likelihood and is retained only for its kurtosis-fidelity advantage on the OoS window.

Penalised-ECM rate sweep. The bracket sensitivity is reported with its table in Section A.1.4; here we report the complementary $1/\nu_k$ shrinkage rate sweep referenced in the body Discussion. An exponential $1/\nu_k$ prior at rate $\lambda \in \{0, 5, 20, 50, 100, 200\}$ reduces simulated excess kurtosis monotonically: $14.30, 11.19, 8.43, 5.41, 3.96, 3.67$. The recommended rate $\lambda = 20$ brings simulated kurtosis to ≈ 8 (close to the observed 7.69) at a 1pp IS KS cost. Only the $\nu_{\min}$ state parameter actually moves ($2.1 \to 4.89$); upper bracket and median remain at 50. The overshoot is therefore a single-state artefact of the two tail regimes hitting the lower bracket under unpenalised ECM.

Walk-forward / rolling-origin OoS for CHMM-N. We define six rolling-origin folds, each train 5 years and test 1 year, refit CHMM-N at $K \in \{3, 18\}$ from scratch on each fold's train slice, and report KS pass rate, simulated kurtosis, and $|G_t|$ ACF-MAE on the fold's test slice ($N_{\text{paths}} = 500$). Across the six folds (Table 33), CHMM-N at $K = 18$ attains median (IQR) KS 67.7% [8.2, 75.0] and $|G_t|$ ACF-MAE 0.0542 [0.0508, 0.0571]; CHMM-N at $K = 3$ attains 62.1% [7.2, 78.4] and 0.0563 [0.0552, 0.0592]. Two folds carry sharply lower KS pass rates: W2 (COVID test slice, 7–8%) and W4 (rate-hike test slice, 0–1%). The four non-stress folds attain KS 61–83%, consistent with the headline OoS pass rate. The headline ranking (CHMM-N at $K = 18 \geq$ CHMM-N at $K^\star = 3$ on KS, both within 0.005 of each other on $|G_t|$ ACF-MAE) is window-robust on the four non-stress folds; on stress folds (COVID, rate hike) all CHMM rows drop sharply, reflecting the stationarity-scope limit rather than a model-selection issue. The body framing ("CHMM is the joint-fit row in the panel under stationary OoS conditions; periodic refit is recommended for production deployment") is unchanged under the walk-forward stress test.

Table 33. Walk-forward / rolling-origin OoS for CHMM-N. Six folds, each train 5 years and test 1 year; $N_{\text{paths}} = 500$ per fold. KS pass rate at $\alpha = 0.05$. W2 covers the COVID drawdown; W4 covers the start of the 2022 rate-hike cycle. The bracketed pairs in the bottom row report $[Q_1, Q_3]$ across the six folds (the empirical first and third quartiles, not a scalar IQR width).

	KS (%)		Kurt sim		ACF-MAE $\lvert G_t \rvert$	
Fold (test window)	$K = 3$	$K = 18$	$K = 3$	$K = 18$	$K = 3$	$K = 18$
W1 (2019)	63.2	82.6	2.51	2.66	0.0572	0.0525
W2 (2020 COVID)	7.2	8.2	2.42	2.50	0.0552	0.0584
W3 (2021)	82.4	74.0	5.73	6.76	0.0602	0.0571
W4 (2022 rate hike)	0.8	0.0	4.72	4.35	0.0592	0.0558
W5 (2023)	78.4	75.0	3.47	2.35	0.0511	0.0508
W6 (2024)	61.0	61.4	2.81	2.86	0.0555	0.0493
median	62.1	67.7	3.14	2.76	0.0563	0.0542
IQR	$[7.2, 78.4]$	$[8.2, 75.0]$	$[2.51, 4.72]$	$[2.50, 4.35]$	$[0.0552, 0.0592]$	$[0.0508, 0.0571]$

Four-family conditional VaR back-test. Extending the regime-conditional VaR construction of the body VaR back-test to all four emission families (CHMM-N, CHMM-t at $\lambda = 20$, CHMM-L, CHMM-GED) at $K \in \{3, 18\}$ and $\alpha \in \{0.01, 0.05\}$ yields sixteen $(K, \alpha, \text{family})$ rows on the OoS window (Table 34). Every row passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly: $p_{\text{cc}} \geq 0.089$ throughout, with all LR_{cc} statistics below the $\chi^2_2(0.05) = 5.991$ critical value. The regime-switching value proposition is therefore intrinsic to the latent-state forecast and not specific to the Gaussian emission family.

Table 34. Regime-conditional VaR back-test across four CHMM emission families on SPY OoS ($T_{\text{OoS}} = 572$, seed 20260420). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. Every $(K, \alpha, \text{family})$ row passes Kupiec, Christoffersen-ind, and Christoffersen-cc at $\alpha = 0.05$.

Family	K	α	breaches	br rate	median $\widehat{\text{VaR}}_t$	LR_{uc}	LR_{ind}	LR_{cc}
CHMM-N	3	0.01	9	1.57%	−4.56	1.62	2.36	3.98
CHMM-N	3	0.05	35	6.12%	−2.87	1.41	0.01	1.42
CHMM-N	18	0.01	9	1.57%	−5.20	1.62	2.36	3.98
CHMM-N	18	0.05	26	4.55%	−3.02	0.26	0.52	0.78
CHMM-t ($\lambda = 20$)	3	0.01	8	1.40%	−5.59	0.82	2.81	3.63
CHMM-t ($\lambda = 20$)	3	0.05	32	5.59%	−2.73	0.41	0.77	1.19
CHMM-t ($\lambda = 20$)	18	0.01	10	1.75%	−6.25	2.65	1.98	4.62
CHMM-t ($\lambda = 20$)	18	0.05	29	5.07%	−3.15	0.01	1.39	1.40
CHMM-L	3	0.01	5	0.87%	−5.45	0.10	4.74	4.84
CHMM-L	3	0.05	26	4.55%	−2.63	0.26	2.23	2.48
CHMM-L	18	0.01	9	1.57%	−6.18	1.62	0.29	1.91
CHMM-L	18	0.05	30	5.24%	−2.80	0.07	1.16	1.23
CHMM-GED	3	0.01	7	1.22%	−5.73	0.27	3.34	3.61
CHMM-GED	3	0.05	31	5.42%	−2.54	0.21	0.96	1.16
CHMM-GED	18	0.01	9	1.57%	−6.55	1.62	2.36	3.98
CHMM-GED	18	0.05	25	4.37%	−3.10	0.50	2.56	3.06

Walk-forward conditional-VaR back-test. Extending the regime-conditional VaR construction to the six rolling-origin folds of Table 33, with CHMM-N refit per fold from scratch on the train slice and forward-filtered through the test slice under fold-IS-fixed parameters, gives twenty-four (fold, K, α) rows in Table 35. The conditional Christoffersen-cc passes at $\alpha = 0.05$ on ten of twelve fold-K combinations (W1, W3, W4, W5, W6 at both $K \in \{3, 18\}$); the two failures concentrate on W2 (COVID, $p_{\text{cc}} \in \{0.011, 0.002\}$ at $K \in \{3, 18\}$). At $\alpha = 0.01$ the construction passes on nine of twelve combinations: W2 fails at both K ($p_{\text{cc}} < 10^{-3}$) and W4 at $K = 18$ fails at $p_{\text{cc}} = 0.022$ on a LR_{ind} tail-event clustering rather than a LR_{uc} coverage miss ($\text{LR}_{\text{ind}} = 7.49$ at $\text{LR}_{\text{uc}} = 0.11$). The pass-rate is therefore 19/24 aggregate at the 5% level and 22/24 if we exclude the two stress folds that the univariate walk-forward already flagged as out-of-distribution by KS in Table 33. The reading is that the conditional construction is window-conditional: under regime shifts of the W2 / W4 magnitude (a regime introduction the IS distribution does not span; the body Discussion, stationarity-scope paragraph), the conditional VaR inherits the same out-of-scope behaviour as the unconditional generators. On the four non-stress folds (W1, W3, W5, W6) plus W4 at $K = 3$ at the 5% level the conditional construction is window-robust, supporting the body framing of the body VaR back-test extending beyond the headline single-window result.

Table 35. Walk-forward regime-conditional VaR back-test for CHMM-N on six rolling-origin folds (train 5y / test 1y; seed 20260420). At each test day t, conditional $\widehat{\text{VaR}}_t(\alpha)$ is the α-quantile of the K-component Gaussian mixture predictive density under fold-IS-fixed parameters, evaluated on the full-timeline filter $\Pr(s_{t+1} \mid \mathcal{F}_t = \text{train} \cup \text{test}[1{:}t-1])$. Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. Bold p_{cc} rows fail at the 5% level.

Fold	K	α	breaches	br rate	median $\widehat{\text{VaR}}_t$	LR_{uc}	LR_{ind}	LR_{cc}	p_{cc}
W1	3	0.01	2	0.80%	−4.01	0.11	0.03	0.15	0.93
W1	3	0.05	15	5.98%	−2.43	0.48	1.18	1.65	0.44
W1	18	0.01	0	0.00%	−5.03	5.05	0.00	5.05	0.08
W1	18	0.05	12	4.78%	−2.66	0.03	0.38	0.40	0.82
W2	3	0.01	16	6.35%	−6.54	32.93	0.87	33.80	**0.00**
W2	3	0.05	23	9.13%	−4.59	7.34	1.71	9.05	**0.01**
W2	18	0.01	14	5.56%	−6.67	25.59	1.56	27.15	**0.00**
W2	18	0.05	25	9.92%	−4.54	10.11	2.56	12.68	**0.00**
W3	3	0.01	6	2.39%	−4.48	3.53	0.30	3.82	0.15
W3	3	0.05	20	7.97%	−2.47	3.98	0.30	4.28	0.12
W3	18	0.01	2	0.80%	−5.57	0.11	0.03	0.15	0.93
W3	18	0.05	15	5.98%	−3.18	0.48	1.18	1.65	0.44
W4	3	0.01	5	2.00%	−7.98	1.96	0.21	2.16	0.34
W4	3	0.05	18	7.20%	−4.16	2.26	1.99	4.24	0.12
W4	18	0.01	2	0.80%	−9.62	0.11	7.49	7.60	**0.02**
W4	18	0.05	10	4.00%	−6.63	0.56	3.80	4.36	0.11
W5	3	0.01	0	0.00%	−6.06	5.01	0.00	5.01	0.08
W5	3	0.05	8	3.21%	−3.90	1.91	0.53	2.44	0.30
W5	18	0.01	1	0.40%	−7.10	1.16	0.01	1.17	0.56
W5	18	0.05	7	2.81%	−3.85	2.96	0.41	3.37	0.19
W6	3	0.01	2	0.80%	−4.68	0.11	0.03	0.15	0.93
W6	3	0.05	13	5.18%	−2.85	0.02	0.15	0.17	0.92
W6	18	0.01	1	0.40%	−5.83	1.19	0.01	1.20	0.55
W6	18	0.05	7	2.79%	−2.98	3.06	0.40	3.46	0.18

Per-ticker $\hat{\lambda}^\star$ sweep. A per-ticker sweep $\lambda \in \{0, 5, 10, 20, 50, 100\}$ on each of the six body tickers (KS-degradation tolerance ≤ 1.5pp from $\lambda = 0$) gives per-ticker optima $\lambda^\star_{\text{SPY}} = 10$, $\lambda^\star_{\text{NVDA}} = 20$, $\lambda^\star_{\text{JNJ}} = 20$, $\lambda^\star_{\text{JPM}} = 10$, $\lambda^\star_{\text{AAPL}} = 10$, $\lambda^\star_{\text{QQQ}} = 0$: $\lambda = 20$ is a reasonable uniform default but per-ticker tuning is recommended whenever a ticker's residual kurtosis at $\lambda = 0$ is within ~ 1 unit of observed.

Cross-asset summary. The full per-construction Pipeline B panel is in Section A.9; for reference, the headline ordering is summarised in Table 36.

Table 36. Cross-asset dependence summary (Pipeline B; CHMM-N marginals at $K = 18$; 200 paths; seed 20260422). Median per-asset KS pass rate (%) and off-diagonal MAE of the simulated correlation matrix vs. observed. Bold marks the column winner. The Student-t copula at $\nu^\star = 6$ is the strongest dependence layer on the two IS columns; on OoS the Gaussian copula attains marginally lower off-diagonal MAE (0.202 vs. 0.209) while the truncated C-vine sits between the elliptical copulas and the SIM baseline. Full per-asset detail is in Section A.9.

	Median IS KS	Median OoS KS	Off-diag MAE IS	Off-diag MAE OoS
Single Index Model	77.0	87.0	0.076	0.252
Gaussian copula	97.8	**87.5**	0.030	**0.202**
Student-t copula ($\nu^\star = 6$)	**98.8**	85.8	**0.027**	0.209
Truncated C-vine (root SPY)	97.2	86.0	0.068	0.235